Stepping Stones or Traps?

The Dynamics of Informal Employment

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Abstract

Does informal employment help workers reach formality, or does it reduce their chances of ever doing so? I answer this question by combining a heterogeneous-agent model with Chile's EPS survey, which records an average of 23 years of formal and informal employment histories. Together, the model and employment histories allow me to isolate and quantify how informality affects whether a new worker ever reaches formality. I find that removing the informal sector increases the probability that a new worker ever reaches formality by 14 percentage points. This increase occurs despite the advantages informal employment offers: workers can search for formal employment at a lower cost while accumulating human capital. However, informal workers have weaker incentives than unemployed workers to search for formal employment and pay the costs of entering formality. Eliminating informal employment strengthens these incentives, but the private welfare losses outweigh the gains from expanding the tax base.

1 Introduction

Informal employment accounts for between 20 and 80 percent of the labor force in low- and middle-income countries (Bobba et al., 2025). Informality broadly encompasses employment relationships that do not comply with labor regulations. As Meghir et al. (2015) and Franco & Quintero (2026) show, these jobs offer workers lower wages and slower wage growth than formal jobs. Yet the lack of compliance spares informal workers from both entry barriers and ongoing costs of remaining formal. Furthermore, the recurring burden of remaining formal can take the form of nonpecuniary costs or the taxes and wage deductions that finance public budgets.[1] The differences in costs and benefits from the informal sector creates comparative advantage for a group of workers, in the sense of Roy (1951), to a group of that will endogenously sort into it.

What is far less is known concerns the long-term implications for workers who opt to work in the informal sector. In part, data limitations have posed a great difficulty in understanding the persistence of informality across the life cycle of workers. Most studies focusing informal workers use short rotating panels and hence are mute about the consequences of being an informal worker over long horizons.[2] Particularly, what is yet unclear is how abosrbing is informality as an employment state. In this paper I study whether informality can provide a stepping stone towards formality or hold workers back from eventually reaching it.

To answer this, I proceed in three steps: First, I build a model of heterogeneous workers with comparative advantage in being employed in either the formal or the informal sector. The model accommodates different explanations of how an informal sector can help or harm eventual access to formal for a cohort of new workers. Moreover, it defines a benchmark to separate changes to the composition of a cohort, to changes in workers' comparative advantage during spell of informal employment. I then proceed to estimate the structural parameters of the model in order to quantify the mechanisms that determine whether informality creates a bridge to formality. And finally, the estimates allow me to evaluate a counterfactual economy without informality. I show that the presence of an informal sector reduces the probability of formalization for new workers. This, however, does not imply that the informal sector creates an economic trap. In fact, the private losses from removing the option value of an informal sector outweigh the fiscal gains of expanding the tax base.

First, Section 2 presents a model of heterogeneous workers who search for islands as in Lucas & Prescott (1974). These islands are either formal or informal, and differ in their access, permanence costs, and wages. Together, these features determine how different workers search and sort between the two types of islands, employment henceforth, and unemployment. Furthermore, employment status also changes the underlying heterogeneity of workers differentially, which I interpret as human capital. Tenure in the informal sector, then, can alter workers' incentives to search for

[1] Registration costs are an example of an entry cost to formality and are studied by de Mel et al. (2013). Examples of recurring costs of formality include Zárate (2026), who shows that commuting frictions act as a nonpecuniary cost of formality, and Olken et al. (2024) and Bernal et al. (2025), who focus on fiscal costs.

[2] Bosch & Maloney (2010) and Cano-Urbina (2015) are among other studies that address similar questions using short panels.

formal employment for better or for worst.

The model possesses three distinct mechanisms that allow me to explain the observed patterns of long-term informal employment. First, the costs of searching for formal employment differ for informal and unemployed workers. Second, informal workers value formality differently than unemployed workers, which is reflected in their search efforts. In that case, if accessing formal employment is costly, then the same cost might deter different workers from search depending on their current labor status. Search costs, then, can alter the search in intensive margin, but entry costs determine the extensive margin. And finally, the dynamics of human capital speccific to informal sector can either increase or decrease comparative advantage relative to the formal sector. The interaction of the three mechanisms is what determines whether informality is an absorbing state or facilitates transitions towards formal employment.

To quantify the three mechanisms, Section 3 presents Chile's worker survey, the Encuesta de Protección Social (EPS). The EPS reconstructs labor histories for formal and informal workers, and allows me to observe on average 23 years of working and unemployment spells. More importantly, the long horizons of the labor history allow me calculate new statistics that shed light about the long-term implications of informal employment. First, I show that the duration distributions of completed spells formal employment, informal employment, and unemployment are ordered in the stochastic dominance sense. The stochastic order extends the same ordering that Bosch & Maloney (2010) documented for mean durations.

Second, the large sample size allows me to calculate the probability of transitioning out of informality towards formal employment, conditioning on completing long spells. I find that an informal worker, after 6 months, has a 7.8% hazard of transitioning to formal employment. For workers who remain over 5 years, the same probability drops to 2%. This, however, should not be interpreted as informality harming the prospects of eventual formalization. The reduction in the hazard can reflect changes in the incentives for formalization, as well as changes in the composition of workers.

In order to interpret the hazard, Section 4 estimates the model. The EPS provides three types of variation that allow me to identify the structural parameters of the model: First, variation in the observed duration of spells, interacted with changes in employment status, disciplines the differences in human capital dynamics, and search and transition frictions. Second, differences in wages for formal and informal workers identify the static returns of employment. And finally, the aggregate allocations of workers discipline the remaining parameters.

The estimated economy anchors the parameters that determine if informality is indeed a stepping stone to formality. First, I find that the raw formal premium is 17.2%, but it shrinks to 9.0% after taxes. However, the main reason why low-skill workers sort into informality is the nonpecuniary costs of formality which acounts for almost half of thei anual wages. Second, I find that informality increases access to formality relative to unemployment by providing lower search costs. Yet, informal workers value formal employment less than unemployed workers, and hence, the vast majority of informal workers are deterred from searching for formal employment. The

reason is not because formality is not a valuable state, but because the gains from transitioning cannot offset the entry costs. And finally, I find that informality provides almost as much human capital accumulation as formal employment, favoring the stepping stone interpretation. Unlike formal workers, however, informal workers are more exposed to idiosyncratic shocks to their human capital.

Finally, Section 5 evaluates a counterfactual economy in which I remove the informal option for workers. On average, the probability of eventual formalization for a new worker increases by 14.08 pp in the absence of an informal sector. This effects happens in spite of informality having search cost and human capital acumulation advantage over unemployment. The explanation then lies in the entry barrier: Informality bridges the gap in value formal employment, and unemployment, to the point that paying the entrey barrier is no worth it. Yet, informality provides option value for new workers. The private value of informality outweights the fiscal gains from expanding the tax base in the absence of it.

Literature Review. This paper contributes to three strands of literature. The first studies the determinants of informality, mostly through the decisions of firms. As Bobba et al. (2025) review, muc of the quantitative work models firms that decide whether to register, whom to hire off the books, and how large to grow when formality is costly (La Porta & Shleifer, 2014; Ulyssea, 2018). Unlike these papers, I study informality from the side of workers, as a state that they enter and leave over their careers.

Second, I take the tools that the macro-labor literature uses to study labor-market transitions (see Castro et al., 2024, for a review) and apply them to a setting in which workers move in and out of informality. This becomes particularly relevant in developing economies, where high labor-market flows reflect a slippery job ladder, as Donovan et al. (2023) document.[3] Following F. Alvarez et al. (2024), who separate structural duration dependence from dynamic selection in a stopping-time model, I build a model of informality in which human capital moves continuously and workers choose where to search. The model is able to quaitify the difference dynamic forces how informality impacts any prospects of formalization..

Finally, I contribute to the literature on the worker side of informality, which asks who sorts into informal jobs and whether they have lasting effects. Magnac (1991) contrasts a segmented market, in which barriers restrict access to formal jobs, with a competitive one, in which workers sort by comparative advantage. Meghir et al. (2015) observe that workers' frequent moves between the sectors are hard to reconcile with comparative advantage, at least as a permanent characteristic, and explain sorting through search frictions. López García (2015), Finamor (2025), and Bobba et al. (2022) explain it through differences in comparative advantage, wealth, or schooling. Lasting effects arise in Seminario-Amez (2021), where informal jobs can help workers climb a job ladder, and in Franco & Quintero (2026), where human capital grows at different rates across sectors. Closest to this paper, Bobba et al. (2021) bring the two together, combining search frictions with

[3]Their paper leaves open how selection interacts with labor-market frictions, which they point to as an avenue for future theories.

human capital that grows faster on formal jobs. In their setting, however, workers' incentives to search and move do not change as a spell unfolds. In this paper, comparative advantage evolves with human capital, and thus, the search incentives change within a spell and differentially across employment states. In this setting, both of Magnac's views then hold, for different workers. The least skilled sort into informality by comparative advantage, as in the competitive view, while workers with a comparative advantage in formal employment are held back by the entry cost, as in the segmented view.

2 The Model

In this section, I develop a model of how workers move across unemployment, informal employment, and formal employment. Human capital determines each worker's comparative advantage across sectors, while search frictions and the cost of entering formality slow their movement toward the sector they prefer. Informal employment changes how human capital evolves, how much workers gain from becoming formal relative to the entry cost, and how efficiently they search for formal employment, so it can either ease or hamper their access to formality. I first set up the environment and the equilibrium, and then study how these three mechanisms shape where workers search and how their exit rates evolve over a spell.

2.1 Production and Wages

Time is continuous and production takes place on a continuum of islands, indexed by $j \in [0, 1]$. Islands are identical in productivity and differ only in their status, formal or informal. Each island produces a homogeneous final good, which trades freely across islands and serves as the numeraire.[4] Workers, unlike goods, cannot move freely across islands. They move only through the costly search described below, so each island is a separate labor market in which competition sets the wage.

Within each of these markets, workers differ only in their human capital. A worker with human capital x supplies x efficiency units of labor, and an island j of status $\ell \in \{i, f\}$ produces

$$y_{\ell j}(t) = A \int_0^\infty x \, n_{\ell j}(x, t) \, \mathrm{d}x, \tag{1}$$

where $n_{\ell j}(x, t)$ is the employment of workers with human capital x on the island and $A > 0$ is a common productivity. Note that the technology in equation (1) exhibits constant returns to scale in efficiency units of labor, and hence the marginal product for every worker depends only on their own human capital, not on who else works there.

[4] Competitive islands with frictional mobility across them follow Lucas & Prescott (1974) and F. Alvarez & Shimer (2011). In those papers, islands differ in product demand or in productivity, whereas here they differ only in status.

Formal and informal islands differ only in the wedge between a worker's marginal product and their wage. On formal islands, wages carry a proportional tax wedge τ. On informal islands, employment avoids the tax, but a fraction $\alpha_0 \in [0, 1)$ of output is lost to concealment and other costs of evading enforcement. Unlike the tax, this loss is not a transfer to the government but a deadweight cost for the economy. Since labor markets within islands are competitive, each worker earns their marginal product net of these wedges:

$$w_f(x) = \frac{A}{1+\tau}\, x, \qquad w_i(x) = A(1-\alpha_0)\, x. \tag{2}$$

Both wages are proportional to human capital, $w_\ell(x) = a_\ell x$, where a_ℓ is the return to human capital on islands of status ℓ. Two properties of these wages shape workers' choices: First, formal employment pays more per unit of human capital when the enforcement loss exceeds the tax wedge, $\alpha_0 > \tau/(1+\tau)$. Second, neither wage depends on how workers are allocated across islands, since the marginal product of each worker does not depend on who else works there.[5] Workers therefore take wages as given when they decide where to search. All islands of the same status pay the same wages, so a worker cares only about the status of their island. I drop the island index from here on, write $n_\ell(x, t)$ for employment on islands of status ℓ, and say that these workers are employed in sector ℓ. A worker on no island is unemployed.

2.2 Workers

The economy contains a unit mass of workers, who search across these islands for employment at the wages each island pays, discount the future at rate $\rho > 0$, and leave the labor force at Poisson rate $\delta > 0$. Each departing worker is immediately replaced by a new unemployed worker, whose human capital is drawn from a distribution $G(x)$ with continuous density $g(x)$ on $\mathbb{R}_+$. At each point in time, a worker's labor market state, $\ell \in \{u, i, f\}$, records whether they are unemployed or employed on an informal or a formal island.

Workers are risk neutral, cannot save, and supply labor inelastically, so their only choices are where to search and when to quit. Each worker also receives a common lump-sum transfer t_t, which, given linear utility, leaves these choices unchanged.[6] An unemployed worker receives government benefits $b \geq 0$, while a worker with human capital x employed in sector $\ell \in \{i, f\}$ earns the wage $w_\ell(x)$. Formal workers also bear a recurring nonpecuniary cost $F > 0$, which summarizes compliance requirements and other non-wage burdens of formality.

Transition Frictions. Workers cannot move at will to the island they prefer, because three frictions slow their movement across labor market states. First, moving into employment requires

[5] A natural extension lets enforcement losses grow with the size of an informal island, since larger informal islands are easier for the government to detect. Informal production then exhibits decreasing returns to scale, and, as in Lucas & Prescott (1974), wages depend not only on workers' human capital but also on the size of their island, so they must be solved jointly with the distribution of workers.

[6] Because the transfer shifts all state values equally, I suppress it until the equilibrium definition.

search. A worker in state $\ell \in \{u, i, f\}$ chooses how hard to search for islands of each status $\ell' \in \{i, f\} \setminus \{\ell\}$, and their search effort delivers opportunities on those islands at rate $\eta_\ell^{\ell'}(x) \geq 0$.[7] Searching harder, however, is increasingly costly, and the cost depends on where the worker searches from:

$$\psi_\ell^{\ell'}(\eta_\ell^{\ell'}) = \frac{\kappa_\ell^{\ell'}}{2}\left(\eta_\ell^{\ell'}\right)^2, \qquad \kappa_\ell^{\ell'} > 0.$$

The parameter $\kappa_\ell^{\ell'}$ captures the search cost that a worker in ℓ faces when looking for islands of status ℓ'. Because this cost varies by origin, the same type of island can be easier to reach from one state than from another.

Second, joining a formal island requires a one-time entry cost $\Psi_f \geq 0$, which captures paperwork, legal requirements, and other costs of formalization, while informal islands carry no such cost, $\Psi_i = 0$. A worker therefore moves into formality only when the gain covers the entry cost, but moves into informality whenever the gain is positive.

Third, employment ends exogenously. Informal and formal employment spells end at rates δ_i and δ_f, sending workers into unemployment. When these rates differ, the two sectors offer different stability, which workers weigh alongside wages. Not every exit from employment is involuntary, however. Workers may also leave their island and quit into unemployment once unemployment is at least as valuable as their current employment, which can happen when their human capital is low and their wage is close to the unemployment benefit.

Human Capital. At a given level of human capital, these payoffs and frictions fix each worker's incentives across states. If human capital were fixed, those incentives, and with them where the worker searches, would never change. Human capital instead evolves according to a process that depends on the worker's current state:

$$\mathrm{d}x_t = \mu_\ell \,\mathrm{d}t + \sigma_\ell \,\mathrm{d}W_t, \qquad \ell \in \{u, i, f\},$$

where W_t is a standard Brownian motion, and human capital cannot fall below zero.[8]

In each state, human capital grows or depreciates at rate μ_ℓ. Unemployed workers lose human capital, $\mu_u < 0$, while formal workers build it, $\mu_f > 0$. For informal employment, I impose only that it is no worse than unemployment and no better than formal employment for accumulation, $\mu_u \leq \mu_i \leq \mu_f$.[9] This restriction accommodates three possibilities: informal workers may lose human capital no faster than unemployed workers, preserve it, or build it no faster than formal workers. Workers also experience idiosyncratic shocks to their human capital, whose volatility $\sigma_\ell > 0$ may differ across states.

[7] The feasible search links are $u \to i$, $u \to f$, $i \to f$, and $f \to i$.

[8] The full process is $\mathrm{d}x_t = \mu_\ell \,\mathrm{d}t + \sigma_\ell \,\mathrm{d}W_t + \mathrm{d}L_t$, where L_t is the minimal nondecreasing process that keeps $x_t \geq 0$ and increases only when $x_t = 0$.

[9] The upper bound is in line with the faster wage growth of formal workers documented by Franco & Quintero (2026).

Value Functions and HJB Equations. Where to search and when to quit depend on how much each state is worth to a worker. Let $U(x)$, $V_i(x)$, and $V_f(x)$ denote the values of unemployment, informal employment, and formal employment for a worker with human capital x. Since every worker enters the labor force unemployed, I start with the value of unemployment, which satisfies the Hamilton–Jacobi–Bellman (HJB) equation

$$\begin{aligned}\rho U(x) = b + \mu_u \partial_x U(x) + \tfrac{\sigma_u^2}{2}\partial_{xx} U(x) - \delta U(x) \\ + \max_{\eta_u^i, \eta_u^f \geq 0} \left\{ \sum_{\ell \in \{i,f\}} \left[\eta_u^\ell(x) \left[V_\ell(x) - U(x) - \Psi_\ell \right] - \psi_u^\ell\left(\eta_u^\ell(x)\right) \right] \right\}.\end{aligned} \tag{3}$$

Equation (3) values unemployment as the benefit a worker collects while waiting plus the option of finding employment. While waiting, the worker collects the benefit b, but their human capital fluctuates with shocks and drifts down, which lowers the value of waiting. Waiting also ends at rate δ, when the worker leaves the labor force. Meanwhile, the worker generates opportunities in each sector and moves there when one arrives. They choose each search intensity by weighing the gain from the destination, net of its entry cost, against the cost of generating opportunities.

Once employed, a worker faces two margins that unemployment lacks, exogenous separations and the option to quit back into unemployment. The value of employment therefore satisfies a variational inequality,

$$\begin{aligned}\rho V_\ell(x) \geq w_\ell(x) - \mathbb{1}_f F + \mu_\ell \partial_x V_\ell(x) + \tfrac{\sigma_\ell^2}{2}\partial_{xx} V_\ell(x) - \delta V_\ell(x) \\ + \delta_\ell \left[U(x) - V_\ell(x) \right] + \max_{\eta_\ell^{-\ell} \geq 0} \left\{ \eta_\ell^{-\ell} \left[V_{-\ell}(x) - V_\ell(x) - \Psi_{-\ell} \right] - \psi_\ell^{-\ell}\left(\eta_\ell^{-\ell}\right) \right\},\end{aligned} \tag{4}$$

subject to $V_\ell(x) \geq U(x)$, where $-\ell$ denotes the other sector.

The employment equation follows the same logic as the unemployment HJB. An informal worker earns $w_i(x)$, and a formal worker earns $w_f(x)$ net of the recurring cost F. While employed, the worker's human capital keeps drifting and fluctuating, and they can search for employment in the other sector. Employment also ends at rate δ_ℓ, when a separation sends them back to unemployment.

Because an employed worker can always quit, the value of employment can never fall below the value of unemployment, and the inequality in (4) marks where this option binds. Whenever $V_\ell(x) > U(x)$, the worker prefers to stay employed, and the inequality holds with equality. Whenever $V_\ell(x) = U(x)$, employment is worth no more than unemployment, and the worker quits. The indifference point $\bar{x}_u^\ell$ separates workers who quit from those who stay.

The same equations determine how hard workers search. For $\ell \in \{i, f\}$, optimal search effort

is

$$\eta_u^{\ell}(x) = \frac{[V_\ell(x) - U(x) - \Psi_\ell]^+}{\kappa_u^{\ell}}, \qquad \eta_\ell^{-\ell}(x) = \frac{[V_{-\ell}(x) - V_\ell(x) - \Psi_{-\ell}]^+}{\kappa_\ell^{-\ell}}, \tag{5}$$

Search responds on two margins. On the extensive margin, a worker searches toward a destination only when the gain from reaching it exceeds its entry cost. On the intensive margin, they search harder when this net gain is larger and when their search cost is lower. With $\Psi_f > 0$, the extensive margin binds for some workers who would rather be formal, since their gain from formal employment is positive but does not cover the entry cost.

These search rules shape a central choice for an unemployed worker who would rather be formal: whether to accept informal employment in the meantime. Accepting it does not end the search for formality, because informal workers can keep searching for formal islands. It does, however, change that search in three ways relative to unemployment: First, human capital changes at rate μ_i instead of μ_u, and hence the gain from formality evolves differently as the spell unfolds. Second, formal opportunities cost κ_i^f to generate instead of κ_u^f, so the same gain can buy more or less search effort. Third, because the worker now gives up a wage rather than benefits, the gain from becoming formal shrinks from $V_f - U$ to $V_f - V_i$, which makes the entry cost harder to cover. These changes last only as long as the spell, since a separation, at rate δ_i, returns the worker to unemployment.

2.3 Distributional Dynamics

Individual choices add up to a distribution of workers across islands and unemployment, which, unlike values and policies, can change over time. Let $m_\ell(x,t)$ denote the density of workers with human capital x in state $\ell \in \{u,i,f\}$ at time t. Four forces move workers across this distribution: changes in human capital, search, separations and quits, and labor-force turnover.

First, accumulation, depreciation, and shocks move workers up and down the human-capital distribution while they remain in the same labor market state. For a density m of workers over human capital, the net number of workers in state ℓ who rise past human-capital level x per unit of time is

$$\mathcal{J}_\ell[m](x,t) = \mu_\ell\, m(x,t) - \frac{\sigma_\ell^2}{2}\partial_x m(x,t). \tag{6}$$

In the first term, workers build human capital and move up the distribution when $\mu_\ell > 0$, and the opposite happens when $\mu_\ell < 0$. How many workers rise past x depends on how fast they move, μ_ℓ, and on how many of them are at x, $m_\ell(x,t)$. The second term captures shocks. A shock is as likely to push a worker up as down, but where workers are concentrated, more of them are pushed out than in, so shocks spread workers toward less populated levels of human capital. To see how these flows change the density, consider the workers in a narrow band of human capital just above x. Drift and shocks move workers into and out of the band at both ends, in either direction, and the flux measures the net upward flow at each end. The band gains workers when the net upward flow is larger at its lower end than at its upper end, that is, when the flux falls with human capital.

This is why drift and shocks change the density at x by $-\partial_x \mathcal{J}_\ell[m_\ell](x,t)$.

Second, search moves workers across labor market states. A worker in state ℓ with human capital x finds an opportunity on an island of status ℓ' at rate $\eta_\ell^{\ell'}(x)$, so the number of workers moving from ℓ to ℓ' at x per unit of time is $\eta_\ell^{\ell'}(x) m_\ell(x,t)$. Since workers carry their human capital with them, the same term is an outflow from ℓ and an inflow into ℓ' at the same x. This holds whether the worker moves from unemployment onto an island or between islands.

Third, employment ends through separations and quits. Workers in sector $\ell \in \{i, f\}$ lose their employment at rate δ_ℓ, which generates an outflow $\delta_\ell m_\ell(x,t)$ from that sector and an equal inflow into unemployment. Workers also quit, leaving employment once their human capital falls to the indifference point $\bar{x}_u^\ell$. Let $q_\ell(x,t)$ denote this quitting flow. It occurs only at the indifference point, where it equals the net number of workers who fall past that point, $-\mathcal{J}_\ell[m_\ell](\bar{x}_u^\ell, t)$, and it is zero when quitting is absent. Like separations, quits move workers into unemployment at their current human capital.

Finally, labor-force exit and replacement change the composition of the population. Workers exit the labor force at rate δ from every state, which removes $\delta m_\ell(x,t)$ from each density. Exiting workers are replaced one for one by entrants, so the labor force has the perpetual-youth structure of overlapping cohorts with a constant size. Entrants start unemployed with human capital drawn from $g(x)$, and since the population has unit mass, entry adds $\delta g(x)$ to unemployment.

Adding up the four forces gives the Kolmogorov forward equations, which track how the density of workers in each state changes over time:

$$\partial_t m_\ell(x,t) = \begin{cases} -\partial_x \mathcal{J}_u[m_u](x,t) - \big[\eta_u^i(x) + \eta_u^f(x) + \delta\big] m_u(x,t) + \delta g(x) \\ \qquad\qquad + \delta_i m_i(x,t) + \delta_f m_f(x,t) + q_i(x,t) + q_f(x,t), & \ell = u, \\ -\partial_x \mathcal{J}_i[m_i](x,t) - \big[\eta_i^f(x) + \delta_i + \delta\big] m_i(x,t) \\ \qquad\qquad + \eta_u^i(x) m_u(x,t) + \eta_f^i(x) m_f(x,t) - q_i(x,t), & \ell = i, \\ -\partial_x \mathcal{J}_f[m_f](x,t) - \big[\eta_f^i(x) + \delta_f + \delta\big] m_f(x,t) \\ \qquad\qquad + \eta_u^f(x) m_u(x,t) + \eta_i^f(x) m_i(x,t) - q_f(x,t), & \ell = f. \end{cases} \tag{7}$$

Because search, separations, and quits only move workers across states, and entry exactly offsets exit, the population remains constant at all times.[10] The densities also deliver the aggregates that determine the equilibrium. In particular, the total number of workers across levels of human capital gives the share of the population in each state, $\mathbf{M}_\ell(t)$.

2.4 Equilibrium

In an equilibrium, every island's labor market clears at the competitive wages in equation (2), workers search and quit optimally at these wages, the distribution of workers evolves as their choices

[10] Appendix A.2 shows that the process preserves mass.

dictate, and the government balances its budget. The following definition states this formally.

Definition 1 (Equilibrium)**.** *An equilibrium consists of wages* $\{w_i(x), w_f(x)\}_{x\geq 0}$*, island employment* $\{n_i(x,t), n_f(x,t)\}_{x\geq 0}$*, worker distributions* $\{m_\ell(x,t)\}_{x\geq 0,\,\ell\in\{u,i,f\}}$*, lump-sum transfers* $\{t_t\}_{t\geq 0}$*, value functions* $(U(x), V_i(x), V_f(x))$*, and policy functions* $\left(\eta_u^i(x), \eta_u^f(x), \eta_i^f(x), \eta_f^i(x)\right)$*, such that:*

(i) Each island's labor market is competitive, so wages satisfy equation (2).

(ii) Workers optimize: value functions and search policies satisfy equations (3)–(5).

(iii) Worker distributions satisfy equation (7), starting from an initial distribution $m_\ell(x,0)$ *and with the corresponding boundary conditions.*

(iv) Each island's labor market clears:

$$n_{\ell j}(x,t) = n_\ell(x,t) = m_\ell(x,t), \qquad \ell \in \{i,f\}, \tag{8}$$

and the population constraint

$$1 - \mathbf{M}_u(t) = \mathbf{M}_i(t) + \mathbf{M}_f(t)$$

holds.

(v) The government balances its budget.

Government Budget. The government sets the tax wedge τ, the enforcement rate α_0, and unemployment benefits b, and each instrument acts on its budget through workers' choices. A higher tax wedge raises revenue per formal worker but lowers the return to formal employment, which can shrink the tax base. More generous benefits raise spending directly and make unemployment more attractive. Stricter enforcement raises no revenue of its own, but by lowering the return to informality it can push workers into the tax base. The lump-sum transfer t_t returns whatever revenue is left after benefits, so the government's flow budget constraint is

$$\tau \int_0^\infty w_f(x) m_f(x,t)\,\mathrm{d}x = b\,\mathbf{M}_u(t) + t_t, \tag{9}$$

Since the population is normalized to one, the transfer each worker receives equals the aggregate transfer, and it becomes a lump-sum tax when benefits exceed revenue. The budget thus pins down the transfer at every instant, given the policy instruments and the distribution of workers.

Stationary Equilibrium. I focus on a stationary equilibrium, in which individual workers keep moving across states and their human capital keeps changing, but the flows into and out of each level of human capital in each state balance. The distribution of workers, and with it the transfer, is then constant over time, and I drop the time argument from here on.

Two features of the model make the stationary equilibrium simple to characterize. First, wages do not depend on how workers are allocated, as Section 2.1 showed, so workers solve their problem at known wages and their choices never feed back into wages. Second, labor-force turnover pins down the distribution that these choices generate. Every worker is eventually replaced by an entrant, so the stationary distribution is an average over entrants' histories and does not depend on where the economy starts. Existence and uniqueness require only two further conditions. Enforcement must leave informal workers part of what they produce, $\alpha_0 < 1$, and entrants must arrive with finite human capital on average, which keeps the wage bill, and with it the budget, finite. The following theorem summarizes these properties.

Theorem 1. *Suppose that* $\alpha_0 < 1$ *so that* $w_i(x) > 0$ *for every* $x > 0$*, and*

$$\int_0^\infty x \, \mathrm{d}G(x) < \infty.$$

Then a stationary equilibrium exists and it is unique.

Proof. See Appendix A.3.

2.5 Dynamic Sorting and Duration Dependence

With the stationary equilibrium in hand, I turn to how informal employment shapes a worker's access to formality. I first define comparative advantage across the three states and show how it shapes where workers search. I then follow cohorts through their spells to see how these choices determine destination-specific hazards.

A worker has a comparative advantage in one state over another when the value of that state is higher at their human capital. Comparative advantage is first shaped by the static payoffs. By Equation (2), a worker with sufficiently low human capital earns less in either sector than the benefit b. Thus, in the absence of any dynamics, their comparative advantage lies in being unemployed. Yet, as human capital increases, unemployment becomes a less attractive state. For employed workers, the comparison depends both on the returns to human capital and the nonpecuniary cost F. Whenever the enforcement loss exceeds the tax wedge τ, workers with enough human capital earn formal wages high enough to offset the nonpecuniary cost, and they favor formal employment. For workers with low human capital, however, formal wages do not offset the nonpecuniary cost, and they favor informality.

Two dynamic forces build on these payoffs. First, employment stability can magnify the net income of being employed. A sector with a lower separation rate δ_ℓ allows the worker to collect their income for longer. Second, human-capital accumulation raises the value of employment, $V_\ell(x)$, because it raises the worker's wage while the spell lasts.

Unemployed workers with more human capital still value unemployment more, even though they earn no return on their human capital and lose some of it as the spell unfolds. Consider a

worker who never leaves unemployment. Their value is $b/(\rho+\delta)$, whatever their human capital. The option of searching and eventually becoming employed, however, raises the value of unemployment above this level, and more so for workers with more human capital. Similarly, an employed worker cannot be worse off than an unemployed worker since quitting is always an alternative. The indifference points $\bar{x}_u^\ell$ mark the level of human capital at which the worker is indifferent between being unemployed and employed in sector $\ell \in \{i, f\}$.

Optimal Search and Sorting. Comparative advantage determines where workers would rather be employed, and their search policies in Equation (5) follow it. Formalization, however, is worth less to an informal worker than to an unemployed worker with the same human capital. Consider, for example, a worker with human capital $x > \bar{x}_u^i$. For this worker, $V_f(x) - V_i(x) < V_f(x) - U(x)$, since $V_i(x) > U(x)$. A smaller gain need not mean less search, however, because search effort depends on the search cost as well as on the gain. If informal workers face the lower search cost, $\kappa_i^f < \kappa_u^f$, they can search harder for formal islands than unemployed workers despite their smaller gain.

Becoming formal also requires paying the entry cost Ψ_f, and unlike the intensive margin, the extensive margin does not depend on the search cost. A worker searches for formal islands only when their gain covers the entry cost, whatever their search cost. As a result, a worker whose gain out of unemployment covers the entry cost can be held back from formality once they become informal. Similar to before, consider an unemployed worker for whom $0 < V_f(x) - V_i(x) < \Psi_f < V_f(x) - U(x)$. If this worker reaches informality before formality, they stop searching for formal islands, even though they would still rather be formal. The search cost κ_ℓ^f can therefore offset the smaller gain from formality only on the intensive margin, and Ψ_f governs the extensive margin.

Human-capital accumulation adds a dynamic dimension to this margin. Search policies only reflect current comparative advantage across sectors. But as a spell unfolds, an unemployed worker's human capital falls faster, on average, than an informal worker's, and with it their incentive to formalize. The informal sector thus sets a trade-off for workers. On the one hand, informality lowers the gain from formalization today. On the other hand, it shields human capital from the depreciation of unemployment. Moreover, if informality provides human capital accumulation, $\mu_i > 0$, the incentives to formalize that it lowers today return in the long run. If instead informality only slows the depreciation of unemployment without offsetting it, $\mu_u < \mu_i \leq 0$, the worker's incentives keep falling, and they may never transition to formality.

Hazards and Duration Dependence. Search frictions make sorting gradual, and because opportunities arrive at random, workers face competing risks. To see this, consider an unemployed worker who is searching for both formal and informal employment. Even if their comparative advantage favors formal employment, they might first find an informal island and move to it. The composition of workers in a state therefore differs at the start of a spell and keeps changing as the spell unfolds.

More explicitly, consider a cohort of workers who start a spell in state ℓ at the same time. Search policies depend only on human capital, so the cohort's human-capital dynamics and exits follow the same laws in equation (7), except that the cohort receives no new entrants. When a worker moves to another state, quits, or exits the labor force, their spell ends and their duration clock resets, so they are no longer part of the cohort. The share of the cohort still in ℓ therefore shrinks as duration d increases. The rate at which members still in the cohort at duration d leave ℓ for destination ℓ' is described by the destination-specific hazard,

$$\lambda_\ell^{\ell'}(d) = \mathbb{E}_d[\eta_\ell^{\ell'}(x)], \tag{10}$$

where $\mathbb{E}_d$ denotes the expectation over the human-capital distribution $\bar{m}_\ell(x, d)$ of the members still in ℓ at duration d. Observe that if search policies did not depend on human capital, the destination-specific hazard would not change with duration.

Nevertheless, search does depend on human capital, and hence a hazard changes with duration for two reasons. First, human capital changes for the surviving members of the cohort as a spell unfolds. Second, the workers who search most leave first, changing the composition of the remaining members of the cohort. To isolate the second force, consider the case where human capital is fixed. Each worker then has the same search intensity throughout the spell, so their own exit rate is constant. The cohort hazard nevertheless falls, because workers who search more intensively leave first and the remaining pool consists increasingly of those who search less. I refer to this force as dynamic selection, and for a hazard between the two sectors, it takes the form,

$$\partial_d \lambda_\ell^{\ell'}(d) = -\operatorname{Var}_d\big[\eta_\ell^{\ell'}(x)\big] \leq 0,$$

so the hazard declines whenever members of the cohort differ in search intensity. The fixed-human-capital case also shows how the destinations of exits change as the spell progresses. To see this, consider a cohort of formal workers, and define

$$\chi_f^i(d) = \frac{\lambda_f^i(d)}{\lambda_f^i(d) + \lambda_f^u(d)} \tag{11}$$

as the share of exits from the spell at duration d that end in informal employment. Whenever human capital is fixed, no formal worker will leave voluntarily for unemployment, and $\lambda_f^u(d)$ is constant and equal to δ_f. By contrast, workers who search for informal employment leave first, and the workers who remain search little or not at all. The informal share of exits $\chi_f^i(d)$ therefore falls as duration increases, and reaches zero whenever there are no workers left searching.

When human capital changes during a spell, however, the search intensity of the members of the cohort changes with it, because human-capital dynamics shift the workers' comparative advantage as the spell unfolds. I refer to this force as structural duration dependence. Its effect on the hazard depends on how human capital changes within the state, which in turn is described by equation (6) applied to the cohort's distribution, $\mathcal{J}_\ell[\bar{m}_\ell](x, d)$. Human-capital dynamics also lead some employed workers to quit into unemployment. Consider an employed worker in sector

ℓ whom a streak of negative idiosyncratic shocks takes down to the indifference point $\bar{x}_u^\ell$, below which unemployment dominates. I denote by $\bar{q}_\ell(d)$ the rate at which members of the cohort in sector ℓ transition voluntarily to unemployment at duration d.[11] The following theorem details how structural duration dependence and dynamic selection, including the selection that quits induce, determine the shape of the hazard.

Theorem 2. *Consider a cohort in a stationary equilibrium. At almost every duration $d \geq 0$ with positive survivor mass, the unemployment hazards satisfy*

$$\partial_d \lambda_u^{\ell'}(d) = \underbrace{\int_0^\infty \partial_x \eta_u^{\ell'}(x) \mathcal{J}_u[\bar{m}_u](x,d)\,\mathrm{d}x}_{\substack{\text{structural duration} \\ \text{dependence}}} \underbrace{-\operatorname{Cov}_d\Big(\eta_u^{\ell'}(x), \eta_u^i(x) + \eta_u^f(x)\Big)}_{\text{dynamic selection}}, \qquad \ell' \in \{i, f\}. \tag{12}$$

For an employment origin $\ell \in \{i, f\}$ with an upper continuation region, the hazard toward the other employment state satisfies

$$\partial_d \lambda_\ell^{\ell'}(d) = \underbrace{\int_{\bar{x}_u^\ell}^\infty \partial_x \eta_\ell^{\ell'}(x) \mathcal{J}_\ell[\bar{m}_\ell](x,d)\,\mathrm{d}x}_{\substack{\text{structural duration} \\ \text{dependence}}} \underbrace{-\operatorname{Var}_d[\eta_\ell^{\ell'}(x)]}_{\text{dynamic selection}} + \underbrace{\bar{q}_\ell(d)[\lambda_\ell^{\ell'}(d) - \eta_\ell^{\ell'}(\bar{x}_u^\ell)]}_{\text{quit selection}}. \tag{13}$$

Proof. See Appendix A.4.

Theorem 2 shows explicitly how each force acts on the instantaneous change of the hazard. First, the structural term weighs how the human capital of the cohort changes, $\mathcal{J}_\ell[\bar{m}_\ell](x,d)$, by how much their search intensities change, $\partial_x \eta_\ell^{\ell'}(x)$. To better understand its sign, consider the case without shocks, $\sigma_\ell = 0$, in which the flux reduces to its deterministic component, $\mu_\ell \bar{m}_\ell(x,d)$. When search intensity increases with human capital, the structural term takes the sign of the drift. Second, the net effect of selection depends on the origin. Out of employment, it works as in the case where human capital is fixed, through the variance of search toward the other sector. Out of unemployment, the two destinations compete for the same workers, and selection works through total exits instead. Unemployed workers who search more in total leave sooner, independently of their destination. Search toward a given sector therefore falls among the remaining members of the cohort when the workers who search most toward it are also those who search most overall.

Quit selection adds a third term to the employment hazards. Workers who reach the quitting threshold leave for unemployment, removing their search intensity from the average. Quits therefore raise the hazard out of employment if those workers search less than the average member of the cohort. Exogenous separations and labor-force exits also shrink the cohort, but because their rates do not depend on human capital, they do not change its composition.

Theorem 2 signs the change in a hazard whenever its forces agree. Consider first an unemployed

[11] In a stationary equilibrium, the rate at which members of the cohort quit is precisely the net flow of workers at the indifference point, and thus, $\bar{q}_\ell(d) = -\mathcal{J}_\ell[\bar{m}_\ell](\bar{x}_u^\ell, d)$. If unemployment is dominated along the entire support of the human capital space, i.e., there is no quitting, then the reflecting barrier at zero implies that $\bar{q}_\ell(d) = 0$.

worker. As the spell unfolds, their human capital depreciates, $\mu_u < 0$, and their comparative advantage and search effort fade with it. Structural duration dependence therefore lowers the hazard at longer durations. Dynamic selection compounds this decline. Since workers with more human capital are the ones who search the most overall, those who remain are the ones who initially search less. Corollary 1 formalizes this intuition in the absence of idiosyncratic shocks.

Corollary 1. *Consider a stationary equilibrium with $\sigma_u = 0$ in which $\partial_x \eta_u^{\ell'}(x) \geq 0$ for each $\ell' \in \{i, f\}$. For an unemployment cohort with finite mean initial human capital, $\partial_d \lambda_u^{\ell'}(d) \leq 0$ for each $\ell' \in \{i, f\}$.*

Proof. See Appendix A.4.

The corollary thus describes how comparative advantage fades over a long unemployment spell, both for each worker and for the cohort as a whole. Each worker who remains unemployed loses human capital, and with it part of their gain from either type of employment, while the workers with the best prospects have already left. The corollary, however, abstracts from idiosyncratic shocks, which can break this monotonicity. For example, a streak of small positive shocks, or a single large idiosyncratic shock, can lift a worker's human capital, and with it their search effort.[12]

Informal-to-Formal Mobility. Unlike unemployment, where both forces push the hazard in the same direction, informal employment need not align them. Informality slows the depreciation of human capital, but that alone need not improve workers' later access to formality. If informal workers still lose human capital during the spell, the logic of Corollary 1 applies. Structural duration dependence and dynamic selection then compound each other, and the hazard decreases. By contrast, if informal workers accumulate human capital, the two forces oppose each other, and the hazard depends on which one dominates. Even then, the hazard can decrease, as Corollary 2 shows.

Corollary 2. *Consider a stationary equilibrium in which $\sigma_i = 0$ and suppose that the comparative advantage of formality over informality increases with human capital so that*

$$\partial_x \eta_i^f(x) \geq 0,$$

with the inequality strict wherever $\eta_i^f(x) > 0$. For a new cohort of informal workers, and any finite duration window $[0, D]$, there is an explicit bound $\bar{\mu}_i(D) > 0$ on informal human capital accumulation such that $0 \leq \mu_i < \bar{\mu}_i(D)$ implies

$$\partial_d \lambda_i^f(d) < 0 \quad \text{for almost every } d \in [0, D].$$

Proof. See Appendix A.4.[13]

[12] Equation (5) shows that nondecreasing employment gains $V_\ell - U$ are sufficient for increasing unemployment search. Appendix A.1 explains when this fails and why it is a reasonable assumption.

[13] The proof also requires regularity conditions on the cohort's initial distribution, such as a finite mean and a share of workers who actively search for formal islands. The explicit bound $\bar{\mu}_i(D)$ is presented in the proof.

Theorem 2 separates the two forces behind this result. When informal employment builds human capital, informal workers' human capital rises as the spell unfolds. Since the comparative advantage of formality increases with human capital, they search harder for formal islands, and structural duration dependence raises the hazard. Dynamic selection pushes the other way, as before, because the informal workers who search hardest leave first. Corollary 2 shows that when accumulation in the informal sector is limited, the search intensity of the remaining members of the cohort does not rise fast enough to offset these departures.

When informal workers also experience idiosyncratic shocks, the same two forces operate, and shocks add two more. First, workers near the quitting threshold $\bar{x}_u^i$ who experience negative shocks quit into unemployment. Those workers were the ones who searched the least, and hence removing them from the cohort increases the hazard. Second, the shocks can also increase the search intensity of informal workers, and hence their contribution to the structural duration dependence term is ambiguous.

This shows why interpreting an informal-to-formal hazard requires caution. A decreasing hazard does not immediately imply that informality harms workers' future prospects of formalization. In fact, as shown by Corollary 2, informality can not only slow down human capital depreciation from unemployment but also provide positive accumulation, and still deliver a decreasing informal-to-formal hazard.

Taking Stock. Three mechanisms determine whether informality is a stepping stone toward formality or lowers a worker's chance of eventually becoming formal. First, informality changes how human capital evolves during a spell. Relative to unemployment, it preserves human capital, $\mu_i \geq \mu_u$, and when it also builds it, $\mu_i > 0$, the gain from formalization widens as the spell unfolds. Without accumulation, the gain stays flat or narrows, and informal workers' incentives to formalize do not recover. Second, at any point in the spell the gain from formalization is smaller for an informal worker than for an unemployed worker with the same human capital. This induces a set of workers for whom paying the entry cost of formality is no longer worth it once they reach informality, and who therefore stop searching for formal employment even though they would rather be formal. This effect, however, operates only on the extensive margin. Finally, on the intensive margin, a lower search cost can lead informal workers to search more intensively than unemployed workers despite their smaller gain, but only once that gain covers the entry cost.

3 Empirical Analysis

In the model, how long workers remain in an employment state and where they go when they leave depend on which workers the state attracts, how their prospects evolve during a spell, and which of them leave first. Measuring these patterns requires following workers through complete spells, often over many years, which short panels cannot do (Bosch & Maloney, 2010). The EPS instead reconstructs each respondent's labor-market history retrospectively, often spanning decades. These

histories serve two purposes: they recover statistical objects that short panels cannot measure, and they allow those objects to be read through the model. I first describe the EPS and how its records map into the model's states, then document durations, hazards, and mobility, and finally interpret these patterns through the model.

3.1 The EPS

The EPS is a longitudinal survey of Chilean adults that combines repeated interviews with retrospective labor-market histories. At each interview, respondents report the timing and characteristics of their employment and non-employment periods. Those who miss an interview can report the intervening months when they return.[14] I construct labor histories from the 2002, 2004, 2006, 2009, and 2015 rounds, excluding the 2012 reports because of interview problems and relying on the 2015 retrospective module to cover the missing years. Appendix B.1 documents the construction of the histories and their coverage across rounds, and Appendix B.2 documents the survey weights.

The analysis sample contains 23,205 respondents with at least one formal or informal employment spell. Table 1 contrasts survey participation with labor-history coverage: the average worker appears in three processed rounds but reports a history spanning 23 years, with similar coverage for men and women. These histories are also nearly continuous, with no gap longer than one month between consecutive spells for 95 percent of workers.[15]

Table 1: EPS sample overview

	Men	Women	All
Workers	12,474	10,731	23,205
Mean age at first interview	38.5	36.9	37.7
Average survey follow-up (rounds)	2.9	3.0	3.0
Average labor-history coverage (years)	23.1	23.3	23.2
Average net attrition per wave (%)	-8.4	-10.9	-9.5

Notes: The table describes respondents with at least one formal or informal employment spell. Worker counts are unweighted. Net attrition uses raw cross-sectional expansion weights, and negative values indicate growth in weighted respondent mass. Source: EPS 2002–2015 rounds; author's calculations.

Measuring Informality. Informality has no single empirical definition. Productive definitions classify jobs by features of the firm, whereas legal definitions classify them by the protection the worker receives, and the two need not agree (Perry et al., 2007). I adopt the legal view: an employee's job is formal only if the worker has a signed employment contract and contributes to

[14] The EPS follows respondents across rounds while periodically adding refreshment samples, so the interview panel is unbalanced and the time between a respondent's interviews varies.

[15] Weighted statistics use the respondent's survey expansion factor, normalized within the respondent's entry cohort and adjusted for differential response across rounds.

the pension or social-security system through that job.[16] This legal view include arrangements that break the law, such as salaried work without mandated contributions, and arrangements that do not, such as self-employment outside the contributory system (Levy, 2008).

These requirements identify the jobs most likely to be subject to the model's tax wedge τ. Workers who contribute to the pension or social-security system are likely to appear in employer–employee administrative records through which payroll taxes are collected. A signed employment contract is legally required and provides similar evidence that the employment relationship is formally documented. I therefore require both indicators to classify an employee's job as formal, treating partial compliance as informality.[17]

The EPS records non-employment spells without distinguishing workers who look for a job from those who do not. I map these spells to the model's unemployment state, which allows workers to choose zero search effort. This mapping is more plausible for workers with strong labor-force attachment. Table 2 shows that women spend 46.5 percent of their labor histories out of work, more than twice the share for men (19.4 percent), and their completed non-employment spells average 44.2 months, compared with 26.6 for men. I therefore restrict the duration and mobility analysis to men ages 15–64, among whom informal jobs account for 25.9 percent of employment.

Table 2: Labor-market attachment by sex

	Men	Women	All
Panel A. Time over the labor history (% of months)			
Formal employment	59.8	39.1	50.4
Informal employment	20.8	14.4	17.8
Non-employment	19.4	46.5	31.9
Panel B. Spells			
Employment spells per worker	5.0	4.1	4.6
Non-employment spells per worker	1.9	2.4	2.1
Mean completed non-employment spell (months)	26.6	44.2	35.4
Workers	12,447	10,713	23,160

Notes: Panel A reports the share of person-months spent in each state: weighted monthly shares, averaged across months from 1980 to 2016, using person-month stock weights. It covers 3,269,821 person-months of men and 2,882,520 of women. Panel B counts cleaned job-level spells that start at ages 15–64. Worker counts are unweighted. Ages assume a January birth month. Source: EPS 2002–2015 rounds; author's calculations.

[16] Article 9 of Chile's Labor Code requires a written employment contract signed by both parties but does not require notarization (Chile, 2003). Self-employed workers, including employers, have no employment contract, so I classify their jobs by contributions alone.

[17] Other studies weigh the two requirements differently: some use contributions alone, others a contract alone, and the narrowest definitions of informality require both to be absent (Perry et al., 2007; Bosch & Maloney, 2010). The gap between my definition and this narrower definition is the share of jobs that meet only one requirement. Appendix Figure D.1 compares my definition with these alternatives and with an official series.

3.2 Long-Run Dynamics of Informal Employment

I organize labor histories into sector-spells that correspond to stays in the model's three states. Consecutive records belong to the same sector-spell when the state is unchanged and each begins within six months of the previous record's end. A change of state or a longer gap starts a new spell, while employer changes within a state do not. Each sector-spell enters the duration and mobility statistics once with a flow weight based on the worker's survey weights, so these statistics describe spell inflows rather than the stock of workers in each state. Appendix B.2 documents the construction, weights, and treatment of gaps and censoring.

Table 3: Sector-spell sample by employment state

	Formal	Informal	Unemployment
Number of sector-spells	24,115	13,633	23,506
Share of completed spells (%)	74.55	81.81	89.26
Mean completed duration (months)	61.92	42.16	26.64
Std. dev. of completed duration (months)	79.07	63.87	36.89
Range of completed durations (months)	1–437	1–432	1–356
Range of observed durations (months)	1–440	1–440	1–435
Workers	11,151	7,577	9,882

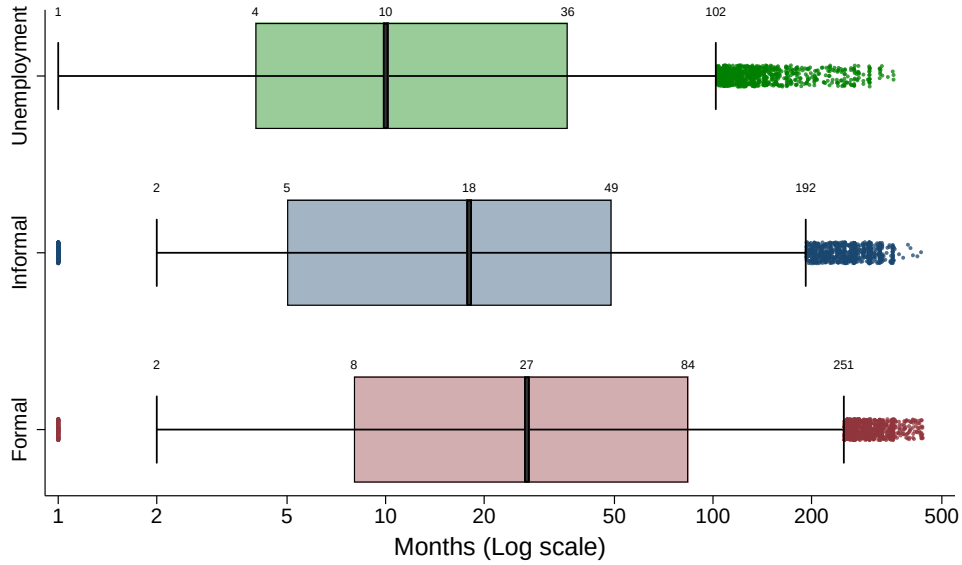
Notes: The table reports sector-spells of men ages 15–64 by origin state. A spell is completed when a later sector-spell is observed for the same worker. Spell counts and completion shares are unweighted, and duration moments use sector-spell flow weights. Source: EPS 2002–2015 rounds; author's calculations.

Spell Duration Distributions. The EPS histories record the end of most sector-spells in the sample. Table 3 describes 61,254 sector-spells from 12,537 men. A spell is completed when a later sector-spell is observed for the same worker, and its duration runs from its recorded start to its recorded end. Under this definition, the histories provide completed durations for 75 percent of formal spells and 89 percent of unemployment spells. These observations allow me to compare persistence across states over complete spells, including stays lasting many years.

Figure 1 shows that informal employment lies between formal employment and unemployment in persistence, with many short spells and a long upper tail. Completed informal spells last 42 months on average, compared with 62 for formal spells and 27 for unemployment spells, as Table 3 shows. This ranking holds across the completed-duration distributions: formal durations first-order stochastically dominate informal durations, which in turn dominate unemployment durations. Half of completed informal spells end within 18 months, yet one in twenty lasts longer than 192 months. For some workers, informal employment therefore persists for many years.

Destination-Specific Hazards. Completed durations summarize how long workers remain in each state. Destination-specific hazards describe how their chances of moving to a particular destination change as a spell unfolds. In the model, equation (10) defines this hazard as an instantaneous exit rate among workers still in the origin state. Because the EPS records spells in

Figure 1: Completed spell duration distributions



Notes: The figure plots flow-weighted distributions of completed sector-spell durations for men ages 15–64. Boxes span the 25th to 75th percentiles, the thick line marks the median, whiskers span the 5th to 95th percentiles, and numbers above each box report these percentiles. Dots show individual spells beyond the whiskers, and Table 3 reports the full range. The sample contains 17,978,11,153, and 20,982 completed formal, informal, and unemployment spells. Source: EPS 2002–2015 rounds; author's calculations.

months, I estimate its discrete counterpart: the probability that a spell in progress at the start of a six-month interval ends with a move to that destination during the interval. Appendix C.2 relates these probabilities to the model's continuous-time hazards.

For each six-month interval, I estimate the hazard as the weighted number of exits to a particular destination divided by the weighted number of spells at risk at the interval's start.[18] For sector-spell n of worker m, let $\ell_{m,n} \in \{f, i, u\}$ denote its origin state, $T_{m,n}$ its observed duration in months, $\ell'_{m,n}$ its observed destination, and $\omega_{m,n}$ its flow weight. Over the interval $I_d = (d-6, d]$, the estimated hazard from ℓ to $\ell' \neq \ell$ is

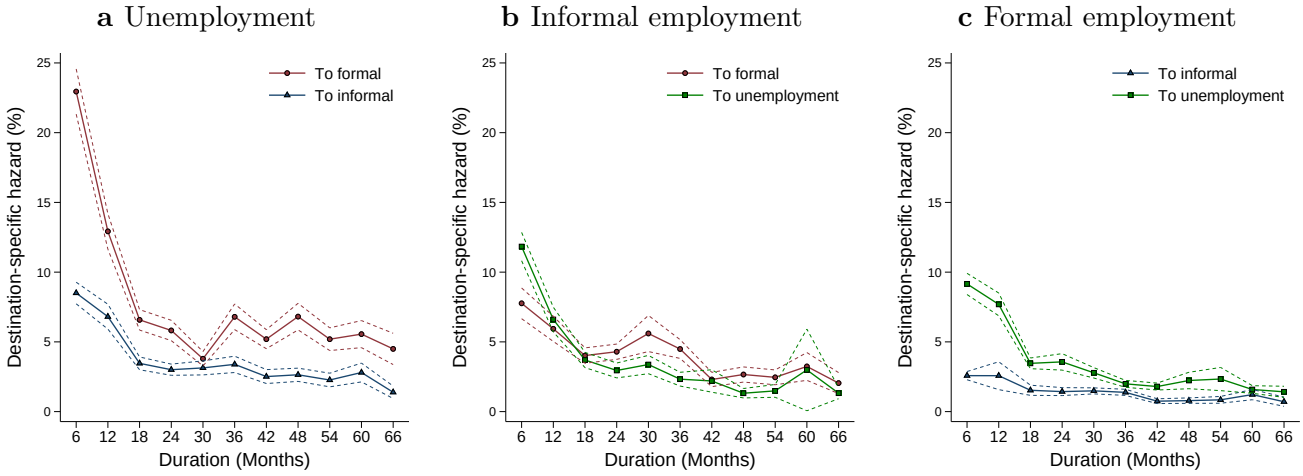
$$\widehat{\lambda}_{\ell}^{\ell'}(d) = \frac{\sum_{m,n} \omega_{m,n} \mathbf{1}\{\ell_{m,n} = \ell,\ T_{m,n} \in I_d,\ \ell'_{m,n} = \ell'\}}{\sum_{m,n} \omega_{m,n} \mathbf{1}\{\ell_{m,n} = \ell,\ T_{m,n} > d-6\}}. \tag{14}$$

A spell still in progress when its history ends stays in the risk set through its last observed interval. When the next sector-spell begins more than six months later, or outside the three states (for example, in unpaid family work), I treat the destination as unknown. The spell leaves the risk set without counting toward either destination.

Figure 2 shows that exit probabilities in the first six months are highest from unemployment, intermediate from informal employment, and lowest from formal employment. First, unemploy-

[18] The baseline intervals end at $d \in \{6, 12, \ldots, 66\}$ months. Appendix A.5 establishes worker-clustered inference.

Figure 2: Destination-specific hazards by origin state



Notes: Each panel plots the flow-weighted destination-specific hazard in Equation (14), in percent. The horizontal axis marks the end of each interval. The sample contains 23,506 unemployment,13,633 informal, and 24,115 formal sector-spells of men ages 15–64 from 12,537 workers. Dashed lines show pointwise 95% CIs based on worker-clustered standard errors. Source: EPS 2002–2015 rounds; author's calculations.

ment spells end mostly in formal employment, with a hazard of 23.0 percent. Second, informal spells end more often in unemployment than in formal employment, with hazards of 11.8 and 7.8 percent, respectively. Finally, formal spells end mostly in unemployment, while direct moves into informality have a hazard of 2.6 percent. Thus, these initial exit probabilities are consistent with Figure 1's ordering of persistence, with informal employment between formal employment and unemployment.

Moreover, these exit probabilities decline at different rates as spells lengthen, with most of the decline occurring in the first 18 months. From unemployment, the hazard into formal employment falls to an average of 5.5 percent between months 18 and 66. Among informal spells, the initially higher hazard into unemployment falls faster than the hazard into formal employment, leaving the two similar after the first year. Employment status alone therefore gives an incomplete account of mobility, which also varies sharply with elapsed spell duration.

Spell-to-Spell Mobility. To summarize where spells lead across durations, I estimate spell-to-spell transition probabilities. These probabilities are the weighted shares of linked exits from each origin that lead to each destination. I calculate these shares directly from linked spells over the full range of observed durations. Thus, they include exits after long spells that extend beyond the horizon shown in Figure 2.

Panel A of Table 4 shows that informal exits divide almost evenly between formal employment and unemployment, at 51 and 49 percent, respectively. By contrast, the other origins have a clear main destination, with 71 percent of formal exits leading to unemployment and 69 percent of unemployment exits leading to formal employment. Thus, when exits are pooled across durations, informal employment leads to formal employment about as often as it leads to unemployment.

Table 4: Spell-to-spell mobility and duration

Panel A. Transition probability (%)				*Panel B. Median duration (months)*			
Origin	Formal	Informal	Unemployment	Origin	Formal	Informal	Unemployment
Formal	—	29.27	70.73	Formal	—	32	22
Informal	51.45	—	48.55	Informal	23	—	11
Unemployment	69.07	30.93	—	Unemployment	9	12	—

Notes: Rows are origin states and columns are destinations. Panel A reports the share of linked exits from each origin that lead to each destination, in percent. Panel B reports the weighted median duration of the origin spell, in months, for exits to that destination. The sample contains 47,396 linked exits of men ages 15–64 from 10,756 workers, and both panels use sector-spell flow weights. Source: EPS 2002–2015 rounds; author's calculations.

Moreover, Panel B of Table 4 shows that the nearly equal shares of informal exits conceal a large difference in completed durations. Informal spells ending in formal employment last 23 months at the median, compared with 11 for those ending in unemployment. Similarly, formal spells ending in informality last 32 months at the median, compared with 22 for those ending in unemployment. By contrast, unemployment spells ending in formal employment are shorter than those ending in informality, with medians of 9 and 12 months, respectively. This ordering of completed durations provides evidence against a version of the model with no human capital dynamics.

In the absence of human capital dynamics, the model predicts the opposite duration ordering. Workers who search most intensively toward the other sector leave first, while the separation rate into unemployment remains constant. Consequently, the job-to-job share of exits in Equation (11) cannot increase with duration. This implies weakly shorter completed durations before job-to-job moves than before unemployment, including at the median. Yet Panel B shows longer spells before job-to-job moves in both sectors. These duration comparisons therefore contradict the model's fixed-human-capital benchmark.

Taken together, these moments provide complementary information about the model's mechanisms. Individually, however, they can be difficult to interpret because different combinations of mechanisms can generate similar patterns. An explanation of the data must therefore account jointly for spell durations, hazard profiles, and exit destinations. I next add wages and previous employment to this comparison, bringing further information about workers' differences and the incentives behind their transitions.

3.3 Wages and Duration Dependence

Wages place an additional restriction on the model's account of mobility. In the model, workers' human capital shapes both what they earn and what they gain from moving between states. The worker heterogeneity and human capital dynamics that explain mobility must therefore also explain wages.

The EPS does not record the full wage path within an employment spell, so I cannot measure wage growth directly. Instead, I construct each spell's average wage from the available reports and

compare these averages across completed spells of different lengths.

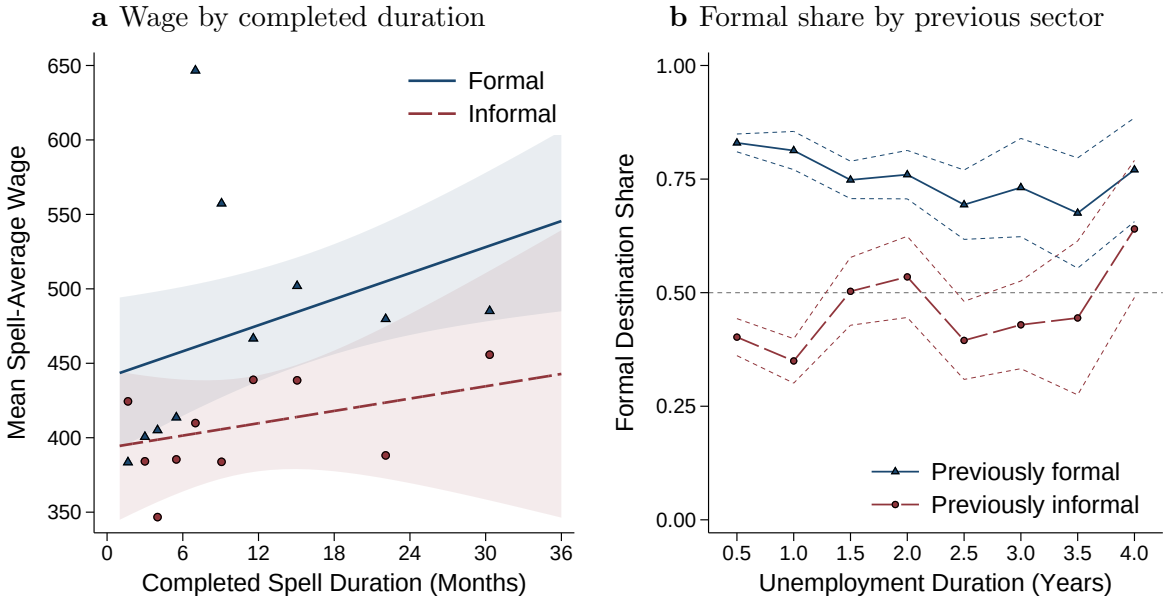
In the model, a spell's average wage combines its entry wage with wage growth and shocks during employment. For a completed spell of duration T in sector ℓ,

$$\overline{w}_\ell = w_0 + a_\ell \left[\mu_\ell T/2 + \sigma_\ell \overline{W}_T\right], \quad \overline{W}_T = \frac{1}{T}\int_0^T W_t \, \mathrm{d}t. \tag{15}$$

Here, w_0 is the wage at entry. Relative to that wage, deterministic wage growth contributes $a_\ell \mu_\ell T/2$ to the spell average, while $a_\ell \sigma_\ell \overline{W}_T$ is the average wage shock over the spell. Because entry wages and average wage shocks can also vary with completed duration, the wage–duration profile need not reflect wage growth alone.

Figure 3a shows that the formal–informal wage gap widens with completed spell duration. The comparison uses spells that begin out of unemployment and last at most 36 months. Among the shortest of these spells, average wages are similar across sectors. As completed duration increases, however, the fitted formal wage profile rises more steeply than the informal profile.[19]

Figure 3: Wage profiles and destinations after unemployment



Notes: Panel (a) plots the flow-weighted mean wage, in real 2010 USD per month, of completed formal and informal sector-spells that begin out of unemployment and last at most 36 months (3,705 and 2,372 spells), against their duration. Bins are deciles of duration in the pooled sample, and lines are linear fits on individual spells. Panel (b) plots the flow-weighted fraction of exits from unemployment that lead to formal jobs, among exits to either employment sector. Shares are calculated by six-month interval separately for spells preceded by formal or informal employment (11,894 and 5,151 spells). Bands and dashed lines are 95% CIs clustered by worker. All samples are men ages 15–64. Source: EPS 2002–2015 rounds; author's calculations.

The steeper formal profile can reflect selection reinforcing wage growth. If lower-wage formal entrants gain more from moving to informality, they search more intensively and tend to leave

[19] Appendix D.1.3 derives the wage–duration regression from Equation (15).

sooner. Longer completed formal spells can then come disproportionately from workers who entered at higher wages. The formal slope therefore contains information about workers' search incentives as well as wage growth.

In informal employment, by contrast, selection can offset wage growth. If higher-wage informal entrants gain more from formal employment, they search more intensively and tend to leave sooner. Their early departures can lower mean entry wages among spells completed at longer durations. Alongside this selection in entry wages, quits after adverse shocks can make average wage shocks vary with completed duration. A relatively flat informal profile can therefore coexist with positive wage growth.

Figure 3b adds evidence on persistence across spells by showing that previous sector predicts destinations after unemployment. In each six-month interval, the formal destination share measures the fraction of exits to employment that lead to formal jobs. This share remains higher for previously formal workers than for previously informal workers throughout the four-year horizon.

In the model, this gap reflects differences in the human-capital distributions of the two groups entering unemployment. At the same human capital, unemployed workers face identical search policies and human-capital dynamics, regardless of their previous sector. The destination gap therefore provides joint information about who enters and leaves each employment sector and how human capital evolves while workers are employed.

The duration comparisons, hazard profiles, and wage evidence jointly restrict explanations of access to formality. Employment spells ending in the other sector last longer than those ending in unemployment, highlighting the importance of human capital dynamics. The steeper formal wage profile further constrains how wage growth and selection interact. These patterns are consistent with the model's interaction of human-capital dynamics, the entry cost of formality, and the cost of generating formal opportunities. The observed profiles, however, do not reveal how much each mechanism contributes. Section 4 therefore estimates the model using wage and mobility moments jointly to quantify the mechanisms' individual contributions.

4 Quantitative Analysis

To measure how much each mechanism contributes, I estimate the model on the EPS employment histories in three steps. First, I set the discount rate and the formal tax wedge from external sources. Second, I estimate the labor-force exit rate directly from the age profile. Third, I estimate the remaining 17 parameters jointly by the generalized method of moments (GMM), targeting 52 independent restrictions on labor-market allocations, wages, and the dynamics of employment spells. I then show how these restrictions separate human-capital dynamics from selection, which the hazards alone cannot do, and how closely the estimated model reproduces them.

4.1 Moments and Estimator

External parameters. I begin by describing the parameters that are set outside of the GMM estimation. First, I set ρ to be the risk-free interest rate of Chile using information from Chile's Central Bank (Banco Central de Chile, 2026). Appendix C.3 details how I construct the parameter.

Second, I use information from the OECD to recover the tax wedge τ. For each country and year, the OECD Taxing Wages database reports the income taxes and social-security contributions that workers and employers pay to the government (OECD, 2026). I use its average tax wedge for a single worker without children earning the average wage, expressed as a share of the employer's labor cost. The wedge excludes workers' contributions to their individual pension accounts, which go to private funds rather than to the government. It therefore measures exactly the revenue that enters the government's budget constraint in Equation (9). Appendix C.3 describes how I map it to the wedge described in the model.

Labor-force exit. The labor-force exit rate δ is the only parameter that I estimate directly from the EPS. In the model, workers leave the labor force at a constant rate and are replaced by entrants. In a stationary population, this constant rate gives the age distribution a convenient functional form: the log share of workers older than a given age falls linearly with age, at slope $-\delta$. I therefore recover δ directly by regressing this log share on age.[20] Appendix C.3 details the sample, weights, and regression.

GMM estimator. I estimate the remaining 17 parameters by GMM:

$$\boldsymbol{\vartheta} \equiv \left(F, b, \alpha_0, \Psi_f, \delta_i, \delta_f, \kappa_u^i, \kappa_u^f, \kappa_i^f, \kappa_f^i, \{\mu_\ell, \sigma_\ell\}_{\ell\in\{u,i,f\}}, \xi_g\right)'.$$

Three sources of variation in the EPS identify them. The first is the allocation of workers across unemployment, informal employment, and formal employment. The second is the wages of formal and informal workers. The third is how fast and where workers move, both as their spells lengthen and in the years after they enter the labor market. Table 5 lists the moments behind each source and the parameters each moment is most informative about, and Section 4.2 develops the identification argument.

I group the moments in Table 5 into families g, one for each profile or set of shares, and estimate $\boldsymbol{\vartheta}$ by minimizing a weighted sum of their losses:

$$\widehat{\boldsymbol{\vartheta}} = \arg\min_{\boldsymbol{\vartheta}} \left[\sum_{g\in\mathcal{G}} \omega_g L_g(\boldsymbol{\vartheta})\right]^{1/2}, \qquad L_g(\boldsymbol{\vartheta}) = \sum_{k=1}^{K_g} c_{gk} \left[\frac{m_{gk}(\boldsymbol{\vartheta}) - \widehat{m}_{gk}}{\frac{1}{2}\left(|m_{gk}(\boldsymbol{\vartheta})| + |\widehat{m}_{gk}|\right)}\right]^2, \tag{16}$$

where ω_g is the weight on family g, and $m_{gk}(\boldsymbol{\vartheta})$ and $\widehat{m}_{gk}$ are the model and data values of its

[20] The estimate absorbs both permanent attrition and actual exits from the labor force. Since the survey cannot distinguish one from the other, a rate that combines both is the consistent object for matching moments computed from the same survey.

Table 5: Parameter and GMM Mapping

Empirical family, L_g	Source	Model counterpart	Parameters
Outside of GMM			
Discount rate	Banco Central de Chile (2026)	ρ	ρ
Tax wedge	OECD (2026)	τ	τ
Age-survival profile	EPS	δ	δ
GMM moments			
Hazard (*origin*$= u$)	EPS	$\lambda_u^i(d),\ \lambda_u^f(d),\ d \in \{12, 24, 48, 54\}$	$\mu_u, \sigma_u, \kappa_u^f/\kappa_u^i, \Psi_f$
Hazard (*origin*$= i$)	EPS	$\lambda_i^f(d),\ d \in \{12, 24, 48, 60\}$	μ_i, σ_i, Ψ_f
Hazard (*origin*$= f$)	EPS	$\lambda_f^i(d),\ d \in \{18, 36, 54\}$	μ_f, σ_f
$u \to f$ share	EPS	$\chi_u^f(d),\ d \in \{18, 36, 54, 72\}$	$\mu_u, \sigma_u, \kappa_u^f/\kappa_u^i, \Psi_f$
$f \to i$ share	EPS	$\chi_f^i(d),\ d \in \{18, 36, 54, 72, 90\}$	$\delta_f \kappa_f^i$
$i \to f$ share	EPS	$\chi_i^f(d),\ d \in \{36, 54, 72, 90\}$	$\delta_i \kappa_i^f$
Wage levels	EPS	$\mathbb{E}[\overline{w}_f], \quad \mathbb{E}[\overline{w}_i]$	$\delta_\ell, \kappa_\ell^{\ell'}, \alpha_0$
Labor shares	EPS	$\mathbf{M}_u, \mathbf{M}_i, \mathbf{M}_f$	b, F
Entry cohort	EPS	$\mathbf{M}_u^E(d), \mathbf{M}_i^E(d), \mathbf{M}_f^E(d),\ d \in \{6, 12, \ldots, 60\}$	$\xi_g, \kappa_u^i, \kappa_u^f$

Notes: The durations targeted in the hazard and destination-share moments were selected based on the standard errors of the empirical moments.

k-th moment. The weights c_{gk} sum to one within each family, so a profile does not count more because it is observed at more durations. Table 7 reports the family weights and the losses at the estimate.[21]

4.2 Identification

Before turning to how each family of moments identifies different parameters, I show that the units in which the model is measured are not identified. Every target is a share, a relative hazard, or a normalized wage. Proposition 1 shows that changing the units of time, human capital, or output rescales the parameters but leaves every target unchanged, so a parameter vector and its rescaled versions are observationally equivalent. The parameters are therefore identified only up to these units, and a normalization that fixes them selects one representative of each equivalence class.

Proposition 1 (Identification up to scale)**.** *Let positive constants c_t, c_x, and c_A rescale time, human capital, and output, with every parameter rescaled according to its units. The stationary equilibrium under the rescaled parameters then generates the same targeted moments, at the same calendar durations, as the original one, so the parameters are identified only up to scale.*

Proof. See Appendix A.6.

The normalization fixes the three numbers that the targeted moments cannot identify: the length of a period, the unit of human capital, and the unit of output. I measure time in years,

[21] I combine a Sobol design with TikTak-style multistart refinement, using pattern search for local optimization (Arnoud et al., 2019).

$a = 12$, human capital relative to the median entrant, $k_g = 1$, and output in units with $A = 1$. Normalizing output so that $A = 1$ has one practical consequence. Because the criterion leaves out the level of formal wages, I can use that level after estimation to recover A. Recovering A, in turn, allows me to report the estimates in 2010 US dollars.

I recover A by matching the mean log formal wage to its EPS counterpart. Equation (2) shows that every wage, formal or informal, is proportional to A, and so is the geometric mean formal wage. Dividing wages by that mean therefore removes A from both wage targets. I can then estimate the model without taking a stance on the units of output.

With the estimated model, I compute unobserved equilibrium outcomes such as $m_f(x)$, the distribution of human capital in the formal sector. Knowing m_f, I can split the observed mean log formal wage, $\overline{\log w}_f$, into three components: the distribution of human capital, the tax wedge, and the level of output,

$$\log A = \overline{\log w}_f + \log(1+\tau) - \mathbf{M}_f^{-1} \int_0^\infty \log x \, m_f(x) \, \mathrm{d}x. \tag{17}$$

Appendix C.3 details the empirical wage scale and the computation.

Unlike A, the remaining parameters cannot be recovered one moment at a time: each family of moments depends on all of them through search policies, selection, and the stationary distribution. The arguments that follow are therefore heuristic rather than a formal proof of identification. For each family, I describe the feature of the data it records and the parameters that have first order effects on the implied moment as displayed in the last column of Table 5. These links are not exclusion restrictions, and they do not establish local or global identification.

Hazard profiles. I begin by targeting each hazard at longer durations relative to its value in the first six months. Recall that the hazard is the average search intensity of the workers still in the spell. Thus, when I normalize the hazard, I remove the common scale set by the search cost $\kappa_\ell^{\ell'}$, as shown by equation (5). Consequently, the moments from employment are not informative about κ_i^f and κ_f^i. For unemployment, the two moments share one denominator, so they discipline the ratio κ_u^f/κ_u^i.

What the moments record is how the hazards change as a spell unfolds. Theorem 2 splits that change into structural duration dependence and dynamic selection. Drift, μ_ℓ, and volatility, σ_ℓ, drive structural duration dependence, because they move survivors across workers with different search intensity. The entry cost Ψ_f shapes dynamic selection, because it shuts off search among workers whose gain from formality falls short of Ψ_f and so widens the differences in search intensity among survivors.

Destination shares. The destination shares out of unemployment add nothing to the hazard moments, but those out of employment do. By equation (11), a destination share divides the hazard into one destination by the sum of the hazards out of the spell. Out of unemployment,

both hazards lead to employment, so the formal share carries the same information as the two unemployment hazard moments. Out of employment, the other destination is unemployment, which a worker reaches either through an exogenous separation or by quitting at the indifference point:

$$\lambda_\ell^u(d) = \delta_\ell + \bar{q}_\ell(d), \qquad \ell \in \{i, f\}.$$

Separations and quits compete with the job-to-job hazard, which equation (5) sets as the average gain from moving divided by $\kappa_\ell^{\ell'}$. The share of exits into the other sector compares the job-to-job hazard with $\delta_\ell + \bar{q}_\ell$, so it pins down δ_ℓ relative to that hazard, that is, the product $\delta_\ell \kappa_\ell^{\ell'}$.

Wage levels. As opposed to the destination shares, the wage moments separate δ_ℓ from $\kappa_\ell^{\ell'}$. Together, δ_ℓ and $\kappa_\ell^{\ell'}$ set how fast a spell ends. The distribution of completed durations is determined by the level of the aggregate exit hazard, which sums three rates: the job-to-job hazard, separations, and quits. Consequently, durations carry the information needed to separate δ_ℓ from $\kappa_\ell^{\ell'}$. The wage moments capture this information because wages grow during a spell. By equation (15), a spell that lasts T pays on average $a_\ell \mu_\ell T/2$ more than its initial wage.

The moments average over the months worked in completed spells, so a spell lasting T contributes T months. Long spells therefore supply more months, and their months pay more. Picking a month at random thus over-samples long spells, and the average wage per month depends on durations through $\mathbb{E}[T^2]/\mathbb{E}[T]$ rather than through the mean duration $\mathbb{E}[T]$. Formally, a spell's earnings are its months times its average wage. Multiplying equation (15) by T and writing the initial wage as $a_\ell x_0$ gives

$$T\,\overline{w}_\ell = a_\ell \Big[T x_0 + \frac{\mu_\ell}{2} T^2 + \sigma_\ell T\,\overline{W}_T \Big].$$

The shocks average out across spells that start with the same human capital and last equally long, as long as completion does not depend on them. Summing earnings and months over completed spells, with expectations over the joint distribution of x_0 and T, therefore gives

$$\frac{\mathbb{E}[T\,\overline{w}_\ell]}{\mathbb{E}[T]} = a_\ell \Big[\frac{\mathbb{E}[T\,x_0]}{\mathbb{E}[T]} + \frac{\mu_\ell}{2} \frac{\mathbb{E}[T^2]}{\mathbb{E}[T]} \Big].$$

Both terms depend on durations, since initial human capital and duration are correlated across spells, but growth depends on them most directly. With μ_ℓ disciplined by the hazard moments, the wage moments pin down $\mathbb{E}[T^2]/\mathbb{E}[T]$, and with it the level of the aggregate exit hazard. Combined with the split from the destination shares, this level separates δ_ℓ from $\kappa_\ell^{\ell'}$. The gap between the formal and informal moments also depends on the informal wage slope $a_i = A(1-\alpha_0)$, so the wage moments discipline the enforcement rate α_0 as well.

Labor-market shares. With α_0 disciplined by the wage gap, the labor shares pin down the benefit b and the recurring cost F. The earlier families restrict the gains from moving only relative to search costs and to one another. They leave the levels of the value functions free, and with them the thresholds at which workers favor one state over another. Because preferences are linear, b and

F shift these levels directly. Each dollar of b raises the value of unemployment by the expected discounted time a worker spends unemployed, and so draws workers into unemployment. The recurring cost F has the opposite effect on the value of formal employment, pushing workers out of it. Each movement shows up in its own share, so the unemployment share is most informative about b and the formal share about F. The informal share follows from adding up, because the three shares sum to one.

Entry-cohort trajectories. The model leaves the entrant distribution general, and for estimation I specialize it to the log-logistic family. With the median normalized to one, its cumulative distribution function is

$$G(x) = \frac{x^{\xi_g}}{1 + x^{\xi_g}}, \qquad x > 0.$$

A single parameter, ξ_g, then governs how dispersed entrants are. The mean of entrants' human capital is finite whenever $\xi_g > 1$, as Theorem 1 requires.[22]

The entry cohort disciplines the two objects that remain: ξ_g and the levels of the two unemployment search costs, whose ratio the hazard moments already fix. New workers start unemployed with human capital drawn from G, and I propagate the cohort with the model's search policies. With the other parameters anchored, the cohort's path depends on the levels of those policies. Through those levels, κ_u^i and κ_u^f set how fast the cohort leaves unemployment. The dispersion ξ_g sets how entrants are spread across search regions, and so shapes how that pace changes and how the cohort splits between sectors over its first five years.

Taken together, the families discipline every parameter, each with variation that the earlier ones leave free. The hazard moments separate structural duration dependence from dynamic selection, the two forces in Theorem 2, but discard the levels of search costs. The destination shares pin down separations relative to the job-to-job hazard. The wage moments add the level of exits, which separates δ_ℓ from $\kappa_\ell^{\ell'}$, and their gap disciplines α_0. The labor shares then anchor the levels of the value functions through b and F. Finally, the entry cohort pins down the dispersion of entrants and the levels of the two unemployment search costs.

4.3 Estimates and Model Fit

With every parameter disciplined, I present the estimates and show how closely the estimated model reproduces the targeted moments. Table 6 reports the estimated parameters. I then compare the model with the data family by family and trace each remaining discrepancy to a feature of the model.

The payoff parameters in Table 6 make comparative advantage follow human capital. Each unit of human capital earns 17.2 percent more in a formal job than in an informal one before taxes, and 9.0 percent more after them, so the tax wedge narrows the formal advantage without closing

[22] The log-logistic arises as the long-run distribution of knowledge in models where agents learn from one another (F. E. Alvarez et al., 2008).

Table 6: Model Parameters

Parameter	Interpretation	Value	Parameter	Interpretation	Value
Panel A. Set outside the GMM criterion					
ρ	Discount rate (annual)	2.60%	τ	Formal tax wedge	7.53%
δ	Labor-force exit hazard (annual)	4.98%			
Panel B. Estimated by GMM					
F	Nonpecuniary cost (USD/year)	521.67	Ψ_f	Formal-entry cost (USD)	6,664.03
b	Unemployment benefit (USD/year)	0.00	ξ_g	Entrant shape and tail	7.81
α_0	Enforcement scale	14.71%	A	Production scale	10,019.37

Panel C. State-Specific Human-Capital and Mobility Parameters (GMM)

Origin state	HC drift $A\mu_\ell$ (USD/year)	Volatility $A\sigma_\ell$ (USD/$\sqrt{\text{year}}$)	Separation δ_ℓ (percent/year)	Cost to informal (USD/year)	Cost to formal (USD/year)
Unemployment	−8,774.62	0.00	—	4,743	1,708
Informal	398.15	466.98	0.10%	—	7.42
Formal	398.32	0.00	0.26%	5,525	—

Notes: Monetary values are in real 2010 USD, and the stationary employed average wage is 9,842 a year. The production scale A is in USD a year per unit of human capital. Search costs are annual flow costs at a common intensity of 0.1 offers a year. †Estimate at the boundary of the parameter space.

it. The nonpecuniary cost F, by contrast, does not scale with human capital, and it outweighs this advantage when human capital is low. In the estimated equilibrium, F amounts to 49.1 percent of the formal wage that workers who favor informality would earn, and the remaining 55.6 percent of workers favor formal employment.

Among those who favor formal employment but work informally, most do not search for it. Only 5.3 percent of informal workers search, and those who do spend on average 0.27 percent of their wage on search. Search is cheap, so the binding constraint is the entry cost Ψ_f, 67.7 percent of a year's average wage. The entry cost holds back workers whose gain from formality falls short of it: essentially none of the unemployed, but 56.5 percent of informal workers.

For these held-back workers, the question is whether informality improves their prospects while they wait. Two estimates in Table 6 suggest it does. First, informal drift is positive and about equal to formal drift, so informal employment shields workers from unemployment, where human capital falls far faster than employment builds it. Second, informal jobs separate less often than formal ones, at 0.10 percent a year against 0.26, so accumulation is rarely interrupted by a return to unemployment. Informal workers, however, remain exposed to idiosyncratic shocks, which formal workers barely face. For a held-back worker, the gain from formality therefore tends to widen as the informal spell lasts, and informality becomes a place to grow into clearing the entry cost.

Model fit. The estimated model closely reproduces the moments that discipline these mechanisms. Table 7 summarizes this fit through the weight of each moment family in the criterion and its loss at the estimate. The criterion at the estimate is 0.17, so the typical model moment misses its data counterpart by about 17 percent. The losses show where that discrepancy comes

from. The model fits the moments out of informal employment almost exactly and misses most on exits from formal employment and on the unemployment hazard into formality. Figures 4 and 5, Table 8, and Figure 6 show what lies behind these losses.

Table 7: GMM Restrictions and Fit

Empirical family g	n_g	Independent restrictions	ω_g	$L_g(\widehat{\boldsymbol{\vartheta}})$
Labor shares	3	2	4/33	0.02
Entry cohort	30	20	8/33	0.03
$u \to f$ share	4	4	2/33	0.00
$f \to i$ share	5	5	2/33	0.12
$i \to f$ share	4	4	2/33	0.00
Wages	2	2	2/33	0.05
$u \to f$ hazards	4	4	3/33	0.01
$u \to i$ hazards	4	4	3/33	0.04
$i \to f$ hazards	4	4	4/33	0.01
$f \to i$ hazards	3	3	3/33	0.02
Total	63	52	33/33	0.17
Jointly estimated structural parameters: $\dim(\boldsymbol{\vartheta}) = 17$				

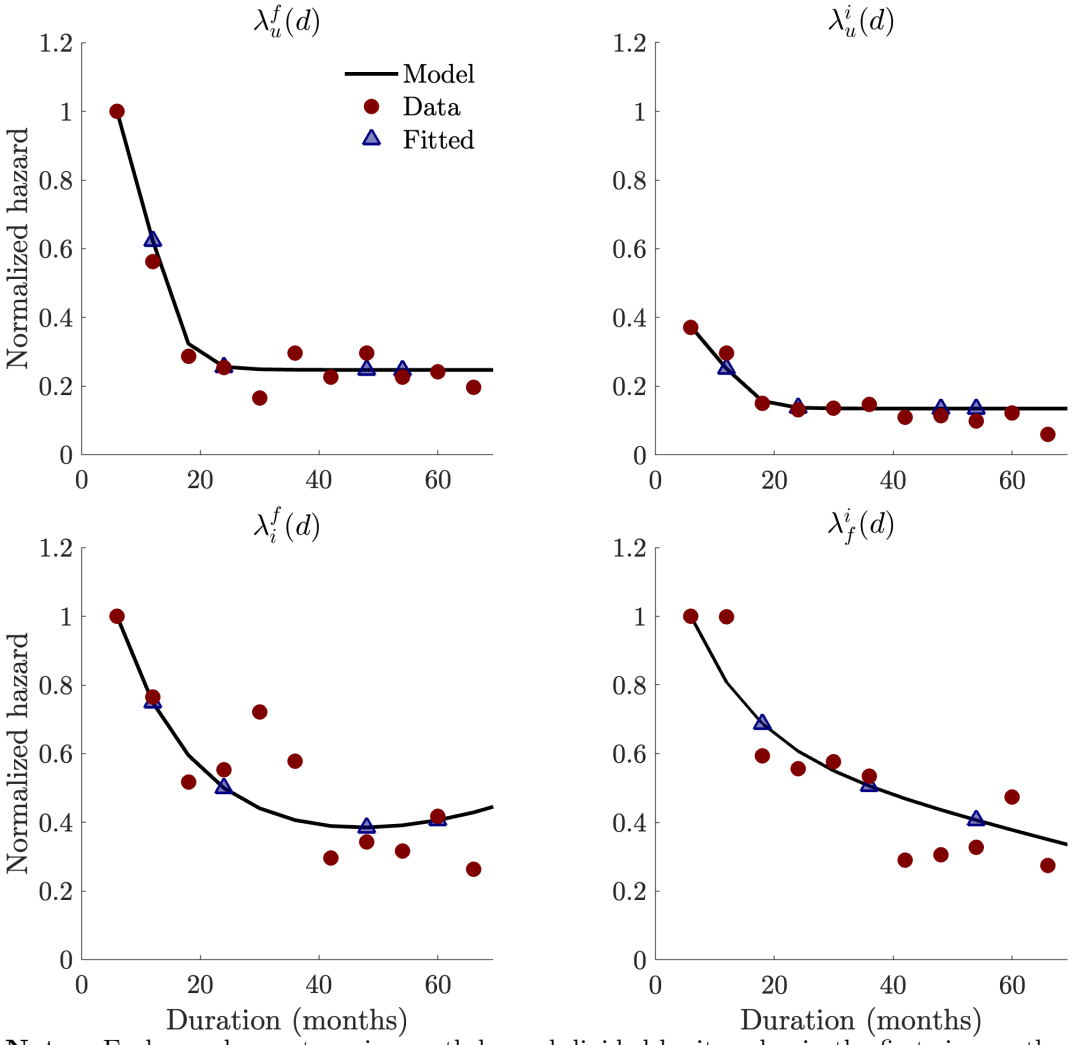
Notes: n_g is the number of targeted objects in family g, and independent restrictions exclude adding-up identities. ω_g is the normalized family weight, and $L_g(\widehat{\boldsymbol{\vartheta}})$ is the family loss at the estimate. The total row reports the criterion in Equation (16).

Figure 4 shows that the model reproduces not only how each hazard changes over a spell but also the forces behind each change. First, the hazards out of unemployment fall steeply in the first two years and then flatten. The steep fall reflects selection, which removes the most active searchers first. The flattening that follows reflects structural duration dependence, because human capital falls so fast in unemployment that the search incentives of those who remain vanish. Second, the hazard from formal to informal employment declines steadily, as selection removes the workers who would rather be informal. Finally, in the model as in the data, the hazard from informal to formal employment falls and then turns up after about four years. Selection again dominates the early fall. The upturn reflects accumulation in informality, which widens the gain from formality as the spell lasts. This upturn is the clearest effect of informal accumulation in the data.

Where spells end tells the same story, as Figure 5 shows. Out of formal employment, selection again drives the pattern. Workers who search for informal jobs leave first, while separations into unemployment continue at a constant rate, so the informal share of exits falls with duration. The formal-origin estimates are noisy at the shortest and longest durations, where the model misses them. By contrast, the formal share of exits from informal employment inherits the U-shape of the informal-to-formal hazard, and with it the effect of accumulation.

Selection and accumulation within spells, in turn, determine the stationary allocation in Table 8, and they account for both its fit and its main miss. The model comes within 0.21 percentage points of the informal share and reproduces the formal–informal wage gap that the estimation targets. The main miss comes from the flat hazards out of unemployment. The long-term unemployed rarely leave, so the model places about 5.1 points too many workers in unemployment, mostly at

Figure 4: Hazard Fit

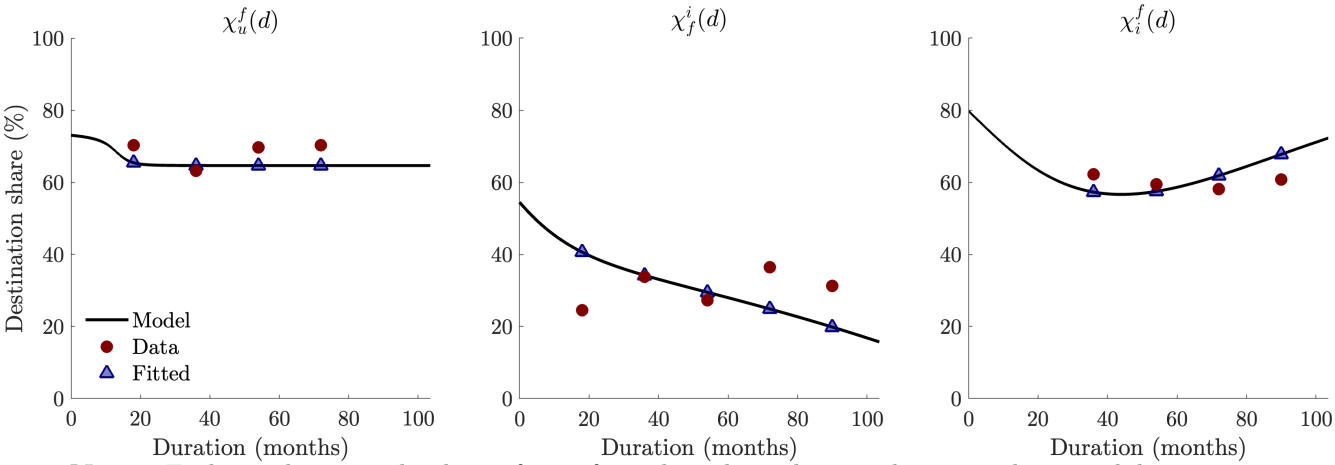


Notes: Each panel reports a six-month hazard divided by its value in the first six months, separately in the model and the data. Both unemployment hazards are divided by the hazard into formal employment. Lines are model profiles, triangles mark the model at targeted durations, and circles are EPS estimates at every six-month interval. Duration is in months.

the expense of formal employment. Long spells also account for the miss in wages. Mean wages exceed their data counterparts by 39.1 percent in formal and 54.5 percent in informal employment. Because the model matches the geometric mean formal wage through A, its higher arithmetic means imply more dispersed wages, as long employment spells keep accumulating human capital.

Figure 6 closes the comparison with a cohort of new workers followed from their first job search. The model reproduces the cohort's informal share throughout. It moves entrants into formal employment somewhat too quickly in the first four years, consistent with hazards out of unemployment that decline less than in the data. The gap narrows to 2.2 percentage points by the fifth year. Appendix B.4 documents the empirical cohort construction and observation rules, with the model counterpart in Appendix C.2. The model therefore reproduces how new workers sort into informal and formal employment over their first years in the labor market.

Figure 5: Destination-Share Fit



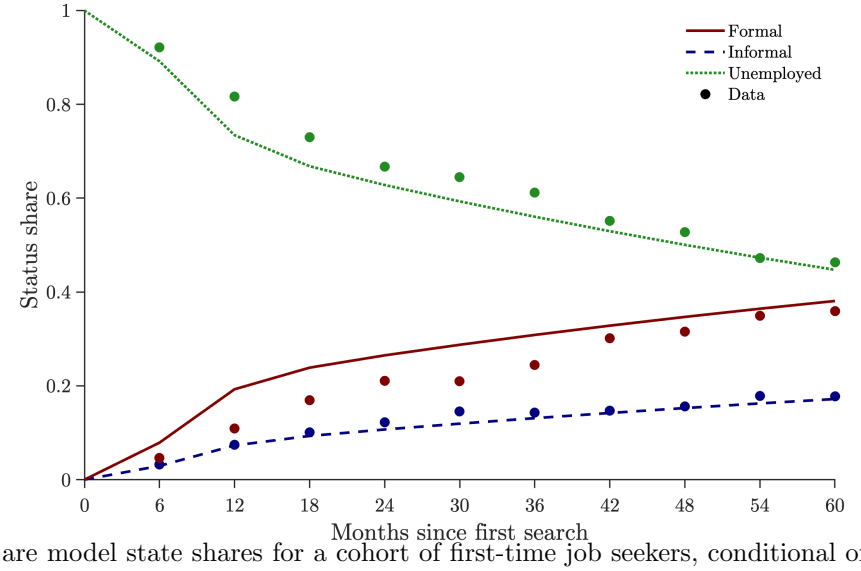
Notes: Each panel reports the share of exits from the indicated origin that go to the named destination, among exits to the other two labor-market states. Lines are model profiles, triangles mark the model at targeted durations, and circles are the EPS targets.

Table 8: Scalar and Wage Moment Fit

Moment	Data	Model	Difference
Panel A. Aggregate moments			
Formal employment (%)	59.83	54.47	-5.36
Informal employment (%)	20.80	21.01	+0.21
Unemployment (%)	19.37	24.52	+5.15
Panel B. Wage moments			
Formal wage mean (USD/year)	7,840	10,903	+39.1%
Informal wage mean (USD/year)	5,655	8,734	+54.5%

Notes: Panel B reports means of spell-average wages over completed spells, in real 2010 USD a year. The data use fully wage-observed, non-left-censored spells satisfying the age and occupation restrictions in Appendix B.3, with stock weights based on observed spell-month exposure. Differences are model minus data, in percentage points in Panel A and in percent in Panel B.

Figure 6: Entry-Cohort Trajectories



Notes: Lines are model state shares for a cohort of first-time job seekers, conditional on remaining in the labor force, and circles are the EPS targets. The horizontal axis measures months since first search.

5 Paths to Formality

With the estimated model, I proceed to assess how informal employment affects new workers' access to formality, and how it translates into economic welfare. I begin by comparing the life cycle of a cohort of new workers in two economies whose only difference is the presence of an informal sector. The difference reveals by how much it changes the probability of formalization for any new worker in the absence of informality. Then, I proceed to quantify the importance of each of the three mechanisms that distinguish informality from unemployment on the probability of formalization. Subsequently, I proceed to evaluate the private value of informality for workers. This in turn allows me to compare the welfare losses of removing the informality option against the fiscal gain generated by expanding the tax base. And finally, I compare the effects of distinct formalization policies on welfare, fiscal gains, and production.

5.1 Access to Formal Employment

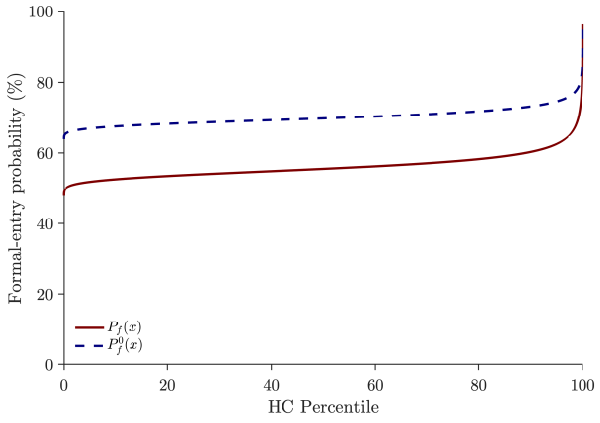
Consider a worker looking for their first job, with human capital x drawn from the entrant distribution G. The worker can only enter the formal sector for the first time at time T_f if they have not left the economy yet. I denote the departure moment of a worker by T_δ and the probability of the worker's eventual formalization as

$$P_f(x) = \Pr(T_f \leq T_\delta \mid x(0) = x, \ell(0) = u)\,. \tag{18}$$

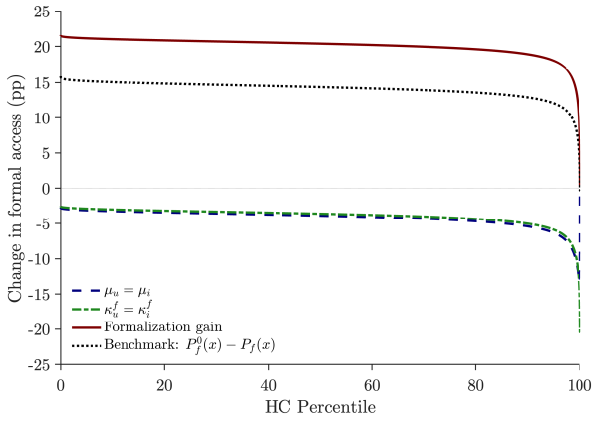
For the same worker, I denote by $P_f^0(x)$ the probability of formalization for a worker in the no-informality economy. Figure 7a shows that the informal option decreases the probability of formalization, $\Delta P_f(x) = P_f(x) - P_f^0(x)$, for the vast majority of the economy.

Figure 7: Informality and Formal Access

a Formal access



b Mechanisms



For the average worker, the probability of formalization jumps from 56.10% to 70.18% when informality is no longer an option. The effects are mostly flat, ranging from an increase of a little over 15 pp for a worker at the bottom to 10 pp for a worker in the 99th percentile of the

human capital entry distribution. The reduction of the effect as one climbs in the human capital distribution is explained by comparative advantage. Workers who were initially at the top of the human capital distribution favor formal employment ex ante and thus still search for formal employment even if they transition initially to informality.

To better understand what drives the reduction in formalization, I isolate each of the mechanisms that give informal workers incentives to search for formal employment. First, I set μ_i to its unemployment counterpart μ_u and, at fixed search policies, compare $P_f(x)$. I do the same comparison using the implied search policies from equalizing the search costs $\kappa_i^f = \kappa_u^f$, and the formal gain that an unemployed worker gets $V_f(x) - V_i(x) = V_f(x) - U(x)$. This partial equilibrium exercise helps me evaluate the effect on $P_f(x)$ when I shut down one of the mechanisms and hold everything else equal.

Figure 7b shows that two of the mechanisms facilitate eventual formalization of new workers, and isolates the one that reduces $P_f(x)$. Both human capital accumulation μ_i and the search cost κ_i^f contribute to the formalization of a new worker. Specifically, when informality loses its human capital accumulation advantage relative to unemployment, the average $P_f(x)$ drops by 4.20 pp. More importantly, all new workers' prospects are dampened, with the more skilled suffering the bigger losses. This shows how informality provides a way to insure the workers' only asset: human capital.

Removing the search advantage of informal employment also reduces the average $P_f(x)$ by 3.91 pp, slightly less than the effect of human capital accumulation. Even with the high search costs from unemployment, informality allows workers to build human capital towards an eventual formalization. By contrast, giving informal workers the gains of formalization increases the average $P_f(x)$ by 20.07 pp. This shows what holds back workers from formalizing once they reach informality: the value of formality is not as appealing to an informal worker as it is to an unemployed worker.

5.2 Welfare and Fiscal Gains

The previous exercise showed that while informality can reduce the probability of eventual formalization, it also has a private value for workers. However, as the share of formal workers grows, so does the tax base. What is unclear is how the welfare losses from removing the private value of informality compare to the fiscal gain of expanding the tax base.

Recall that the government levies taxes to subsidize unemployment benefits and the rest is rebated to the workers as a lump-sum transfer. Since all workers enter the market as unemployed, and preferences are linear in consumption, the welfare value can be expressed as the sum of the value of unemployment and the present discounted value of the lump-sum transfers,

$$W(x) = U(x) + \frac{t}{\rho + \delta}, \qquad \Delta W(x) = \Delta U(x) + \frac{\Delta t}{\rho + \delta}. \tag{19}$$

In turn, the same formula allows me to decompose the welfare effects of removing the informal option into two terms: the private value of the informal option and the fiscal gain. Table 9 presents the welfare decomposition for three different groups of workers. First, I consider the workers for whom formal employment was dominated by informality, or $V_f(x) - V_i(x) < 0$. Second, I study the workers who find formality more favorable than informality, but not enough to offset the entry cost, $0 \leq V_f(x) - V_i(x) < \Psi_f$. And finally, I consider workers who would search for formal employment from every other employment status.

Table 9: Formal Access and Welfare of New Workers

Group	Share (%)	P_f^0	P_f	ΔP_f	$\Delta\tilde{U}$	$\Delta\tilde{W}$	Transfer (%)
Informality preferred	0.35	65.21	49.59	−15.62	+11.57	+10.57	−9.45
Formality preferred, entry cost not covered	97.11	69.99	55.78	−14.21	+11.18	+10.53	−6.13
Searching from informality	2.54	78.36	69.37	−8.99	+10.38	+10.06	−3.21
All entrants	**100.00**	**70.18**	**56.10**	**−14.08**	**+11.14**	**+10.51**	**−6.00**

Notes: Share, P_f^0, and P_f are in percent, and $\Delta P_f = P_f - P_f^0$ is in percentage points. $\Delta\tilde{U}$ and $\Delta\tilde{W}$ report private and total welfare changes in consumption-equivalent percent relative to consumption in the economy with informality. Transfer (%) is the discounted change in common transfers as a percentage of the group's net total welfare change. Changes are with informality minus without informality, and group averages use the entrant distribution G.

For the vast majority of the economy, formality is the more valuable employment status. Yet only a tiny minority would be willing to search for formal employment if they were to reach informality first. This further explains the result in Figure 7b. Informality provides enough value relative to unemployment that most informal workers do not find it worth paying the entry cost of formality.

Consequently, the option value of informality is sufficient to outweigh the fiscal gains from increasing the tax base. On average, the private value of the informal option is valued at 11.14% in consumption-equivalent terms relative to the benchmark economy. By contrast, the fiscal losses of allowing an informal sector only account for 6% of the total welfare change.

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A Theoretical Appendix

A.1 Worker Problem and Optimal Policies

This subsection establishes the properties of the stationary worker values that the rest of the appendix uses. The values grow at most linearly and are nondecreasing, Lipschitz, and convex in human capital. These properties then determine where workers quit and how values and search behave at high human capital.

Wages are stationary, $w_\ell(x) = a_\ell x$ with the coefficients in equation (2), so I suppress calendar time. Both coefficients are positive, since $\alpha_0 < 1$, and I write $\bar{a} = \max\{a_i, a_f\}$. Results that need a formal wage advantage, $a_f > a_i$, state it.

The proofs compare each value with the payoff of a particular plan. A plan chooses search intensities and quitting dates from the information available to date.[23] Write $W_\ell(x;\pi)$ for the expected lifetime payoff of an admissible plan π in state $\ell \in \{u, i, f\}$ at human capital x, and $W_\ell(x)$ for its supremum, so that $W_u = U$, $W_i = V_i$, and $W_f = V_f$. Let τ_δ be the date of labor-force exit.

Bounds and shape. The first result supplies the bounds that the later comparisons use. Intuitively, a worker can always secure the benefit by remaining unemployed, while no plan earns more than the highest wage slope applied to accumulated human capital. Values therefore lie between the value of permanent unemployment, $b/(\rho+\delta)$, and a line whose slope $\bar{a}/(\rho+\delta)$ is the value of earning the highest wage slope on a unit of human capital until labor-force exit.

Lemma 1 (Value bounds and transversality)**.** *The stationary worker values have the following properties.*

(a) There is a finite constant B, independent of search costs, such that for every $\ell \in \{u, i, f\}$,

$$\frac{b}{\rho+\delta} \le W_\ell(x) \le \frac{\bar{a}}{\rho+\delta}x + B, \qquad x \ge 0. \tag{A.1}$$

(b) The formal value satisfies

$$V_f(x) \ge \frac{b}{\rho+\delta} + \frac{a_f x - F - b}{\rho+\delta+\delta_f}. \tag{A.2}$$

(c) For every worker value W_ℓ and every admissible plan, $\mathbb{E}_x[e^{-\rho T}|W_\ell(x(T))|\mathbb{1}\{\tau_\delta > T\}] \to 0$ as $T \to \infty$. The same limit holds when survival is restricted to the current spell.

[23] An admissible plan has nonnegative, predictable search intensities and quitting dates that are stopping times. Cumulative intensities are finite on finite horizons, transitions are nonexplosive, and discounted payoffs and costs are integrable. The Brownian motion, search-arrival drivers, separation clocks, and exit clock are mutually independent on a complete, right-continuous filtered probability space.

Proof. Fix an initial state ℓ, initial human capital x, and an admissible plan π. All three parts rest on one estimate: expected human capital rises at most linearly in time, uniformly over admissible plans. The proof derives this estimate first. Integrating it against the discount factor $e^{-(\rho+\delta)s}$ gives the upper bound in (a), and evaluating it at a date T gives (c). The lower bounds in (a) and (b) come from two simple plans.

To separate human-capital dynamics from survival, evaluate the plan along histories without labor-force exit. Survival then enters payoffs only through the effective discount rate $\rho + \delta$. Let $j(s)$ be the pretransition labor-market state in this exit-free process. The process is independent of τ_δ and agrees with the worker's state on $\{\tau_\delta > s\}$. Admissibility gives nonexplosion and a predictable pretransition state, and human capital is continuous across transitions. Define

$$z(s) = \int_0^s \mu_{j(r)}\,\mathrm{d}r + M(s), \qquad M(s) = \int_0^s \sigma_{j(r)}\,\mathrm{d}W_r, \qquad k(s) = -\inf_{0\le r\le s} z(r).$$

The one-dimensional Skorokhod representation of the reflected path then gives $x(s) = z(s) + \max\{x, k(s)\}$.

Write $\bar\mu = \max_j |\mu_j|$ and $\bar\sigma = \max_j \sigma_j$, where $j \in \{u, i, f\}$. The maximum in this representation is at most $x + \bar\mu s + \sup_{r\le s} |M(r)|$, and the drift part of $z(s)$ is at most $\bar\mu s$. Since $\mathbb{E}[M(s)] = 0$, and Doob's inequality with $\mathbb{E}[M(s)^2] \le \bar\sigma^2 s$ gives $\mathbb{E}[\sup_{r\le s} |M(r)|] \le 2\bar\sigma\sqrt{s}$, taking expectations gives, uniformly over admissible plans,

$$\mathbb{E}_x[x(s)] \le x + 2\bar\mu s + 2\bar\sigma\sqrt{s}. \tag{A.3}$$

Upper bound in (a). Search, formal-employment, and transition costs are nonnegative, so flow payoffs are at most $b + \bar a x(s)$. Because labor-force exit is independent and occurs at rate δ, discounting at ρ until exit is equivalent to discounting at $\rho + \delta$. With equation (A.3), this gives

$$\begin{aligned} W_\ell(x;\pi) &\le \int_0^\infty e^{-(\rho+\delta)s}\big[b + \bar a\,\mathbb{E}_x[x(s)]\big]\,\mathrm{d}s \\ &\le \int_0^\infty e^{-(\rho+\delta)s}\big[b + \bar a(x + 2\bar\mu s + 2\bar\sigma\sqrt{s})\big]\,\mathrm{d}s = \frac{\bar a}{\rho+\delta}x + B. \end{aligned}$$

Since $\rho + \delta > 0$, the constant B is finite. It depends only on benefits, wage and human-capital coefficients, and $\rho + \delta$, so it is the same for every initial state and plan and does not depend on search costs. Taking suprema over plans gives the upper bound for each value.

Lower bounds in (a) and (b). Each lower bound comes from a simple feasible plan. An unemployed worker who never searches receives b until labor-force exit, which is worth $b/(\rho + \delta)$. An employed worker can quit immediately and then follow the same plan. Taking suprema gives the lower bound in (a). For (b), consider a formal worker who never searches, keeps the job until exogenous separation, and then remains unemployed without searching. Since $\mu_f > 0$ and reflection only adds human capital, expected human capital during the formal spell is at least x. Discounting the formal spell at $\rho+\delta+\delta_f$ and the subsequent unemployment at $\rho+\delta$ gives equation

(A.2).

Transversality. Part (a) provides a finite C with $|W_\ell(x)| \le C(1+x)$ for every ℓ. Independence of exit and equation (A.3) then give

$$\begin{aligned}\mathbb{E}_x\left[e^{-\rho T}\left|W_\ell(x(T))\right|\mathbb{1}\{\tau_\delta > T\}\right] &= e^{-(\rho+\delta)T}\mathbb{E}_x[|W_\ell(x(T))|] \\ &\le Ce^{-(\rho+\delta)T}\left(1 + x + 2\bar{\mu}T + 2\bar{\sigma}\sqrt{T}\right) \longrightarrow 0.\end{aligned}$$

Restricting survival to the current spell only reduces this nonnegative term, so the same limit holds. □

Monotonicity, Lipschitz continuity, and convexity all rest on one construction. Note that a worker can imitate the search and quitting decisions of a worker who starts with different human capital. The two workers then follow the same labor-market path and pay the same costs, so their payoffs differ only through wages.

Proposition 2 (Monotonicity of worker values)**.** *Each worker value is nondecreasing in human capital: $W_\ell(y) \ge W_\ell(x)$ for every $\ell \in \{u, i, f\}$ and $y \ge x \ge 0$.*

Proof. Fix an initial state ℓ, levels of human capital $y \ge x$, and an admissible plan π at x. It suffices to construct an admissible plan at y whose payoff is at least $W_\ell(x;\pi)$, because taking the supremum over π then gives $W_\ell(y) \ge W_\ell(x)$. The plan at y imitates π. The proof first defines the imitation and checks that it is admissible, then shows that it follows the same labor-market path with weakly more human capital.

Imitation. A worker starting from y can reconstruct, from x and the primitive shocks observed to date, the history that π would have generated, and then take the search intensities and quitting decisions that π prescribes. On histories outside this construction, the plan specifies zero search and immediate quitting if employed. Copied intensities are predictable, and copied quitting times are stopping times. Because the two workers share the arrival drivers and separation clocks and take the same actions, their next transition is the same. By induction and nonexplosion, the labor-market paths coincide on every finite horizon. Search costs, entry costs, and transition dates coincide, so cost integrability is inherited from π, and equation (A.3) gives integrability of flow income. Hence the imitating plan is admissible.

Payoff comparison. Let $j(s)$ be the common labor-market path, and define $z(s)$ and $k(s)$ along it as in the proof of Lemma 1. The two reflected paths are $x(s;x) = z(s) + \max\{x, k(s)\}$ and $x(s;y) = z(s) + \max\{y, k(s)\}$, so

$$0 \le x(s;y) - x(s;x) \le y - x. \tag{A.4}$$

The worker starting from y thus holds weakly more human capital at every date and earns weakly higher wages, while benefits and costs coincide. Consequently, the imitating plan yields at least $W_\ell(x;\pi)$, which is what was needed. □

Lemma 2 (Global Lipschitz continuity and convexity)**.** *For every $\ell \in \{u, i, f\}$, the value W_ℓ is convex, and*

$$0 \le W_\ell(y) - W_\ell(x) \le \frac{\bar{a}}{\rho + \delta}(y - x), \qquad y \ge x \ge 0.$$

In particular, W_ℓ has one-sided derivatives on $(0, \infty)$, with $0 \le \partial_x^- W_\ell(x) \le \partial_x^+ W_\ell(x) \le \bar{a}/(\rho + \delta)$.

Observe that the Lipschitz constant is the slope of the upper bound in Lemma 1: an extra unit of human capital is worth at most the highest wage slope, earned on that unit until labor-force exit.

Proof. Both properties follow from the imitation in the proof of Proposition 2. An imitating plan reproduces the labor-market path and all costs, so payoffs differ only through wages, and wages differ only through human capital. For the Lipschitz bound, a worker with less human capital imitates one with more, and equation (A.4) limits the wage gap. For convexity, workers at two endpoints imitate a worker at an intermediate point, and convexity of the reflected path in initial human capital does the rest.

Lipschitz continuity. Fix ℓ, $y \ge x$, and an admissible plan π_y at y. The imitation yields an admissible plan $\tilde{\pi}_x$ at x with the same labor-market path and costs as π_y, and equation (A.4) keeps the human-capital gap in $[0, y - x]$. Since wage slopes, including $a_u = 0$, lie in $[0, \bar{a}]$,

$$W_\ell(y; \pi_y) - W_\ell(x; \tilde{\pi}_x) \le \bar{a}(y - x)\mathbb{E}\left[\int_0^{\tau_\delta} e^{-\rho s}\, \mathrm{d}s\right] = \frac{\bar{a}}{\rho + \delta}(y - x).$$

Replacing $W_\ell(x; \tilde{\pi}_x)$ by the larger $W_\ell(x)$ and taking the supremum over π_y gives the upper bound. Proposition 2 gives the lower bound.

Convexity. Fix $x_0, x_1 \ge 0$, $\lambda \in [0, 1]$, and $x_\lambda = \lambda x_1 + (1 - \lambda)x_0$. For an admissible plan π_λ at x_λ, workers starting at x_0 and x_1 imitate it. This gives admissible plans $\tilde{\pi}_0$ and $\tilde{\pi}_1$ with the same labor-market path and costs. Along this path, $z(s)$ and $k(s)$ are common, and $x \mapsto z(s) + \max\{x, k(s)\}$ is convex, so $x(s; x_\lambda) \le \lambda x(s; x_1) + (1 - \lambda)x(s; x_0)$ for every s. Flow income in each state is affine in human capital with a nonnegative slope, and all costs are common. Integrating along paths and taking expectations therefore gives

$$\begin{aligned} W_\ell(x_\lambda; \pi_\lambda) &\le \lambda W_\ell(x_1; \tilde{\pi}_1) + (1 - \lambda)W_\ell(x_0; \tilde{\pi}_0) \\ &\le \lambda W_\ell(x_1) + (1 - \lambda)W_\ell(x_0). \end{aligned}$$

Taking suprema over π_λ proves convexity. Finally, a convex function whose increments lie in $[0, \bar{a}(y - x)/(\rho + \delta)]$ has one-sided derivatives in $[0, \bar{a}/(\rho + \delta)]$, ordered as stated. □

Quitting. For $\ell \in \{i, f\}$, the employment surplus $D_\ell = V_\ell - U$ is the gain from holding a job in ℓ rather than being unemployed. Because workers can quit, $D_\ell \ge 0$, and the quitting and

continuation sets are

$$\mathcal{S}_\ell = \{x \geq 0 : D_\ell(x) = 0\}, \qquad \mathcal{C}_\ell = \{x \geq 0 : D_\ell(x) > 0\}.$$

Whether quitting is confined to low human capital depends on how the surplus changes with human capital. A higher level of human capital raises the value of a job in ℓ through the wage and through human-capital accumulation, which is never slower than in unemployment. The value of unemployment rises as well, because unemployed workers search for jobs whose wages increase with human capital. When the gain in the job value is at least as large as the gain in the unemployment value, D_ℓ is nondecreasing in human capital. I call this property increasing differences between employment and unemployment and maintain it where stated.[24] Under increasing differences, unemployment search $\eta_u^\ell = [D_\ell - \Psi_\ell]^+/\kappa_u^\ell$ is nondecreasing, so unemployed workers search toward ℓ only above some level of human capital.

A worker at a quitting threshold is indifferent between quitting and staying employed, so the surplus is zero there. The first result shows that the slope of the surplus is zero as well. This argument, like the ones that follow, differentiates the stationary HJB equations.[25]

Proposition 3 (Value matching and smooth pasting)**.** *Suppose $\bar{x}_u^\ell > 0$ is a quitting threshold: points immediately to its left belong to $\mathcal{S}_\ell$, and points immediately to its right belong to $\mathcal{C}_\ell$. Then value matching and smooth pasting hold:*

$$V_\ell(\bar{x}_u^\ell) = U(\bar{x}_u^\ell),$$
$$\lim_{x \downarrow \bar{x}_u^\ell} \partial_x V_\ell(x) = \partial_x U(\bar{x}_u^\ell).$$

Proof. Both claims concern the surplus D_ℓ at $\bar{x}_u^\ell$. Value matching says that D_ℓ vanishes there, and smooth pasting says that its slope from the continuation side vanishes too. The first claim is immediate. Continuity makes $\mathcal{S}_\ell$ closed, and $\bar{x}_u^\ell$ is a limit point of $\mathcal{S}_\ell$, so $D_\ell(\bar{x}_u^\ell) = 0$.

The second claim requires an argument. By the mean-value theorem, the finite continuation-side limit of $\partial_x D_\ell$ equals the right derivative of D_ℓ at $\bar{x}_u^\ell$. This derivative is nonnegative, because $D_\ell \geq 0$ and $D_\ell(\bar{x}_u^\ell) = 0$. Suppose, to reach a contradiction, that it is positive. Then a worker at $\bar{x}_u^\ell$ who delays quitting for a short time h gains from favorable shocks, which carry the worker into the region where the surplus is positive. This gain is of order $\sqrt{h}$, while the flow and discounting costs of the delay are of order h. For small h, delay would therefore beat quitting. The rest of the proof makes this comparison precise.

A positive right derivative gives $c > 0$ and a radius r, with $0 < r < \bar{x}_u^\ell/2$ and $(\bar{x}_u^\ell, \bar{x}_u^\ell + r) \subseteq \mathcal{C}_\ell$, such that $D_\ell(x) \geq c(x - \bar{x}_u^\ell)$ for $x \in (\bar{x}_u^\ell, \bar{x}_u^\ell + r)$. The delay strategy is as follows. Starting at $\bar{x}_u^\ell$, the worker postpones quitting and does not search until time h or until human capital leaves $(\bar{x}_u^\ell - r, \bar{x}_u^\ell + r)$, whichever comes first, unless separation or labor-force exit occurs earlier. Let τ_r be

[24] Sufficient conditions on the primitives are available from the author upon request.

[25] I take as given $U \in C^3((0,\infty))$ and $V_\ell \in C^3(\mathcal{C}_\ell \cap (0,\infty))$, with finite continuation-side derivative limits at finite boundaries and a zero derivative where human capital is reflected at zero.

the date human capital leaves this interval and τ_{δ_ℓ} the date of exogenous separation from ℓ. After the delay, the worker continues optimally, so continuation is valued at the corresponding worker value, and labor-force exit pays zero. Let J_h be the value of this strategy. Since the strategy is feasible, $J_h \le V_\ell(\bar{x}_u^\ell)$.

To compare J_h with $U(\bar{x}_u^\ell)$, split the employment continuation value as $U + D_\ell$. The unemployment value changes only by $O(h)$, and the surplus carries the gain. Apply Itô's lemma to U up to h, τ_r, or an earlier exogenous event. On the neighborhood, wages, U, and its derivatives are bounded, and the stopped martingale has zero expectation. Flow income, changes in U, discounting, and exogenous events therefore contribute $O(h)$, and for some finite C,

$$\begin{aligned} J_h - U(\bar{x}_u^\ell) \ge & -Ch \\ & + \mathbb{E}\big[e^{-\rho h} D_\ell(x(h)) \mathbb{1}\{\tau_r > h,\ \min\{\tau_{\delta_\ell}, \tau_\delta\} > h\}\big]. \end{aligned}$$

Dropping the terminal surplus on paths with $\tau_r \le h$ preserves the bound, since that surplus is nonnegative. On the event $\{\tau_r > h,\ \min\{\tau_{\delta_\ell}, \tau_\delta\} > h\}$, the worker is still in state ℓ, and $r < \bar{x}_u^\ell/2$ keeps human capital away from zero. Hence $x(h) - \bar{x}_u^\ell = \mu_\ell h + \sigma_\ell W_h$.

It remains to bound the surplus term from below. The drift contributes only $O(h)$, whereas $\mathbb{E}[W_h^+] = \sqrt{h/(2\pi)}$. Exogenous events have probability $O(h)$, and leaving the neighborhood before h has exponentially small probability, so excluding these paths changes the Brownian gain by $o(\sqrt{h})$. The local surplus bound therefore gives

$$\mathbb{E}\big[e^{-\rho h} D_\ell(x(h)) \mathbb{1}\{0 < x(h) - \bar{x}_u^\ell < r,\ \tau_r > h,\ \min\{\tau_{\delta_\ell}, \tau_\delta\} > h\}\big] \ge \frac{c\sigma_\ell}{\sqrt{2\pi}} \sqrt{h} + o(\sqrt{h}).$$

For small h, the gain of order $\sqrt{h}$ dominates the loss of order h, so $V_\ell(\bar{x}_u^\ell) \ge J_h > U(\bar{x}_u^\ell)$. This contradicts value matching. Hence the continuation-side slope of D_ℓ is zero, which is smooth pasting. □

The next result shows that quitting, when it occurs, is confined to a bounded interval of low human capital. Its proof uses the generator of human capital in state ℓ, $\mathcal{L}_\ell = \mu_\ell \partial_x + (\sigma_\ell^2/2)\partial_{xx}$.

Proposition 4 (Classification of quitting sets)**.** *Suppose increasing differences hold in both sectors. Fix $\ell \in \{i, f\}$ and suppose $\sigma_\ell^2/\sigma_u^2 > 1 - a_\ell/\bar{a}$. There is a tolerance $\epsilon_\ell > 0$, which does not depend on $\kappa_\ell^{-\ell}$, such that, if $\kappa_u^{-\ell}/\kappa_\ell^{-\ell} > \sigma_\ell^2/\sigma_u^2 - \epsilon_\ell$, the quitting set is either empty or a bounded lower interval $[0, \bar{x}_u^\ell]$. Thus the continuation set is connected, and when quitting occurs, its threshold is unique.*

Observe that the volatility restriction holds automatically in a sector whose wage slope equals $\bar{a}$, so it has content only in the sector with the lower slope. The search-cost restriction is best read as a bound on search from employment: searching toward the other sector cannot be too costly for an employed worker relative to an unemployed one, once costs are scaled by the variance ratio. Finally, note that the proposition separates two cases at the bottom of the state space. A zero

threshold means that workers quit at zero human capital, whereas an empty quitting set means that human capital is reflected there.

Proof. Fix $\ell \in \{i, f\}$. There are two claims: $\mathcal{S}_\ell$ is a lower interval, and it is bounded. The first follows from increasing differences. Because $D_\ell \geq 0$ is nondecreasing, $D_\ell(y) = 0$ implies $D_\ell(x) = 0$ for every $x \leq y$. For the second, the proof constructs a comparison function Γ_ℓ that is nonpositive at every positive quitting point and positive above an explicit level X_ℓ. No quitting point can then exceed X_ℓ. A final step shows that Γ_ℓ increases at a rate bounded away from zero below X_ℓ.

The comparison function combines the two HJB equations. Write $F_\ell = \mathbb{1}\{\ell = f\}F$ for the employment flow cost. Substituting $V_\ell = U + D_\ell$ into the employment HJB and eliminating U'' with the unemployment HJB gives

$$\begin{aligned}\Gamma_\ell(x) = a_\ell x - F_\ell - \frac{\sigma_\ell^2}{\sigma_u^2} b + \Big(\frac{\sigma_\ell^2}{\sigma_u^2} - 1\Big)(\rho + \delta)U(x) + \Big(\mu_\ell - \frac{\sigma_\ell^2}{\sigma_u^2}\mu_u\Big)U'(x) \\ + \left(\frac{\kappa_u^{-\ell}}{\kappa_\ell^{-\ell}} - \frac{\sigma_\ell^2}{\sigma_u^2}\right) \frac{\big([D_{-\ell}(x) - \Psi_{-\ell}]^+\big)^2}{2\kappa_u^{-\ell}}.\end{aligned} \tag{A.5}$$

The last term contains $[D_{-\ell} - \Psi_{-\ell}]^+$, the net gain in the unemployed worker's search rule toward the other employment state, equation (5). By Lemma 2 and increasing differences, this gain is nondecreasing and Lipschitz with constant $\bar{a}/(\rho + \delta)$.

At a positive quitting point y, the smooth function U touches V_ℓ from below, and a short feasible continuation cannot improve on $U(y)$. The dynamic-programming supersolution inequality with U as the test function, after optimizing search and substituting the unemployment HJB, therefore gives $\Gamma_\ell(y) \leq 0$.

Two bounds on the unemployment value control Γ_ℓ. Lemma 1 gives $b/(\rho+\delta) \leq U(x) \leq \bar{a}x/(\rho+\delta) + B$, where B does not depend on search costs. Since $U'' \geq 0$, $\mu_u \leq 0$, and $0 \leq U' \leq \bar{a}/(\rho + \delta)$, the unemployment HJB also bounds the value of search toward the other employment state:

$$0 \leq \frac{\big([D_{-\ell}(x) - \Psi_{-\ell}]^+\big)^2}{2\kappa_u^{-\ell}} \leq (\rho + \delta)U(x) - b - \mu_u U'(x) \leq \bar{a}x + K, \tag{A.6}$$

with $K = (\rho + \delta)B - b - \mu_u\bar{a}/(\rho + \delta)$. The construction of B in Lemma 1 gives $(\rho + \delta)B > b$, so $K > 0$. The estimate extends to zero by continuity.

Bounded quitting sets. The bound on quitting comes from the linear growth of Γ_ℓ. The volatility restriction makes the slope margin $\gamma_\ell = a_\ell - (1 - \sigma_\ell^2/\sigma_u^2)^+\bar{a}$ positive. Define

$$X_\ell = \frac{2\{b + F_\ell + [1 + (1 - \sigma_\ell^2/\sigma_u^2)^+]K\}}{\gamma_\ell}, \qquad \epsilon_\ell = \min\left\{\frac{\gamma_\ell}{2\bar{a}}, \frac{\gamma_\ell(\rho + \delta)\sqrt{\kappa_u^{-\ell}}}{2\bar{a}\sqrt{2(\bar{a}X_\ell + K)}}\right\}.$$

Neither constant depends on $\kappa_\ell^{-\ell}$, so the search-cost restriction is not circular. That restriction makes the coefficient of the last term in Γ_ℓ larger than $-\epsilon_\ell$. To bound the remaining terms, use

$U \geq b/(\rho+\delta)$ when $\sigma_\ell \geq \sigma_u$ and $U \leq \bar{a}x/(\rho+\delta)+B$ when $\sigma_\ell < \sigma_u$. The drift term is nonnegative because $\mu_\ell - \mu_u\sigma_\ell^2/\sigma_u^2 \geq 0$ and $U' \geq 0$. Together with equation (A.6), these bounds give

$$\Gamma_\ell(x) \geq (\gamma_\ell - \bar{a}\epsilon_\ell)x - b - F_\ell - \left[(1-\sigma_\ell^2/\sigma_u^2)^+ + \epsilon_\ell\right]K.$$

Since $\epsilon_\ell \leq \gamma_\ell/(2\bar{a}) \leq 1/2$, the right side is positive for $x > X_\ell$. Every positive quitting point satisfies $\Gamma_\ell \leq 0$ and therefore lies in $[0, X_\ell]$. If $\mathcal{S}_\ell$ is nonempty, it is closed and downward closed, so it has a largest point $\bar{x}_u^\ell \in [0, X_\ell]$ and $\mathcal{S}_\ell = [0, \bar{x}_u^\ell]$. Otherwise $\mathcal{S}_\ell$ is empty.

Growth of the comparison function. On the continuation set, Γ_ℓ bounds the forcing in the surplus equation:

$$\begin{aligned}(\rho+\delta+\delta_\ell)D_\ell - \mathcal{L}_\ell D_\ell = \Gamma_\ell - \frac{\sigma_\ell^2}{\sigma_u^2}\frac{([D_\ell - \Psi_\ell]^+)^2}{2\kappa_u^\ell} \\ - \frac{([D_{-\ell} - \Psi_{-\ell}]^+)^2 - ([V_{-\ell} - V_\ell - \Psi_{-\ell}]^+)^2}{2\kappa_\ell^{-\ell}} \leq \Gamma_\ell. \qquad \text{(A.7)}\end{aligned}$$

Both subtracted terms are nonnegative. The first is unemployed search toward ℓ. The second compares the net gain from reaching the other sector out of unemployment with the same gain out of employment, and $D_\ell \geq 0$ makes the first gain at least as large. It remains to show that

$$\Gamma_\ell(y) - \Gamma_\ell(x) \geq \frac{\gamma_\ell}{2}(y-x), \qquad 0 \leq x < y \leq X_\ell.$$

By the same case split, comparing σ_ℓ with σ_u, the income and value-level terms of Γ_ℓ increase by at least $\gamma_\ell(y-x)$. Convexity of U makes the drift term nondecreasing. Only the search term can decrease, and only when its coefficient is negative. On $[0, X_\ell]$, equation (A.6) gives $[D_{-\ell} - \Psi_{-\ell}]^+ \leq \sqrt{2\kappa_u^{-\ell}(\bar{a}X_\ell + K)}$. Because the gain is nondecreasing and Lipschitz, its square rises by at most $2\bar{a}\sqrt{2\kappa_u^{-\ell}(\bar{a}X_\ell + K)}\,(y-x)/(\rho+\delta)$. With a coefficient above $-\epsilon_\ell$, the second term in ϵ_ℓ limits the decline of the search term to $(\gamma_\ell/2)(y-x)$, which proves the bound. □

Upper-tail behavior. At high human capital, wages dominate benefits and fixed costs, so a strict formal wage advantage governs behavior there. Values then grow linearly, but surpluses and search rates grow only like $\sqrt{x}$, because quadratic search costs make a value gain of order $\sqrt{x}$ worth order x in search.

Proposition 5 (Upper-tail values and search policies)**.** *Suppose $a_f > a_i$ and $\kappa_i^f(a_f - a_i) < a_f\kappa_u^f$, and let $d = \sqrt{2\kappa_i^f(a_f - a_i)}$. Then there is a unique $q^\star > 0$ satisfying*

$$\frac{(q^\star)^2}{2\kappa_u^i} + \frac{(q^\star + d)^2}{2\kappa_u^f} = a_f, \qquad \text{(A.8)}$$

and the following limits hold as $x \to \infty$.

(a) Values grow at the formal slope, and surpluses grow like $\sqrt{x}$:

$$\frac{U(x)}{x},\ \frac{V_i(x)}{x},\ \frac{V_f(x)}{x} \longrightarrow \frac{a_f}{\rho+\delta}, \tag{A.9}$$

$$\frac{V_i(x)-U(x)}{\sqrt{x}} \longrightarrow q^\star, \qquad \frac{V_f(x)-U(x)}{\sqrt{x}} \longrightarrow q^\star + d. \tag{A.10}$$

Hence $(V_f - V_i)/\sqrt{x} \to d$, and both employment continuation sets contain an upper tail.

(b) Search from formal toward informal employment stops: $\eta_f^i(x) = 0$ for all sufficiently large x. Every other search intensity grows like its net gain divided by its cost: $\eta_u^i(x)/\sqrt{x} \to q^\star/\kappa_u^i$, $\eta_u^f(x)/\sqrt{x} \to (q^\star + d)/\kappa_u^f$, and $\eta_i^f(x)/\sqrt{x} \to d/\kappa_i^f$.

Note that the constant d has a direct interpretation. In the upper tail, an informal worker earns the lower wage but searches for formal jobs, and to first order the value of that search, $d^2x/(2\kappa_i^f)$, exactly makes up the wage gap $(a_f - a_i)x$. This is why informal employment shares the formal slope $a_f/(\rho+\delta)$. The search-cost restriction, in turn, keeps the informal surplus from vanishing. Without it, the informal surplus could grow more slowly than $\sqrt{x}$, and unemployed search toward informal jobs would become negligible relative to search toward formal ones.

Proof. The proof identifies three numbers: the common limit $\mathcal{A}$ of the ratios $W_\ell(x)/x$, and the limits p_i and p_f of the normalized surpluses $D_i(x)/\sqrt{x}$ and $D_f(x)/\sqrt{x}$. It proceeds in three steps. First, surpluses grow at most like $\sqrt{x}$, so these normalizations are bounded and $\mathcal{A}$ exists. Second, passing to the limit in the HJB system along sequences $x_n \to \infty$ yields an algebraic system for $\mathcal{A}$, p_i, and p_f. Third, solving that system under the wage ordering and the search-cost restriction gives $(\rho+\delta)\mathcal{A} = a_f$, $p_i = q^\star$, and $p_f = q^\star + d$. Neither assumption is used in the first two steps.

Values and surpluses. The first step bounds the surpluses. For $x > 0$, convexity gives a supporting slope $s_x \in [\partial_x^- U(x), \partial_x^+ U(x)]$. The line $U(x) + s_x(z-x)$ touches U from below and has zero curvature, so the unemployment supersolution inequality gives

$$\frac{D_i(x)^2}{2\kappa_u^i} + \frac{([D_f(x) - \Psi_f]^+)^2}{2\kappa_u^f} \le (\rho+\delta)U(x) - b - \mu_u s_x \le \bar{a}x + C_0$$

for some finite C_0. The last inequality uses $0 \le s_x \le \bar{a}/(\rho+\delta)$, the upper bound on U in Lemma 1, and $\mu_u \le 0$. Since $D_i, D_f \ge 0$ and $\Psi_f \ge 0$, it follows that, for some finite C,

$$D_i(x) + D_f(x) \le C(1 + \sqrt{x}), \qquad x \ge 0. \tag{A.11}$$

Continuity extends this bound to zero.

With this bound, the value ratios converge. For each W_ℓ, convexity makes $[W_\ell(x) - W_\ell(0)]/x$ nondecreasing, and Lemma 2 keeps it below $\bar{a}/(\rho+\delta)$. Thus $W_\ell(x)/x$ has a finite limit. Since equation (A.11) gives $D_\ell(x)/x \to 0$, the three limits coincide. Their common value $\mathcal{A}$ satisfies $\mathcal{A} \le \bar{a}/(\rho+\delta)$.

The second step passes to the limit in the HJB system. Equation (A.11) also makes $D_\ell(x)/\sqrt{x}$ bounded, so along any sequence $x_n \to \infty$ a subsequence, not relabeled, satisfies $D_\ell(x_n)/\sqrt{x_n} \to p_\ell \geq 0$ for $\ell \in \{i, f\}$. On every bounded interval of shifts z, Lemma 2 makes the variations of $W_\ell(x_n + z)/x_n$ and $D_\ell(x_n + z)/\sqrt{x_n}$ of order x_n^{-1} and $x_n^{-1/2}$, respectively. The convergence to $\mathcal{A}$ and p_ℓ is therefore locally uniform in z.

On each fixed compact interval, the surpluses are bounded, so the maximizing search intensities in equation (5) are bounded as well. The short-horizon dynamic-programming comparison then applies, giving the unemployment equation and the employment supersolution inequalities, with equality on continuation. Translate these inequalities by x_n and divide by x_n. The normalized search gains, and hence their squares, converge locally uniformly. Benefits and flow costs contribute b/x_n and F/x_n, separation contributes $\delta_\ell D_\ell/x_n$, and the entry cost enters the normalized gain as $\Psi_f/\sqrt{x_n}$. All of these terms vanish. Stability of viscosity inequalities, applied to the constant limiting value profile, then gives[26]

$$\begin{aligned}
(\rho+\delta)\mathcal{A} &= \frac{p_i^2}{2\kappa_u^i} + \frac{p_f^2}{2\kappa_u^f}, \\
(\rho+\delta)\mathcal{A} &\geq a_i + \frac{([p_f - p_i]^+)^2}{2\kappa_i^f}, \\
(\rho+\delta)\mathcal{A} &\geq a_f + \frac{([p_i - p_f]^+)^2}{2\kappa_f^i}.
\end{aligned} \tag{A.12}$$

If $p_\ell > 0$, the employment obstacle is inactive on every translated compact set for sufficiently large n, so the corresponding inequality holds with equality.

The third step solves this system, and only here do the assumptions enter. Under the wage ordering, $\bar{a} = a_f$. The bound on $\mathcal{A}$ and the formal-employment inequality in (A.12) therefore give $(\rho+\delta)\mathcal{A} = a_f$ and $p_i \leq p_f$. Since $a_f > 0$, the unemployment equation then implies $p_f > 0$. The search-cost restriction rules out $p_i = 0$. In that case, the unemployment equation would give $p_f^2 = 2\kappa_u^f a_f$, and the informal-employment inequality would require $\kappa_i^f(a_f - a_i) \geq a_f \kappa_u^f$, the opposite of the search-cost restriction. Thus $p_i, p_f > 0$, and both employment inequalities hold with equality.

The informal-employment equation now gives $p_f - p_i = d$. With $q^\star = p_i$, the unemployment equation becomes (A.8). Its left-hand side is continuous and strictly increasing in $q^\star \geq 0$, equals $d^2/(2\kappa_u^f)$ at zero, and tends to infinity. Hence a unique positive solution exists exactly when $d^2 < 2a_f\kappa_u^f$, which is the search-cost restriction. Every convergent subsequence of the bounded vector of normalized surpluses has the limit $(q^\star, q^\star + d)$, so the full vector converges. Since both limits are positive, both surpluses are eventually positive, and both continuation sets contain an upper tail.

Search policies. Since $V_f - V_i \sim d\sqrt{x}$ with $d > 0$, the net gain from moving from formal to informal employment is eventually negative, so $\eta_f^i(x) = 0$ for all sufficiently large x. Dividing

[26] See Lemma 6.1 and Remarks 6.2–6.4 of Crandall et al. (1992).

the remaining search rules in equation (5) by $\sqrt{x}$, and using $\Psi_f/\sqrt{x} \to 0$, gives the three stated limits. □

A.2 Worker Distributions and Population Accounting

This subsection turns the worker policies of Appendix A.1 into worker distributions. Throughout, the policies are the stationary ones characterized there, and I maintain increasing differences. The first result shows that each worker's state is well defined under these policies, the second that the population is conserved, and the third constructs the unique stationary distribution used in the equilibrium and sorting arguments.

The worker process. Each worker's state at date t is the pair $(\ell(t), x(t))$ of labor-market state and human capital, evolving as in Section 2. Within a spell in state ℓ, human capital follows the reflected diffusion $\mathrm{d}x = \mu_\ell \,\mathrm{d}t + \sigma_\ell \,\mathrm{d}W + \mathrm{d}L$. The worker moves to ℓ' at rate $\eta_\ell^{\ell'}(x)$, separates from employment at rate δ_ℓ (with $\delta_u = 0$), quits on first contact with $\mathcal{S}_\ell$, and leaves the labor force at rate δ. Each departing worker is replaced by an unemployed entrant whose human capital is drawn from g. Write $m_\ell(\mathrm{d}x, t)$ for the mass of workers in state ℓ, the measure counterpart of the density $m_\ell(x,t)$. Three facts from Appendix A.1 govern these movements. First, search intensities grow at most linearly in human capital. Each value gain in the search rules (5) is a difference of two values whose slopes lie in $[0, \bar{a}/(\rho+\delta)]$ by Lemma 2, so every intensity satisfies $\eta_\ell^{\ell'}(x) \le C(1+x)$ for a finite C. Second, by increasing differences, each quitting set is a closed lower interval. Search toward a quitting set is zero, because the destination value equals U there while the origin value is at least U and entry costs are nonnegative. A quitting set equal to the whole state space therefore leaves its sector empty. Otherwise $\mathcal{S}_\ell$ is empty or $[0, \bar{x}_u^\ell]$. Third, workers enter employment only in the continuation set and human capital moves continuously, so an employed worker quits exactly when human capital first falls to $\bar{x}_u^\ell$. Two conventions complete the description. Initial employment in a quitting set is reassigned to unemployment at date zero. Quitting at zero makes zero an absorbing employment boundary, whereas a sector without quitting keeps reflection there.

Lemma 3 (Worker histories and transition counts)**.** *Under these policies, each worker's state is well defined at every date, and transitions do not accumulate in finite time. If the entrant density and the initial distribution have finite first moments, then on every bounded time interval the population's expected human capital is finite, and so are the expected numbers of search transitions, separations, and quits.*

Proof. The proof constructs the worker state until human capital first reaches a level R, shows that this date tends to infinity with R, and then counts events.

Fix an initial state, suppress labor-force exit, and let τ_R be the first date at which $x(t) \ge R$. Before τ_R, the linear bound caps the total search rate, $\sum_{\ell'} \eta_{\ell(t)}^{\ell'}(x(t)) \le C(1+R)$, and separations occur at rate at most $\max_\ell \delta_\ell$. Search and separation arrivals before τ_R are therefore dominated by a Poisson process with a finite rate, so only finitely many occur on a bounded interval. Quits

cannot outnumber employment entries plus any initial quit. The worker state is thus uniquely defined on $[0, \tau_R)$. Applied to the path stopped at τ_R, the reflected-path representation in the proof of Lemma 1 gives $\sup_{s\le t} x(s) \le x + 2\bar{\mu}t + 2\sup_{s\le t}|M(s)|$. Doob's and Markov's inequalities then yield

$$\mathbb{P}(\tau_R \le t) \le \frac{x + 2\bar{\mu}t + 4\bar{\sigma}\sqrt{t}}{R} \longrightarrow 0 \qquad \text{as } R \to \infty.$$

Hence $\tau_R \to \infty$, so the worker state is well defined at every date, with finitely many transitions on bounded intervals. Labor-force exit and replacement restart it from an entrant draw.

For the counts, fix T and assume the moment conditions. Let $N(T)$ be the number of search transitions on $[0, T]$. Its expectation is the search rate integrated over the population and over time, so the linear bound gives

$$\mathbb{E}[N(T)] = \int_0^T \sum_\ell \int \sum_{\ell'} \eta_\ell^{\ell'}(x)\, m_\ell(\mathrm{d}x, s)\, \mathrm{d}s \le C \int_0^T \sum_\ell \int (1 + x)\, m_\ell(\mathrm{d}x, s)\, \mathrm{d}s.$$

The population at date s consists of initial workers and entrants who arrived since, at most $\mathbf{M}(0) + \delta s$ workers in all. By equation (A.3), each worker's expected human capital rises by at most $2\bar{\mu}s + 2\bar{\sigma}\sqrt{s}$. Therefore

$$\sum_\ell \int x\, m_\ell(\mathrm{d}x, s) \le \sum_\ell \int x\, m_\ell(\mathrm{d}x, 0) + \delta s \int_0^\infty x g(x)\, \mathrm{d}x + \big[\mathbf{M}(0) + \delta s\big]\big(2\bar{\mu}s + 2\bar{\sigma}\sqrt{s}\big).$$

The right side is finite because the initial distribution and g have finite first moments, so $\mathbb{E}[N(T)]$ is finite. Separations number at most $\max_\ell \delta_\ell \int_0^T \mathbf{M}(s)\, \mathrm{d}s$ in expectation, and quits cannot outnumber employment entries plus those initially employed. □

Population accounting. The next result shows that these transitions conserve the population. They move workers across states without creating or destroying them, so only labor-force exit and entry change the total.

Proposition 6 (Population conservation)**.** *Suppose the entrant density and the initial distribution have finite first moments, and let* $\mathbf{M}(t)$ *be the total population, the sum of the state masses* $\mathbf{M}_\ell(t)$*. Then the worker distributions satisfy the forward equations* (7) *in the weak sense, and*

$$\mathbf{M}(t) = 1 + [\mathbf{M}(0) - 1]e^{-\delta t}.$$

In particular, a unit initial population remains one.

Proof. Both claims follow from accounting for each state's inflows and outflows. Fix a state ℓ and a smooth compactly supported test function φ, with $\varphi' = 0$ at a reflecting boundary. Apply Itô's formula, spell by spell, to $\varphi(x(t))$ while the worker is in ℓ, and take expectations, which Lemma 3 makes finite. Movement of human capital within ℓ contributes the generator, and each transition

into or out of ℓ contributes φ evaluated at the human capital where it occurs:

$$\int \varphi \, m_\ell(\mathrm{d}x, t) - \int \varphi \, m_\ell(\mathrm{d}x, 0) = \int_0^t \int \mathcal{L}_\ell \varphi \, m_\ell(\mathrm{d}x, s) \, \mathrm{d}s + \mathrm{In}_\ell(\varphi, t) - \mathrm{Out}_\ell(\varphi, t). \tag{A.13}$$

The outflow term integrates φ against search at rates $\eta_\ell^{\ell'}$, separation at rate δ_ℓ, labor-force exit at rate δ, and quits. The inflow term integrates φ against search from other states and, for unemployment, against separations, quits, and entrants arriving at rate δg. These are the flows in equation (7), with quits counted by their expected measures $q_\ell(\mathrm{d}x, \mathrm{d}s)$, so equation (A.13) is the weak form of equation (7).

Now set $\varphi = 1$ and add over states. The generator term vanishes because $\mathcal{L}_\ell 1 = 0$. Every search, separation, and quit is an outflow from one state and an inflow into another, so these terms cancel in the sum, and only entry and exit remain:

$$\mathbf{M}(t) - \mathbf{M}(0) = \delta \int_0^t \big[1 - \mathbf{M}(s)\big] \, \mathrm{d}s.$$

Localization and the finite flows of Lemma 3 justify the constant test function. Solving this equation gives the stated law. □

Stationary worker distribution. Under stationary policies, the population renews itself: each departing worker is replaced by an entrant drawn from g. The stationary law is therefore an average of entrant histories over the exponential age distribution, and that average does not depend on the initial population.

Proposition 7 (Stationary worker distribution)**.** *Under the stationary policies, the worker distribution has the following properties.*

(a) There is a unique stationary distribution, with sectoral densities and unit total mass.

(b) If the entrant density has a finite first moment, so does the stationary distribution. It satisfies the stationary forward equations in the weak sense. Aggregate search, separation, and quitting flows are finite, and total employment output is strictly positive and finite.

(c) Under the moment condition in (b), the densities are continuous, and twice continuously differentiable away from quitting thresholds. As human capital approaches a threshold from above, each density's derivative and flux have finite limits. The fluxes vanish at infinity. Let Q_ℓ be the aggregate stationary quit rate from sector ℓ. At a quitting threshold, the employment density vanishes and the quit rate equals the flux into the threshold:

$$\lim_{x \downarrow \bar{x}_u^\ell} m_\ell(x) = 0, \qquad Q_\ell = -\lim_{x \downarrow \bar{x}_u^\ell} J_\ell(x) = \frac{\sigma_\ell^2}{2} \lim_{x \downarrow \bar{x}_u^\ell} m_\ell'(x) \geq 0.$$

In a sector without quitting, the flux vanishes at zero, $J_\ell(0) = 0$.

Proof. The stationary law is the distribution of an entrant's state at an independent exponential age. The proof first shows that this age average is invariant and unique, which gives (a). Under the moment condition, Lemma 3 makes moments and flows finite, so Proposition 6 applies to the invariant law. A separate argument gives positive output, completing (b). Finally, the weak stationary equations yield the regularity and boundary conditions in (c).

Construction and uniqueness. Let $P_t(B)$ be the probability that an entrant, before any labor-force exit, belongs to a measurable set B of worker states at age t. Average over the exponential age distribution:

$$m(B) = \delta \int_0^\infty e^{-\delta t} P_t(B)\, \mathrm{d}t. \tag{A.14}$$

This law has unit mass. After replacement is restored, workers with no replacement over $h > 0$ contribute

$$e^{-\delta h}\delta \int_0^\infty e^{-\delta t} P_{t+h}(B)\, \mathrm{d}t = \delta \int_h^\infty e^{-\delta t} P_t(B)\, \mathrm{d}t.$$

Workers whose most recent replacement occurred during this interval contribute $\delta \int_0^h e^{-\delta t} P_t(B)\, \mathrm{d}t$. Their sum is $m(B)$, proving invariance.

Couple workers from arbitrary initial distributions using a common replacement clock. At their first replacement, use the same entrant draw and subsequent shocks. Their probabilities of belonging to any set B at date h then differ by at most $e^{-\delta h}$. Letting $h \to \infty$ shows that any two invariant laws coincide.

Before labor-force exit, the quadratic variation of human capital accumulates at rate at least $\min_j \sigma_j^2 > 0$. The occupation-time formula gives zero time in any Lebesgue-null set of human-capital levels. The age mixture therefore assigns such sets zero stationary probability in each state, giving densities m_ℓ. This proves (a).

Moments, flows, and output. Under the moment condition in (b), average equation (A.3) over entrants and ages:

$$\sum_\ell \int_0^\infty x m_\ell(x)\, \mathrm{d}x \le \int_0^\infty x g(x)\, \mathrm{d}x + C\delta \int_0^\infty e^{-\delta t}(t + \sqrt{t})\, \mathrm{d}t < \infty.$$

The invariant law thus has a finite first moment, and Lemma 3 gives finite search, separation, and quit flows. Stationarity makes expected quit counts proportional to interval length, which defines finite stationary quit-rate measures. Proposition 6 applies to the invariant law, and stationarity makes the left side of equation (A.13) zero, giving the stationary weak forward equations.

Employment output is finite by the first-moment bound. If both search intensities out of unemployment vanished everywhere, then $U = b/(\rho + \delta)$. The formal-value floor in Lemma 1(b) would then make $V_f - U - \Psi_f > 0$ at large x, contradicting zero formal search. Hence an employment-entry intensity is positive on an open interval in $(0, \infty)$, where the destination's surplus is positive and entry does not trigger quitting.

Choose a compact interval inside this region on which the entry rate is bounded away from zero. A bounded set of entrant human-capital levels has positive probability. Positive unemployment volatility gives positive probability of reaching the interval along a bounded path before any search arrival. Locally bounded rates then give positive probability of entry and a short employment stay inside continuation, with human capital bounded away from zero. Averaging this event over ages gives strictly positive employment output, completing (b).

Density regularity and boundary conditions. Each employment sector reflects at zero without quitting, is absorbed at a threshold $\bar{x}_u^\ell \geq 0$, where all of its quits occur, at rate Q_ℓ, or retains no workers and has zero density. Extend employment densities by zero below positive thresholds. The stationary weak equations express distributional flux derivatives as integrable flows plus finite quit measures, giving locally bounded-variation fluxes on $(0, \infty)$. Since $\frac{\sigma_\ell^2}{2} m_\ell' = \mu_\ell m_\ell - J_\ell$ by equation (6), the densities are locally absolutely continuous. Away from thresholds, continuous search policies and g give continuous flux derivatives and hence twice continuously differentiable densities. Finite one-sided flux limits give finite density-derivative limits. Integrability of the flows near zero also gives finite limits as $x \downarrow 0$.

With these limits in hand, the threshold conditions follow from computing the boundary contribution in two ways. Integrating the stationary interior equation over $(\bar{x}_u^\ell, \infty)$ against a smooth test function φ gives the lower-boundary contribution

$$\lim_{x \downarrow \bar{x}_u^\ell} \left[\varphi(x) J_\ell(x) + \frac{\sigma_\ell^2}{2} \varphi'(x) m_\ell(x) \right].$$

The quit outflow in equation (A.13) gives $-Q_\ell \varphi(\bar{x}_u^\ell)$ instead. The test value and derivative can be chosen independently and $\sigma_\ell > 0$, so matching coefficients gives the zero density limit and the quit-flux identity. The zero density limit and nonnegativity imply a nonnegative derivative limit. The argument applies equally at a threshold of zero. Without quitting, zero reflects and $J_\ell(0) = 0$. The weak unemployment equation, tested with $\varphi_u'(0) = 0$ and arbitrary $\varphi_u(0)$, gives $J_u(0) = \sum_{\ell : \mathcal{S}_\ell = \{0\}} Q_\ell$. Quits at zero thus add mass at the boundary even though each unemployed worker's path reflects there. At a positive threshold, where quits enter unemployment, the unemployment flux jumps up by the quit rate while its density stays continuous. Integration by parts gives the stationary flux equations.

Finally, finite transition flows make J_ℓ' integrable on an upper tail, so J_ℓ has a finite limit at infinity. Integrability of m_ℓ gives a sequence $R_n \to \infty$ with $m_\ell(R_n) \to 0$. Integrate the flux-density relation from a fixed point to R_n and divide by R_n. The density and integrated-mass terms vanish, so the average flux, and hence its limit, is zero. □

Finally, note that the accounting of Proposition 6 extends to the transition path on which the main text defines equilibrium, where policies may vary over time. Appendix A.1 does not characterize time-varying policies, so the conclusions of Lemma 3 must be assumed for them. Given those conclusions, the proof of Proposition 6 applies unchanged. A positive time-varying threshold $\bar{x}_u^\ell(t)$ adds one step when continuation is connected and the threshold moves in a C^1

way.[27] Differentiating the moving lower limit gives

$$Q_\ell(t) = \lim_{x \downarrow \bar{x}_u^\ell(t)} \left[\dot{\bar{x}}_u^\ell(t) m_\ell(x,t) - J_\ell(x,t)\right] = - \lim_{x \downarrow \bar{x}_u^\ell(t)} J_\ell(x,t),$$

where the final equality uses the zero density limit, obtained as in Proposition 7(c). Integrating the generator terms by parts gives the interior equations in (7). Their absorbing boundary and the quit term q_ℓ describe the same employment loss, so the loss is counted once.

A.3 Stationary Equilibrium

This subsection proves Theorem 1 and then gives conditions under which comparative advantage sorts workers with high human capital into formal employment.

Recall that zero profits alone set wages (2), so workers solve their problem at known wages. The proof follows the equilibrium in the same direction. Wages determine the worker problem, its policies determine the distribution, and the distribution determines labor demand and the transfer. It suffices to show that each step has exactly one outcome. As in Appendix A.2, I maintain increasing differences.

Proof of Theorem 1. By condition (i) of Definition 1, wages are those of equation (2), with $a_f = A/(1+\tau)$ and $a_i = A(1-\alpha_0)$. Both are positive because $\alpha_0 < 1$.

Consider first the workers. Each value is the supremum of lifetime payoffs over plans, so U, V_i, and V_f are uniquely determined, and Lemma 1 makes them finite. The search rules (5) give each worker a unique optimal intensity. Quitting is optimal exactly where the surplus D_ℓ vanishes, so the values also determine the quitting sets. Condition (ii) therefore has one solution.

Given these policies, Proposition 7(a) gives a unique stationary distribution with unit mass, which is condition (iii) together with the population constraint. Market clearing then sets each sector's labor demand equal to its employment density. Firms produce under constant returns at zero profits, so they are willing to hire that amount at these wages.

Notice how the finite mean enters. By Proposition 7(b), the stationary distribution then has a finite first moment, so tax revenue $\tau \int w_f m_f$ is finite and the budget (9) determines a unique finite transfer. The transfer shifts all values equally and leaves the policies and the distribution unchanged. Every component of the equilibrium is therefore determined, and uniquely so. □

The stationary allocation also varies continuously with the informal wage. Hold formal wages and the other primitives fixed, and let the informal wage coefficient $a_i = A(1-\alpha_0)$ vary over $(0, A]$. I take quitting to be well behaved as a_i varies: each sector keeps one of the three regimes of Appendix A.2, and positive quitting thresholds vary continuously with a_i.

[27] I take as given the forward-solution regularity, finite density and flux limits at the threshold, and differentiation under the integral that this step needs.

Lemma 4 (Continuity in wages)**.** *At each $a_i \in (0, A]$, let $m_\ell(x|a_i)$ be the stationary densities of Proposition 7. If the entrant density has a finite first moment, sectoral output $H_\ell(a_i) = \int_0^\infty x\, m_\ell(x|a_i)\,\mathrm{d}x$ is continuous in a_i for $\ell \in \{i, f\}$.*

Proof. Let $a_i^n \to a_i$. The proof follows the allocation from values to output: policies converge, worker histories converge, and the convergence survives the passage to output.

Policies converge first, because values do. The imitation argument of Lemma 2 applies with wages in place of human capital: a worker can follow the same plan at both wages, so human-capital paths and costs coincide and only informal wage income differs. Bounding expected discounted human capital as in the proof of Lemma 1 and taking suprema over plans gives, for $a_i'' > a_i'$ and a finite C,

$$0 \le W_j(x|a_i'') - W_j(x|a_i') \le C(1+x)(a_i'' - a_i'), \qquad j \in \{u, i, f\}. \tag{A.15}$$

Values at a_i^n therefore converge to values at a_i, locally uniformly in human capital, and the search rules (5) turn this into locally uniform convergence of search policies. By assumption, positive quitting thresholds converge as well.

To compare histories, I run workers at both wages from the same initial human capital, Brownian motion, and Poisson arrivals. On a finite horizon, truncate rates at $\min\{x, R\}$ for an R above all finite thresholds. Rates are then bounded, so search and separation events at both wages can be drawn by thinning common Poisson arrivals at that bound. Each arrival becomes an event when an independent uniform mark falls below the worker's current rate relative to the bound. Each search and separation event converges, because the marks almost surely avoid the limiting acceptance levels. Quits converge too. Positive employment volatility makes the path cross below a threshold immediately after first contact, so the contact date moves continuously. In sectors without quitting, reflection gives continuous dependence on starting dates and initial human capital, and a sector that retains no workers receives no search, as shown in Appendix A.2. Induction over the finitely many events gives convergence of human capital and labor-market states away from event dates. The escape bound in the proof of Lemma 3 removes the truncation. Finally, evaluating histories at an independent exponential age τ_δ, which almost surely avoids event dates, gives convergence of stationary laws through the age mixture (A.14).

Convergence of stationary laws does not yet give convergence of output, because human capital is unbounded. Uniform integrability closes the gap. Let $x^n(\tau_\delta)$ be human capital at that age along the history before labor-force exit, and $x(0)$ the common entrant draw. The reflected-path representation and Doob's inequality, used as in the proof of Lemma 1, bound $\sup_n \mathbb{E}[|x^n(\tau_\delta) - x(0)|^2]$, so

$$\sup_n \mathbb{E}[x^n(\tau_\delta)\mathbb{1}\{x^n(\tau_\delta) > R\}] \le 2\mathbb{E}[x(0)\mathbb{1}\{x(0) > R/2\}] + \frac{C'}{R} \longrightarrow 0.$$

Truncating human capital within each sector and then removing the truncation gives convergence of H_i and H_f. □

Sectoral comparative advantage. Recall from Section 2.5 that comparative advantage rests on the formal advantage $V_f - V_i$. Pay and stability govern its slope, and the recurring cost F shifts its level. The following proposition gives conditions under which the two values cross exactly once. The slope comparisons must favor formality, and F must be large enough that informality is the more valuable sector at low human capital.

Proposition 8 (Sectoral comparative advantage)**.** *Suppose increasing differences hold in both sectors,* $a_f \geq a_i$*, and*

$$\frac{a_f}{\sigma_f^2} \geq \frac{a_i}{\sigma_i^2}, \qquad \frac{\mu_f}{\sigma_f^2} \geq \frac{\mu_i}{\sigma_i^2}, \qquad \frac{\delta_f}{\sigma_f^2} \leq \frac{\delta_i}{\sigma_i^2},$$

with the separation inequality strict when $a_f = a_i$*. If* $F > \bar{F}$*, where* $\bar{F}$ *is the explicit bound in equation* (A.17) *and depends only on the primitives, there is a unique threshold* $\bar{x}_i^f$ *such that* $V_f(x) > V_i(x)$ *exactly when* $x > \bar{x}_i^f$*.*

Proof of Proposition 8. Let $E = V_f - V_i$ be the formal advantage. The proof has three steps. First, the cost bound makes E negative at the lowest human capital at which workers keep both jobs. Second, the proposition's three comparisons make the forcing term nondecreasing and unbounded. Third, a minimum principle shows that E can fall only at first and then rises without bound, so it crosses zero once. I begin by collecting the bounds that the argument inherits and deriving the equation for E.

Bounds and the equation for E*.* Since $a_f \geq a_i$, Lemmas 1 and 2 give convex values with

$$\frac{b}{\rho+\delta} \leq W_j(x) \leq \frac{a_f x + b}{\rho+\delta} + B_0, \qquad 0 \leq W_j' \leq \frac{a_f}{\rho+\delta},$$

where $B_0 = a_f[2\bar{\mu}/(\rho+\delta)^2 + \bar{\sigma}\sqrt{\pi}/(\rho+\delta)^{3/2}]$ is the constant from the proof of Lemma 1. In particular, $D_i(x) \leq a_f x/(\rho+\delta) + B_0$ and $|E'| \leq a_f/(\rho+\delta)$. By increasing differences, each nonempty employment continuation region is an upper interval.

Two facts about high human capital are needed later: formal employment continues there, and the values share the slope $a_f/(\rho+\delta)$. Equation (A.11) bounds the normalized surpluses $D_\ell(x)/\sqrt{x}$, and along any convergent subsequence their limits p_i and p_f satisfy the limit system (A.12), with common value slope $\mathcal{A} \leq a_f/(\rho+\delta)$. The formal-employment inequality of that system gives $(\rho+\delta)\mathcal{A} \geq a_f$ and $p_i \leq p_f$. Hence $(\rho+\delta)\mathcal{A} = a_f$, and the unemployment equation gives $p_f^2 \geq 2a_f/(1/\kappa_u^i + 1/\kappa_u^f) > 0$, so D_f is positive for all sufficiently large x.

Where both jobs continue, subtract σ_f^2/σ_i^2 times the informal HJB from the formal HJB. The informal value's curvature cancels, leaving a second-order equation for E:

$$\mathcal{H}(E) - \mu_f E' - \frac{\sigma_f^2}{2} E'' = \Phi(x), \tag{A.16}$$

where

$$\mathcal{H}(z) = (\rho+\delta+\delta_f)z + \frac{\sigma_f^2}{\sigma_i^2}\,\frac{([z-\Psi_f]^+)^2}{2\kappa_i^f} - \frac{([-z]^+)^2}{2\kappa_f^i},$$

$$\Phi(x) = \Big(a_f - \frac{\sigma_f^2}{\sigma_i^2}a_i\Big)x + \Big(\frac{\sigma_f^2}{\sigma_i^2}-1\Big)(\rho+\delta)V_i(x) - F + \Big(\mu_f - \frac{\sigma_f^2}{\sigma_i^2}\mu_i\Big)V_i'(x) + \Big(\frac{\sigma_f^2}{\sigma_i^2}\delta_i - \delta_f\Big)D_i(x).$$

The function $\mathcal{H}$ collects discounting, separation, and search, and Φ is the forcing term.

Informal advantage at the bottom. This step shows that E is negative where both jobs first continue. The argument bounds the formal value from above by the informal value plus an allowance for formal continuation, on an interval that reaches beyond the region where both jobs can be quit. Two quantities enter the cost bound. The first, R_0, bounds that region. The second, Λ_f, sets the allowance for continuation in formal employment:

$$R_0 = \frac{b}{a_i} + \sqrt{\frac{(\sigma_u^2-\sigma_i^2)^+ a_f}{(\rho+\delta)a_i}}, \qquad \Lambda_f = \frac{\mu_f + \sqrt{\mu_f^2 + 2(\rho+\delta+\delta_f)\sigma_f^2}}{2(\rho+\delta+\delta_f)}.$$

The cost bound is

$$\begin{aligned} F > \bar{F} := {} & \Big(a_f - \frac{\sigma_f^2}{\sigma_i^2}a_i\Big)(R_0+\Lambda_f) + \Big(\mu_f - \frac{\sigma_f^2}{\sigma_i^2}\mu_i\Big)\frac{a_f}{\rho+\delta} \\ & + \Big(\frac{\sigma_f^2}{\sigma_i^2}\delta_i - \delta_f\Big)\Big[B_0 + \frac{a_f(R_0+\Lambda_f)}{\rho+\delta}\Big] + \Big(\frac{\sigma_f^2}{\sigma_i^2}-1\Big)b \\ & + \Big(\frac{\sigma_f^2}{\sigma_i^2}-1\Big)^+\big[a_f(R_0+\Lambda_f) + (\rho+\delta)B_0\big]. \end{aligned} \tag{A.17}$$

Consider first why R_0 bounds the region in which both jobs can be quit. Suppose both are quit throughout $[0,R]$. Search gains are then zero, and the unemployment HJB and the informal quitting inequality give

$$a_i x - b + (\mu_i - \mu_u)U' + \frac{\sigma_i^2 - \sigma_u^2}{2}U'' \le 0.$$

The drift term is nonnegative. If $\sigma_i \ge \sigma_u$, the curvature term is also nonnegative, so $R \le b/a_i$. Otherwise, integrating $(\sigma_u^2-\sigma_i^2)U''/2 \ge a_i x - b$ from b/a_i to R, when $R > b/a_i$, and using the slope bound gives $a_i(R-b/a_i)^2 \le (\sigma_u^2-\sigma_i^2)a_f/(\rho+\delta)$. In either case, $R \le R_0$.

With R_0 in hand, I claim that $E \le 0$ on an interval $[0,R]$ that reaches beyond it. Let

$$\gamma = a_f - \frac{\sigma_f^2}{\sigma_i^2}a_i + \Big(\frac{\sigma_f^2}{\sigma_i^2}-1\Big)^+ a_f + \Big(\frac{\sigma_f^2}{\sigma_i^2}\delta_i - \delta_f\Big)\frac{a_f}{\rho+\delta}.$$

It is positive. If $\sigma_f < \sigma_i$, then $a_f \ge a_i > a_i\sigma_f^2/\sigma_i^2$. If $\sigma_f > \sigma_i$, the third term is positive. If $\sigma_f = \sigma_i$,

either $a_f > a_i$, or the wage slopes are equal and the proposition requires $\delta_i > \delta_f$. The value bounds above, using the upper bound on V_i if $\sigma_f \geq \sigma_i$ and the lower bound if $\sigma_f < \sigma_i$, give

$$\Phi(x) \leq \gamma(x - R_0 - \Lambda_f) + \bar{F} - F.$$

Choose $R > R_0$ with $\gamma(R - R_0) < F - \bar{F}$, so that $\Phi(x) < \gamma(x - R - \Lambda_f)$.

To show that $E \leq 0$ on $[0, R]$, I bound the formal value from above by the informal value plus an allowance for continuing in formal employment beyond R. With $\lambda_- = -2(\rho+\delta+\delta_f)\Lambda_f/\sigma_f^2 < 0$, the allowance is

$$\Xi_R(x) = \begin{cases} 0, & x \leq R, \\ \dfrac{\gamma}{\rho + \delta + \delta_f}\left[(x - R) + \dfrac{e^{\lambda_-(x-R)} - 1}{-\lambda_-}\right], & x > R. \end{cases}$$

It is nonnegative, has value and derivative zero at R, and grows linearly. Direct substitution gives $(\rho + \delta + \delta_f)\Xi_R - \mathcal{L}_f\Xi_R = \gamma(x - R - \Lambda_f)$ above R, while below R, $\Xi_R = 0$ and $\Phi < 0$. The candidate $V_i + \Xi_R$ dominates both unemployment and informal employment, so its search gain toward informal employment is zero. Multiplying the informal HJB inequality by σ_f^2/σ_i^2 and dropping its nonnegative search term gives

$$\begin{aligned}(\rho + \delta + \delta_f)(V_i + \Xi_R) - \mathcal{L}_f(V_i + \Xi_R) - a_f x + F - \delta_f U \\ \geq -\Phi + (\rho + \delta + \delta_f)\Xi_R - \mathcal{L}_f\Xi_R \geq 0,\end{aligned}$$

so $V_i + \Xi_R$ is a supersolution for the formal value. It also dominates what a formal worker receives on quitting or on moving to informal employment. At zero, $\lim_{x\downarrow 0} V_\ell'(x) = 0$, because a positive right derivative would make a short delay in quitting profitable, as in the proof of Proposition 3. A verification argument therefore gives $V_f \leq V_i + \Xi_R$, hence $E \leq 0$ on $[0, R]$.[28]

It remains to make this ranking strict where both jobs first continue. If $D_i(R) = 0$, then $E(R) \leq 0$ also gives $D_f(R) = 0$, and increasing differences would make both jobs quit throughout $[0, R]$, contradicting $R > R_0$. Thus $D_i(R) > 0$, and both employment continuation sets are nonempty. Let x_0 be their lower common boundary. If $x_0 > R$, it is a formal quitting threshold, and $E(x_0) = -D_i(x_0) < 0$. If $x_0 \leq R$, then $E(x_0) \leq 0$. Equality would require $x_0 < R$ and, by smooth pasting or the zero derivatives at zero, $\lim_{x\downarrow x_0} E'(x) = 0$. Equation (A.16) would then give $\lim_{x\downarrow x_0} E''(x) = -2\Phi(x_0)/\sigma_f^2 > 0$, contradicting $E \leq 0$ just above x_0. Hence $E(x_0) < 0$ in every case, and both employment HJBs hold with equality on (x_0, ∞).

The forcing term. This step shows that Φ is nondecreasing and unbounded above. Multiplied by σ_f^2, the proposition's comparisons read $a_f \geq a_i\sigma_f^2/\sigma_i^2$, $\mu_f \geq \mu_i\sigma_f^2/\sigma_i^2$, and $\delta_i\sigma_f^2/\sigma_i^2 \geq \delta_f$. The last two terms of Φ are then nondecreasing, because V_i is convex and D_i is nondecreasing. If $\sigma_f \geq \sigma_i$,

[28] Proposition 3 makes V_i continuously differentiable at positive quitting thresholds, and Ξ_R is continuously differentiable at R, so Itô's formula up to the end of a formal spell has no local-time terms. Lemma 1 removes the localizations in time and human capital.

the first two terms are nondecreasing as well. If $\sigma_f \leq \sigma_i$, rewrite them as

$$\frac{\sigma_f^2}{\sigma_i^2}(a_f - a_i)x + \left(1 - \frac{\sigma_f^2}{\sigma_i^2}\right)(\rho + \delta)\left[\frac{a_f}{\rho + \delta}x - V_i(x)\right],$$

whose bracket is nondecreasing by the slope bound. Hence Φ is nondecreasing in either case.

To see that Φ is unbounded above, suppose first that $a_f > a_i$. The common value slope, the bounded value derivatives, and $D_i(x)/x \to 0$ give $\Phi(x)/x \to (a_f - a_i)\sigma_f^2/\sigma_i^2 > 0$. If instead $a_f = a_i$, the comparison of wage slopes gives $\sigma_f \leq \sigma_i$, and the proposition requires $\delta_i\sigma_f^2/\sigma_i^2 > \delta_f$. The informal-employment inequality in (A.12), together with $(\rho + \delta)\mathcal{A} = a_f$, gives $p_f \leq p_i$, so every limit of the normalized surpluses has $p_i = p_f > 0$ by the bound $p_f^2 \geq 2a_f/(1/\kappa_u^i + 1/\kappa_u^f)$. Hence $D_i(x) \to \infty$, and the positive coefficient $\delta_i\sigma_f^2/\sigma_i^2 - \delta_f$ makes $\Phi(x) \to \infty$ as well.

A single crossing. Finally, this step shows that E can fall only at first and then rises without bound, so it crosses zero once. The tool is a minimum principle for E'. The search terms are continuously differentiable, with $\mathcal{H}'(E) = \rho + \delta + \delta_f + (\sigma_f^2/\sigma_i^2)\eta_i^f + \eta_f^i > 0$, so differentiating (A.16) gives

$$\mathcal{H}'(E)E' - \mu_f E'' - \frac{\sigma_f^2}{2}E''' = \Phi' \geq 0 \qquad \text{on } (x_0, \infty). \tag{A.18}$$

First, E cannot be bounded above on any upper tail. If $E \leq M$ there, equation (A.16) and the bound on E' give

$$\frac{\sigma_f^2}{2}E'' = \mathcal{H}(E) - \Phi - \mu_f E' \leq \mathcal{H}(M) - \Phi + \frac{\mu_f a_f}{\rho + \delta} \longrightarrow -\infty,$$

which would make E' unbounded below, contradicting its bound.

Now consider where E decreases. A bounded component of $\{E' < 0\}$ whose endpoints lie in common continuation would contain an interior negative minimum of E', where $E'' = 0$ and $E''' \geq 0$. That would make the left side of (A.18) strictly negative. Hence E can decrease only on a component that adjoins x_0 or extends to infinity. An unbounded component would leave E bounded above on a tail, which was just ruled out. So E can decrease only on an initial bounded interval. Beyond it, $E' \geq 0$, and the strong minimum principle applied to (A.18) makes $E' > 0$, because an interior zero would force E' to vanish throughout and make E constant on a tail. If E never decreases, the same argument gives $E' > 0$ throughout common continuation.

The increasing branch cannot remain bounded above, so $E(x) \to \infty$. Since $E(x_0) < 0$ and any initial decline only lowers E further, continuity and the strict increase that follows give a unique $\bar{x}_i^f > x_0$ at which $E = 0$, with $E > 0$ exactly above it in common continuation. Below x_0 there is no crossing. Informal employment cannot be quit there while formal employment continues, because that ordering of the upper continuation regions would give $V_i(x_0) = U(x_0) \leq V_f(x_0)$, contradicting $E(x_0) < 0$. Below common continuation, therefore, either both jobs are quit, with $E = 0$, or only informal employment continues, with $E = -D_i < 0$. The strict formal advantage $V_f > V_i$ thus holds exactly for $x > \bar{x}_i^f$ on the whole state space. Notice that E also vanishes wherever both jobs

are quit, so $\bar{x}_i^f$ is the unique crossing of E, not its unique zero. □

The cost bound has a simple reading. Where workers first keep both jobs, formal employment still offers a higher volatility-normalized wage, faster accumulation, and more stability than informal employment, and F must outweigh all three over the range $[0, R_0 + \Lambda_f]$ on which the comparison is made. The first three terms of $\bar{F}$ price these advantages. The remaining terms adjust for unequal employment volatilities and vanish when the volatilities are equal.

Note that the single crossing also shapes search. By the search rule (5), informal-to-formal search is active on a single upper interval, strictly increasing wherever positive, and unbounded as human capital rises. Its cutoff $\hat{x}_i^f$ lies above $\bar{x}_i^f$ when $\Psi_f > 0$ and coincides with it when $\Psi_f = 0$.

A.4 Sorting and Duration Dependence

This subsection proves Theorem 2 and then derives its consequences for unemployment and informal-to-formal mobility.

The decomposition involves the slopes of search rates and their covariances over survivors, so search rates must be Lipschitz with finite second moments. By Lemma 2 and the search rules (5),

$$|\eta_\ell^{\ell'}(y) - \eta_\ell^{\ell'}(x)| \leq \frac{\bar{a}}{(\rho+\delta)\kappa_\ell^{\ell'}}|y - x|, \qquad [\eta_\ell^{\ell'}(x)]^2 \leq C(1+x) \tag{A.19}$$

for some finite C. The second bound uses equation (A.11) and $V_\ell \geq U$, which bounds every job-to-job gain by the corresponding gain from unemployment. Since the cohort starts with finite mean human capital, Lemma 3 keeps that mean finite at every finite duration, and with it these second moments.

Hazard decomposition. Recall that the hazard toward ℓ' is the average search intensity of the workers still in the spell. As the spell lengthens, that average can change in two ways. Survivors drift, through accumulation and shocks, toward levels of human capital at which they search more or less, and the mix of survivors shifts as some of them leave. The proof follows both changes through the cohort's forward equation, and an integration by parts sorts them into the theorem's three terms. Unlike the stationary densities of Proposition 7(c), cohort densities change with duration, and for this calculation I take them to be well behaved.[29]

Proof of Theorem 2. Fix an origin ℓ and a duration with $S_\ell(d) > 0$. To cover unemployment with the same calculation, set $\mathcal{C}_u = \mathbb{R}_+$ and $\bar{q}_u = 0$. Let $r_\ell(x)$ denote the rate at which survivors at x leave through search, separation, and labor-force exit.

[29] Precisely: differentiation under the mass and search-weighted integrals is permitted, the flux terms are integrable with finite limits at the threshold, and $\bar{J}_\ell$ and $\eta_\ell^{\ell'}\bar{J}_\ell$ vanish at infinity.

The cohort receives no inflows, so in the per-state accounting (A.13) only outflows remain: survivors leave at rate r_ℓ inside $\mathcal{C}_\ell$ and through quits at the threshold. For the unnormalized density $S_\ell \bar{m}_\ell$, this gives the forward equation, and integrating it over $\mathcal{C}_\ell$ gives the survival law:

$$\begin{aligned} \partial_d(S_\ell \bar{m}_\ell) &= -S_\ell \partial_x \bar{J}_\ell - r_\ell S_\ell \bar{m}_\ell, \\ -\frac{S'_\ell(d)}{S_\ell(d)} &= \mathbb{E}_d[r_\ell] + \bar{q}_\ell(d). \end{aligned}$$

Since the hazard (10) divides search-weighted survivor mass by survival, its derivative combines the two:

$$\begin{aligned} \partial_d \lambda_\ell^{\ell'}(d) &= \frac{1}{S_\ell(d)} \int_{\mathcal{C}_\ell} \eta_\ell^{\ell'}(x) \partial_d[S_\ell(d) \bar{m}_\ell(x,d)] \, \mathrm{d}x - \frac{S'_\ell(d)}{S_\ell(d)} \lambda_\ell^{\ell'}(d) \\ &= -\int_{\mathcal{C}_\ell} \eta_\ell^{\ell'}(x) \partial_x \bar{J}_\ell(x,d) \, \mathrm{d}x - \mathrm{Cov}_d(\eta_\ell^{\ell'}, r_\ell) + \bar{q}_\ell(d) \lambda_\ell^{\ell'}(d). \end{aligned}$$

Notice that the last line already has the theorem's shape. The flux term records how survivors move, the covariance records who leaves, and quits enter through $\bar{q}_\ell$. Integrating the flux term by parts moves the derivative onto the search rate:

$$-\int_{\mathcal{C}_\ell} \eta_\ell^{\ell'}(x) \partial_x \bar{J}_\ell(x,d) \, \mathrm{d}x = \int_{\mathcal{C}_\ell} \partial_x \eta_\ell^{\ell'}(x) \bar{J}_\ell(x,d) \, \mathrm{d}x - \bar{q}_\ell(d) \eta_\ell^{\ell'}(\bar{x}_u^\ell).$$

The flux vanishes at infinity. At a quitting threshold it equals $-\bar{q}_\ell$, and under reflection at zero it is zero, so the boundary term disappears whenever no one quits. The integral is structural duration dependence.

It remains to reduce the covariance and collect the quit terms. In unemployment, $r_u = \eta_u^i + \eta_u^f + \delta$. The constant δ drops out of the covariance, and with $\bar{q}_u = 0$ this gives equation (12). In employment, the only search is toward the other sector, so $r_\ell = \eta_\ell^{\ell'} + \delta_\ell + \delta$ and the covariance becomes $\mathrm{Var}_d[\eta_\ell^{\ell'}]$. The boundary term removes the search intensity of the workers who quit. The quit term of the survival law raises the weight of those who remain by $\bar{q}_\ell \lambda_\ell^{\ell'}$. Together they form the quit-selection term in equation (13). □

Unemployment duration. Recall that without shocks, human capital cannot rise during unemployment. The corollary follows by signing the two terms of Theorem 2.

Proof of Corollary 1. Fix a destination $\ell' \in \{i, f\}$. With $\sigma_u = 0$, human capital moves only through the drift, $x(d) = (x(0) + \mu_u d)^+$, and survivors who reach zero stay there. The flux in equation (12) is therefore $\mu_u \bar{m}_u$ above zero, and the theorem gives, for almost every duration,

$$\partial_d \lambda_u^{\ell'}(d) = \mu_u \mathbb{E}_d\big[\partial_x \eta_u^{\ell'}(x) \mathbb{1}\{x > 0\}\big] - \mathrm{Cov}_d\big(\eta_u^{\ell'}(x), \eta_u^i(x) + \eta_u^f(x)\big).$$

The finite initial mean keeps both terms finite. The structural term is nonpositive because $\mu_u \leq 0$ and search is nondecreasing. The covariance is nonnegative by Chebyshev's association inequality,

because $\eta_u^{\ell'}$ and $\eta_u^i + \eta_u^f$ are both nondecreasing in human capital. Both forces therefore lower the hazard. □

Informal-to-formal mobility. Recall that informal workers accumulate human capital, so structural duration dependence now pushes the formal hazard up. The proof shows that slow accumulation lets dynamic selection prevail, and it gives an explicit bound on the drift.

Proof of Corollary 2. Fix $D > 0$. There are two claims: the hazard strictly declines on $[0, D]$, and it stays bounded away from zero. Since $V_i - U = \kappa_u^i \eta_u^i$ is nondecreasing, informal continuation is an upper interval. With $\sigma_i = 0$, human capital only rises, along $x(d) = x(0) + \mu_i d$, so no survivor reaches the quitting threshold. Theorem 2 then has no quit term and gives, for almost every duration,

$$\partial_d \lambda_i^f(d) = \mu_i \mathbb{E}_d\big[\partial_x \eta_i^f(x)\big] - \mathrm{Var}_d\big[\eta_i^f(x)\big]. \tag{A.20}$$

Notice that the two forces now pull in opposite directions. By equation (A.19), formal search rises with human capital at slope at most $\bar{a}/[(\rho+\delta)\kappa_i^f]$, so accumulation raises a worker's search at rate at most $\mu_i \bar{a}/[(\rho+\delta)\kappa_i^f]$. It suffices to show that the variance stays above this rate throughout the window. The floor will then follow from the decline.

Selection acts on survivors who still differ in formal search. Workers who search slowly leave slowly, so the natural group to follow is the entrants whose initial search lies below some level η_*. The survival of this group is easy to bound, and under slow accumulation its search stays spread out. Nondegenerate initial human capital, a positive active share, and strictly increasing search on the active branch make initial search $\eta_i^f(x(0))$ nondegenerate. The level η_* can therefore be chosen so that the group has positive mass p and positive dispersion s, where s^2 is the variance of initial search within the group. The drift bound is then chosen so that accumulation neither erases this dispersion nor outweighs it:

$$\bar{\mu}_i(D) = \frac{(\rho+\delta)\kappa_i^f}{\bar{a}} \min\left\{ \frac{s}{2D},\ \frac{ps^2}{4} e^{-(\eta_* + s/2)D} \right\}. \tag{A.21}$$

With the group fixed, the bound falls with the horizon and tends to zero, which is the horizon dependence noted in the main text.

Suppose $\mu_i < \bar{\mu}_i(D)$. By the first term of the bound, accumulation raises each worker's search by less than $s/2$ over the window. The group's search therefore keeps a dispersion above $s/2$ and stays below $\eta_* + s/2$. Each member then survives formal-search exits to D with probability at least $e^{-(\eta_* + s/2)D}$. Separation and labor-force exit strike every worker at the same rate, so they cancel from survivor averages. A variance is the smallest mean squared deviation about any center, and survival probabilities are at most one, so

$$\mathrm{Var}_d\big[\eta_i^f(x)\big] \geq \frac{ps^2}{4} e^{-(\eta_* + s/2)D} > \frac{\mu_i \bar{a}}{(\rho+\delta)\kappa_i^f},$$

where the last inequality is the second term of the bound. Selection therefore outweighs accumulation at almost every duration in $[0, D]$. Since equation (A.19) makes the hazard locally absolutely continuous, the hazard strictly declines.

Finally, the hazard stays positive. Entrants who start out searching keep searching, because their human capital only rises and formal search is nondecreasing, and they survive with positive probability. Hence $\lambda_i^f(D) > 0$, and the decline extends this floor to all of $[0, D]$. □

A.5 Hazard Estimation

This subsection establishes consistency, asymptotic normality, and worker-clustered inference for the interval estimator of Section 3. Appendix C.2 relates its population counterpart to the model's continuous-time hazards.

Estimator and population counterpart. A worker can contribute several spells, and those spells need not be independent, so the arguments below treat workers as the sampling unit. Fix an origin ℓ, a destination $\ell' \neq \ell$, and an interval $I_d = (d - \Delta, d]$ of Δ months, six in the baseline. For worker m, let $Z_{m,Y}$ be the weighted number of spells that move from ℓ to ℓ' during I_d, and $Z_{m,A}$ the weighted number at risk at its start:

$$Z_{m,Y} = \sum_n \omega_{m,n} \mathbf{1}\{\ell_{m,n} = \ell,\ T_{m,n} \in I_d,\ \ell'_{m,n} = \ell'\},$$
$$Z_{m,A} = \sum_n \omega_{m,n} \mathbf{1}\{\ell_{m,n} = \ell,\ T_{m,n} > d - \Delta\}. \tag{A.22}$$

Summing over the G workers writes the estimator (14) as a ratio of worker totals, which defines its population counterpart:

$$\widehat{\lambda}_\ell^{\ell'}(d) = \frac{\sum_{m=1}^G Z_{m,Y}}{\sum_{m=1}^G Z_{m,A}}, \qquad p_\ell^{\ell'}(d) = \frac{\mathbb{E}[Z_{m,Y}]}{\mathbb{E}[Z_{m,A}]}, \tag{A.23}$$

provided the denominators are positive. Write p for this population object. It is an interval exit probability, not the instantaneous hazard $\lambda_\ell^{\ell'}(d)$ of equation (10).

Sampling properties. The estimator is a ratio of two averages across workers, so the law of large numbers and the central limit theorem for independent workers deliver its sampling properties.

Proposition 9 (Consistency and asymptotic normality)**.** *Suppose the worker totals* $(Z_{m,Y}, Z_{m,A})$ *are independent and identically distributed across workers, with* $\mathbb{E}[Z_{m,A}] > 0$ *and finite second*

moments, while a worker's own spells may be arbitrarily dependent. Then, as $G \to \infty$,

$$\widehat{\lambda}_\ell^{\ell'}(d) \xrightarrow{p} p,$$
$$\sqrt{G}\big(\widehat{\lambda}_\ell^{\ell'}(d) - p\big) \xrightarrow{d} \mathcal{N}\left(0, \frac{\mathrm{Var}(Z_{m,Y} - pZ_{m,A})}{(\mathbb{E}[Z_{m,A}])^2}\right).$$

Proof. Write $\overline{Z}_Y = G^{-1}\sum_m Z_{m,Y}$ and $\overline{Z}_A = G^{-1}\sum_m Z_{m,A}$. The worker-level law of large numbers implies that these averages converge to their expectations. A positive expected number of spells at risk makes the denominator positive with probability approaching one, so their ratio converges to p.

For the limit distribution, center each worker's exits at p times their spells at risk. The estimation error is then an average of these centered contributions, divided by the average number at risk:

$$\sqrt{G}\big(\widehat{\lambda}_\ell^{\ell'}(d) - p\big) = \frac{1}{\overline{Z}_A}\frac{1}{\sqrt{G}}\sum_{m=1}^{G}\big(Z_{m,Y} - pZ_{m,A}\big).$$

The summands have mean zero by the definition of p and finite variance by the second-moment condition. The worker-level central limit theorem gives their normal limit. Slutsky's theorem replaces $\overline{Z}_A$ with $\mathbb{E}[Z_{m,A}]$, yielding the stated variance. □

The same worker-level contributions yield a consistent variance estimator after replacing the population probability with its sample counterpart:

$$\widehat{\mathrm{Var}}\Big(\widehat{\lambda}_\ell^{\ell'}(d)\Big) = \frac{\sum_{m=1}^{G}\big[Z_{m,Y} - \widehat{\lambda}_\ell^{\ell'}(d)Z_{m,A}\big]^2}{\left(\sum_{m=1}^{G} Z_{m,A}\right)^2}. \tag{A.24}$$

Corollary 3 (Worker-clustered variance)**.** *Under the conditions of Proposition 9,*

$$G\,\widehat{\mathrm{Var}}\Big(\widehat{\lambda}_\ell^{\ell'}(d)\Big) \xrightarrow{p} \frac{\mathrm{Var}(Z_{m,Y} - pZ_{m,A})}{(\mathbb{E}[Z_{m,A}])^2}.$$

Proof. It suffices to show that replacing p by its consistent estimate leaves the limit of the average squared contribution unchanged. Expanding the squared fitted contributions produces the average of $(Z_{m,Y} - pZ_{m,A})^2$ and two additional terms containing $\widehat{\lambda}_\ell^{\ell'}(d) - p$ and its square. Finite second moments make the averages of $(Z_{m,Y} - pZ_{m,A})Z_{m,A}$ and $Z_{m,A}^2$ bounded in probability. Consistency therefore makes both additional terms vanish. The law of large numbers gives

$$\frac{1}{G}\sum_{m=1}^{G}\big[Z_{m,Y} - \widehat{\lambda}_\ell^{\ell'}(d)Z_{m,A}\big]^2 \xrightarrow{p} \mathbb{E}[(Z_{m,Y} - pZ_{m,A})^2] = \mathrm{Var}(Z_{m,Y} - pZ_{m,A}).$$

The average worker exposure converges to $\mathbb{E}[Z_{m,A}] > 0$, so division by its square yields the stated limit. Multiplication by a correction factor converging to one preserves this limit. □

A.6 Identification up to Scale

This subsection proves Proposition 1. A change of units relabels time, human capital, and output without changing the economy, much as a rotation or a translation moves a figure without changing its shape. I prove the proposition by showing that, written in the new units, each equation of the model is the original equation multiplied by a positive constant: $c_A c_t$ for each HJB and c_t/c_x for each forward equation. A solution in the old units is then a solution in the new ones. The targeted moments compare objects measured in the same units, so they cannot depend on which units are chosen. Parameter vectors related by a change of units therefore form an equivalence class, and the normalization selects one member of each class.

Table A.1: Units of the model's objects

Object	Measured in	Multiplied by
Rates $\rho, \delta, \delta_i, \delta_f$ and search intensities $\eta_\ell^{\ell'}$	per unit of time	c_t
Drift μ_ℓ	human capital per unit of time	$c_x c_t$
Volatility σ_ℓ	human capital per square root of time	$c_x\sqrt{c_t}$
Values U, V_ℓ and the entry cost Ψ_f	output	c_A
Flow payoffs F, b, wages, output, and enforcement	output per unit of time	$c_A c_t$
Search-cost coefficients $\kappa_\ell^{\ell'}$	output times time	c_A/c_t
Production scale A	output per unit of human capital and time	$c_A c_t/c_x$
Human capital, thresholds, and the median k_g	human capital	c_x
Densities m_ℓ and g	per unit of human capital	$1/c_x$

Proof of Proposition 1. Fix positive constants c_t, c_x, and c_A, and write relabeled objects with a tilde. A worker at date t with human capital x has new coordinates $\tilde{t} = t/c_t$ and $\tilde{x} = c_x x$, and a model period lasts $\tilde{a} = c_t a$ months, so $\tilde{a}\tilde{d} = ad$ and calendar durations are unchanged. Every other object is relabeled according to what it measures, and Table A.1 lists the factors. Two of them deserve a word. Volatility scales by $c_x\sqrt{c_t}$ because the variance of a Brownian increment grows linearly with elapsed time. Search-cost coefficients scale by c_A/c_t because a quadratic cost of a rate must still be a flow payoff. For the log-logistic entrant density, relabeling replaces the median k_g by $c_x k_g$, and the dimensionless primitives τ, ε, α_0, θ, and ξ_g are unchanged.

It remains to show that the relabeled objects form an equilibrium of the relabeled economy and generate the same targeted moments. I check each equation of the model in turn, beginning with the law of motion of human capital, on which the others rest. Notice that each step uses nothing but the units in Table A.1.

Consider first the law of motion. That is, the relabeled human capital must follow the law of motion with the drift and volatility of Table A.1. Brownian motion rescales with the clock:

$W(t) = \sqrt{c_t}\,\tilde{W}(\tilde{t})$ for a standard Brownian motion $\tilde{W}$ in the new clock. Hence

$$\mathrm{d}\tilde{x} = c_x\,\mathrm{d}x = c_x c_t \mu_\ell\,\mathrm{d}\tilde{t} + c_x\sqrt{c_t}\,\sigma_\ell\,\mathrm{d}\tilde{W} = \tilde{\mu}_\ell\,\mathrm{d}\tilde{t} + \tilde{\sigma}_\ell\,\mathrm{d}\tilde{W},$$

as required, and reflection at zero is unchanged.

Next, consider the worker problem at a fixed conjectured informal share. That is, the relabeled values must satisfy each HJB with the relabeled primitives. The wage wedges in equation (2) depend only on that share, so wages scale with the production scale, $\tilde{a}_\ell\tilde{x} = (c_A c_t/c_x)a_\ell \cdot c_x x = c_A c_t w_\ell(x)$, like every other flow payoff. Every term of an HJB then scales by the same factor. For the drift and diffusion terms,

$$\begin{aligned}\tilde{\mu}_\ell\,\partial_{\tilde{x}}\tilde{V}_\ell &= c_x c_t \mu_\ell \cdot \frac{c_A}{c_x}\partial_x V_\ell = c_A c_t\,\mu_\ell \partial_x V_\ell,\\ \frac{\tilde{\sigma}_\ell^2}{2}\partial_{\tilde{x}\tilde{x}}\tilde{V}_\ell &= \frac{c_x^2 c_t \sigma_\ell^2}{2}\cdot\frac{c_A}{c_x^2}\partial_{xx}V_\ell = c_A c_t\,\frac{\sigma_\ell^2}{2}\partial_{xx}V_\ell,\end{aligned}$$

and the discount, separation, and time-derivative terms behave the same way, because rates scale by c_t and values by c_A. So do the search terms, since with $\tilde{\eta} = c_t\eta$,

$$\tilde{\eta}\,c_A\Delta V - \frac{\tilde{\kappa}_\ell^{\ell'}}{2}\tilde{\eta}^2 = c_A c_t\left[\eta\Delta V - \frac{\kappa_\ell^{\ell'}}{2}\eta^2\right],$$

where ΔV includes the applicable fixed entry cost. Dividing each relabeled HJB by $c_A c_t$ recovers the original one, so the relabeled values solve the relabeled worker problem. The first-order condition then gives $\tilde{\eta}_\ell^{\ell'}(\tilde{x},\tilde{t}) = c_t\eta_\ell^{\ell'}(x,t)$, and value matching, smooth pasting, and the formal-entry equality hold with every threshold multiplied by c_x.

With policies in hand, turn to the distribution of workers. That is, the relabeled densities must solve the forward equations with the relabeled policies. The relabeled flux is

$$\tilde{J}_\ell = \tilde{\mu}_\ell\tilde{m}_\ell - \frac{\tilde{\sigma}_\ell^2}{2}\partial_{\tilde{x}}\tilde{m}_\ell = c_x c_t\mu_\ell\cdot\frac{m_\ell}{c_x} - \frac{c_x^2 c_t\sigma_\ell^2}{2}\cdot\frac{\partial_x m_\ell}{c_x^2} = c_t J_\ell.$$

Every term of each forward equation therefore scales by c_t/c_x: the time derivative, the flux term, each search, separation, and exit flow, the entry flow $\tilde{\delta}\tilde{g} = (c_t/c_x)\delta g$, and the quit flows at thresholds. The relabeled densities solve the relabeled forward equations, and each state holds the same mass of workers at the same calendar date.

It remains to check that the conjectured share is the one the relabeled economy generates. Sectoral output scales by $c_A c_t$, since

$$\tilde{A}\int_0^\infty \tilde{x}\,\tilde{m}_\ell(\tilde{x})\,\mathrm{d}\tilde{x} = \frac{c_A c_t}{c_x}A\int_0^\infty c_x x\,m_\ell(x)\,\mathrm{d}x = c_A c_t A\int_0^\infty x\,m_\ell(x)\,\mathrm{d}x.$$

The informal share is therefore unchanged and equal to the conjectured one, so the relabeled allocation is an equilibrium. Enforcement, $\tilde{\alpha}_t = \alpha_0\tilde{Y} = c_A c_t\alpha_t$, and the government budget,

including the lump-sum transfer, rescale by the same factor.

Finally, consider the targeted moments. Each compares objects measured in the same units, so it cannot depend on which units are chosen. Benefits and output both scale by $c_A c_t$, so $b\mathbf{M}_u/Y$ is unchanged. Completed-spell wages scale like the formal-wage level that normalizes them, so normalized means and variances are unchanged. Hazards scale by c_t, so destination shares and log ratios of hazards at two calendar durations are unchanged. Because $\tilde{a}\tilde{d} = ad$, every profile evaluated at a given number of calendar months is unchanged, and the same mapping of densities and durations preserves the three entry-cohort state shares.

The normalization now follows by choosing the units. Taking

$$c_t = \frac{12}{a}, \qquad c_x = \frac{1}{k_g}, \qquad c_A = \frac{c_x}{c_t A}$$

gives $\tilde{a} = 12$, $\tilde{k}_g = 1$, and $\tilde{A} = 1$, and in particular

$$(\tilde{F}, \tilde{b}) = \frac{c_x}{A}(F, b), \qquad \tilde{\Psi}_f = \frac{c_x}{c_t A}\Psi_f, \qquad \tilde{\kappa}_\ell^{\ell'} = \frac{c_x}{c_t^2 A}\kappa_\ell^{\ell'}.$$

Every parameter vector is thus equivalent to one with $a = 12$, $k_g = 1$, and $A = 1$, the representative the estimation uses. Once a and k_g are fixed, an absolute wage expressed in a fixed empirical output unit determines the remaining output scale A through the inversion in equation (17). □

Note that the argument never solves the model. It only tracks what each object measures, which is why fixing the units costs the estimation nothing.

B Data Appendix

B.1 EPS Sources and Sample Construction

I combine retrospective EPS reports into monthly labor histories, apply sample restrictions, and assess the resulting coverage.

Sources and retrospective reports. The retrospective modules of the 2002, 2004, 2006, 2009, and 2015 EPS waves record work and nonwork periods. Each report gives start and end months, together with employment category, contract and contribution information, earnings, hours, occupation, industry, firm size, job term, and region. Respondents can describe several periods in one interview and revisit periods reported in earlier waves.

Missing a wave therefore need not leave a gap in a worker's history. Table B.1 shows that returning respondents account for a substantial share of later interviews, alongside first-time respondents and those observed in successive processed waves. In Panel C, weighted outflows use origin-wave expansion factors and inflows use target-wave factors. Because these factors can change between interviews, net attrition subtracts both inflows and continuing respondents' weight revisions from outflows, with revisions measured as target minus origin.

I rely on later reports to cover the years around the 2012 wave, whose substantive responses I exclude because of interview problems. Only its identifiers and expansion factors enter participation and flow accounting, while the 2015 module supplies retrospective reports for the intervening years. Table B.2 reports participation around 2012 in Panels A and B and coverage of January 2010–December 2014 among 2015 respondents in Panel C.

Construction of labor histories. Before pooling these reports, I harmonize identifiers, spell types, job characteristics, and reporting waves and put all dates on a common monthly calendar. I also correct the reversed 2002 work and first-time-search labels and remove reports with invalid month codes, sentinel years, or end dates before their start dates. At this stage, I classify employment within each wave using the rules in Appendix B.2.

To reconcile overlapping and duplicate reports, I score their completeness by counting observed earnings, hours, two firm-size fields, occupation, industry, contract, contributions, region, and job term. A formal or informal work report with at least eight of these ten fields is classified as rich work. In flagged histories, rich work takes overlapping months from nonwork reports with no observed characteristics. I split surrounding records and trim one-sided overlaps, using completeness and rich-work status to resolve remaining conflicts while preserving retained work classifications. For exact same-span duplicates, I give priority to work, then rich formal or informal work, greater completeness, and the reporting wave closest to the start. Each retained piece keeps its original identifiers and dates. This lets me count affected workers and records in Panels B and C of Table B.3, treating a source report as dropped only if none of its pieces remains.

Table B.1: EPS respondent composition and subsequent participation

	2002	2004	2006	2009	2015
Panel A. Composition of respondents observed in each wave					
Respondents observed	15,365	14,899	14,856	13,096	14,413
First-time respondents (%)	100.0	7.8	0.0	0.0	13.3
Continuous respondents (%)	0.0	92.2	78.6	66.3	40.8
Returning respondents (%)	0.0	0.0	21.4	33.7	45.9
Panel B. Subsequent status of respondents from each processed origin wave					
Observed next wave (%)	78.4	87.4	78.0	53.7	—
Missed next, returned later (%)	14.4	6.1	6.4	0.0	—
Not observed again (%)	7.2	6.5	15.6	46.3	—

Panel C. Weighted respondent flows Flow	2002–04	2004–06	2006–09
Outflow	21.6	12.4	19.8
First-observed inflow	29.0	0.0	0.0
Return inflow	0.0	12.3	14.0
Weight revision	26.8	14.4	7.0
Net attrition	-34.2	-14.3	-1.2

Notes: Counts are unweighted. Panels A and B use respondent weights. Panel C reports flows in % of the origin-wave stock using raw expansion factors. Negative net attrition denotes growth. The sample includes men and women eligible under Table B.3. Source: EPS 2002–2015, author's calculations.

I address errors identified through manual inspection with fixed worker-specific rules, coded to run in the prescribed record order. These rules recover fields, remove duplicated or implausible reports, adjust boundaries, split records, and remove isolated formality changes attributed to reporting errors. In making these corrections, I close only specified local gaps through boundary adjustments and leave other unreported months unassigned, excluding histories that cannot be reconciled.

Once reports have been reconciled, I consolidate those with matching spell types and employment states when they overlap or meet the survey-boundary condition, and combine consecutive unemployment and other non-employment reports. Each consolidated spell retains the first start, last end, contributing waves, modal categorical characteristics, and mean reported earnings and hours. A formal–informal change always starts a new spell, and the final histories must consist of positive-duration, nonoverlapping spells. Subsequent sector-spell linkage and censoring follow the rules in Appendix B.2.

Analysis sample. The cleaned-history sample includes workers with observed sex and birth year, at least one formal or informal employment spell, and a positive weight from the first processed interview. Applying these restrictions after excluding unreliable histories leaves 23,205 men and women, as reported in Panel A of Table B.3. Because reports can be removed or consolidated without excluding their workers, the record counts in Panels B and C are reported separately.

Table B.2: 2012 survey participation and retrospective labor-history coverage

Classification	Workers	Unweighted (%)	Weighted (%)
Panel A. Subsequent participation of 2009 respondents			
Neither 2012 nor 2015	2,895	22.1	30.6
2012 only	2,124	16.2	15.7
2015 only	2,316	17.7	18.4
Both 2012 and 2015	5,761	44.0	35.3
Panel B. Prior participation of 2015 respondents			
Neither 2009 nor 2012	5,928	41.1	54.5
2009 only	2,316	16.1	16.4
2012 only	408	2.8	4.7
Both 2009 and 2012	5,761	40.0	24.4

Panel C. Coverage of 2010–2014 in accepted histories of 2015 respondents	Workers	Mean coverage (%)	Complete coverage (%)	Mean months last reported in 2015
Observed in 2009 and 2015	8,077	100.0	99.9	60.0
Earlier returner, absent in 2009	1,172	100.0	100.0	60.0
First processed in 2015	5,164	96.2	83.4	57.7

Notes: Counts and coverage statistics are unweighted. Participation percentages use 2009 respondent weights in Panel A and 2015 weights in Panel B. Coverage concerns January 2010–December 2014 among 2015 respondents of both sexes. Complete coverage requires all 60 months. Source: EPS 2002–2015, author's calculations.

The duration and mobility analyses then restrict these histories to men ages 15–64, whereas Tables B.1–B.4 cover the broader cleaned-history sample. Additional wage-specific observation rules are given in Appendix B.3, with weighting described in Appendix B.2.

Table B.3: EPS sample construction and labor-history cleaning

Panel A. Worker-level sample construction

Sample stage	Workers retained	Workers removed	Share of initial workers (%)
Enter spell-history cleaning	23,321	—	100.00
After unreliable-history exclusions	23,315	6	99.97
Valid person record and baseline weight	23,315	0	99.97
At least one formal or informal spell	23,205	110	99.50
Cleaned-history sample	23,205	0	99.50

Panel B. Spell-record cleaning and boundary adjustments

Cleaning operation	Records affected	Workers affected	Record sample remaining (%)
Spell records entering cleaning	192,990	23,321	100.00
Source spell records dropped	756	273	99.61
Records created by splitting	44	40	99.63
Boundaries moved to resolve overlaps	426	220	99.63
Boundaries moved to close gaps	6	6	99.63
Spell records after cleaning	192,278	23,315	99.63

Panel C. Consolidation of cleaned records into final spells

Cleaning operation	Records affected	Workers affected	Record sample remaining (%)
Date inconsistencies removed	320	316	99.47
Short overlaps removed	907	766	99.00
Compatible records combined	23,485	10,611	86.83
Additional records removed	14	14	86.82
Adjacent non-employment combined	1,500	1,290	86.04
Same-span duplicates removed	18	18	86.03

Cleaned-history sample: 165,696 cleaned spells

Notes: Unweighted worker and record counts for men and women. Percentages use initial workers in Panel A and initial source records in Panels B and C. Panel B's operation counts may overlap. Source: EPS 2002–2015, author's calculations.

Coverage and construction checks. The unweighted statistics in Table B.4 show that 95 percent of workers have no inter-spell gap longer than one month. Observed time nevertheless excludes every uncovered month, including those permitted by the continuity criterion. The resulting histories have a median of 26.4 observed years and 6 spells.

For 2015 respondents, the 2010–2014 window is nearly fully covered whether they were interviewed in 2009 or returned after missing that wave, as shown in Panel C of Table B.2. Those first processed in 2015 have more gaps, though their mean coverage remains close to the full window. The last column measures how much of this window falls in spells last reported in 2015, averaging these months across respondents.

Table B.4: Continuity and depth of cleaned EPS labor histories

Panel A. Continuity of cleaned labor histories

Diagnostic	Events	Workers	Share of workers (%)
Workers in the cleaned-history sample	—	23,205	100.0
Gaps longer than one month	1,302	1,161	5.0
Complete histories (gaps of 0–1 month)	—	22,044	95.0
Overlapping adjacent spells	0	0	0.0

Panel B. Depth of cleaned labor histories across workers

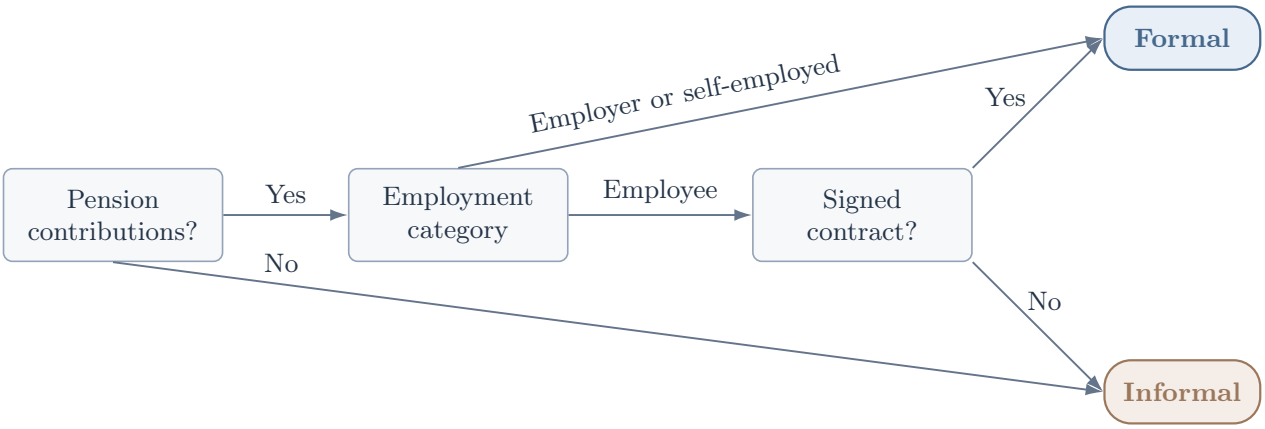
History characteristic	Mean	25th percentile	Median	75th percentile
Time covered by spells (years)	24.0	14.2	26.4	34.3
Cleaned spells per worker	7.1	4.0	6.0	9.0
Gap length above one month	17.9	8.0	12.0	12.0

Notes: Unweighted statistics for men and women. Complete histories allow at most one uncovered month between adjacent spells. Gap-length statistics condition on longer gaps. Observed years exclude uncovered months. Source: EPS 2002–2015, author's calculations.

B.2 Labor-Market States and Sector-Spells

Employment classification. The EPS questions on pension contributions, employment category, and signed contracts supply the inputs to the baseline classification in Figure B.1. Contribution responses 1–6 indicate contributions to a pension system and 7 explicit noncontribution. Employment categories 1 and 2 identify employers and the self-employed, whose classification uses contributions alone. For employees, contract codes 1–3 indicate a signed contract and 4 its absence in 2002, whereas later waves use 1 and 2–3, respectively. Unknown responses supply no evidence of noncompliance, although an employee record is informal whenever either requirement is explicitly absent, even if the other is unreported. Other unresolved working records remain unclassified, while nonwork and first-time search records are assigned to unemployment. Unpaid family work and other special employment categories, including Army or police records, remain in the history but are excluded from the formal–informal comparison.

Figure B.1: Classification of employment spells



Regular employment spells with explicit yes/no responses.

Sector-spell construction. I join consecutive same-state records when the next start month is no more than six months after the previous end, keeping armed-forces or unclassifiable occupation records (code 10) separate. A state change or a longer gap starts a new sector-spell. Because dates are inclusive, a record ending in January and one beginning in July meet this condition despite five uncovered months between them. These months enter elapsed sector-spell duration, measured from the first start through the last end, whereas the observed-time coverage in Appendix B.1 counts only months assigned to records.

The next sector-spell identifies an exit destination only when its start is within six months of the origin's end and it satisfies the destination restrictions. I construct these links from the full chronology before excluding records by age or occupation, so an excluded record cannot be bypassed to create a link to a later spell. Age is evaluated at the start of each constituent job record, and a sector-spell is age-eligible if at least one starts between ages 15 and 64. The mobility statistics then use only links between formal, informal, and unemployed states.

Observing any subsequent sector-spell marks the origin duration as completed, including when a long gap or destination exclusion prevents recording a usable exit. I treat the final observed spell as right-censored at the end of its recorded history. At the other end of the window, spells beginning in January 1980 are potentially left-censored, as flagged in Table B.5. They remain in the baseline, with the exclusion exercise reported in Figure D.2.

Table B.5: Sector-spell construction and censoring diagnostics

	Formal	Informal	Unemployment
All sector-spells	24,115	13,633	23,506
Completed sector-spells	17,978	11,153	20,982
Potentially left-censored January 1980 starts	3,456	1,380	772

Notes: Unweighted sector-spell counts for men satisfying the age and occupation restrictions above.
Source: EPS 2002–2015, author's calculations.

Respondent weights. Within the labor-history sample, I rescale the EPS expansion factors by their mean across observed respondent-wave records in the same entry cohort. The cohort is defined by the first processed interview, and its normalization uses all available waves. For interviews after entry, I also multiply the rescaled factor by the inverse predicted probability of response. A weighted logit models response using the target and previous observed waves, years since that observation, entry cohort, and previously observed sex, age, education, and region. Respondents may miss an interview and return later. Entry interviews receive no response adjustment, while predicted probabilities for later interviews are bounded between 0.01 and 0.99. The resulting weights are relative analysis weights.

Spell and month weights. Each cleaned job spell inherits the respondent weight from its last contributing reporting wave. For duration and mobility statistics, I assign each sector-spell the duration-weighted average of its component job-spell weights. Component durations count only recorded months, and the resulting flow weight applies once per sector-spell. For stock statistics, I multiply each job spell's weight by its observed duration to obtain its spell-month weight. When expanding histories to one row per month, I divide this weight by the spell's length, so each observed month carries its reporting-wave weight once.

B.3 Wage Measurement

Wage reports and units. I construct a standardized monthly wage equivalent from reported monthly net labor income and weekly hours. Income values at or below 10 CLP or at or above 100 million CLP are treated as missing, as are weekly hours outside 0–168. Before consolidating repeated reports, I deflate each income report to 2010 Chilean pesos with its reporting wave's annual CPI and convert it to U.S. dollars at the 2010 annual-average market exchange rate. This assumes income is reported in contemporaneous nominal units. I then average real income and hours across repeated reports of the same job. For positive income and hours, the hourly wage divides real monthly income by 4.35 times weekly hours. Multiplying it by 40×4.35 gives the monthly equivalent for a common 40-hour week. The 2002 wave supplies no income report, although a later report can supply wage information for a consolidated job that includes 2002.

Wage spells and averages. I combine contiguous job records in the same employment state into a wage sector-spell, allowing employer changes within that state. An uncovered month starts a new wage spell, unlike the six-month link used for mobility in Appendix B.2. Every component job must have a valid positive wage. The wage sample contains men ages 15–64 at the start of the wage sector-spell and excludes the entire spell if any component has occupation code 10 (armed forces or unclassifiable occupations). For a spell in sector ℓ, let w_h be component job h's monthly wage equivalent and L_h its observed duration in months. The arithmetic average corresponding to $\overline{w}_\ell$ in Section 3.3 is

$$\overline{w}_\ell = \frac{\sum_h L_h w_h}{T}, \qquad T = \sum_h L_h.$$

This assigns each component wage to its recorded months without recovering the within-job wage path or the wage at entry. Survey weights apply when averaging across spells, as described in Appendix B.2, and do not enter this within-spell time average.

B.4 First-Time Job-Search Cohort

I identify labor-market entrants whose recorded histories begin with an explicit report of first-time job search. I mark this report before consolidation and preserve the marker when search records merge with other nonemployment records. Entry is dated from the start of the resulting initial nonemployment spell, rather than the first EPS interview. The cohort contains men from the labor-history sample in Appendix B.1, which requires at least one report of regular work somewhere in the history. No additional entry-age restriction is applied.

I follow these entrants for five years, reconstructing their formal, informal, and nonemployment states from the recorded spells. A worker remains in the follow-up sample if every six-month interval contains some observed time in one of these states. Months without an observed state contribute neither a state nor exposure. I hold each worker's survey weight fixed at the weight assigned to the entry spell and pool observed person-months with these weights to calculate the six-month state shares. The coverage requirement retains 122 of 130 male entrants. Among those retained, recorded states cover 99.97 percent of potential person-month exposure across the 61 monthly observations from entry through month 60, using the fixed entry weights.

C Estimation and Numerical Methods

C.1 Solving the Stationary Equilibrium

Given model parameters, I solve for the informal output share that reproduces itself through wages, workers' optimal choices, and their stationary distribution.

Equilibrium algorithm. The outer loop solves the scalar fixed point in equation (**??**). For a candidate informal output share $\mathcal{I}$, the calculation proceeds as follows:

1. Compute formal and informal wages from equation (2).
2. Solve the worker problem at these wages, obtaining U, V_i, and V_f, the search policies in equation (5), and the employment continuation regions.
3. Solve the stationary KFE under these policies and normalize the worker population to one.
4. Compute the implied informal output share,

$$T(\mathcal{I}) = \frac{\sum_{j=1}^{N_x} x_j m_i(x_j;\mathcal{I})\,\Delta x}{\sum_{j=1}^{N_x} x_j[m_i(x_j;\mathcal{I}) + m_f(x_j;\mathcal{I})]\,\Delta x},$$

   and update $\mathcal{I}$ using the residual $\mathcal{I} - T(\mathcal{I})$.

Because $\mathcal{I}$ is an output share, the aggregation weights employment densities by human capital. The root search maintains bounds within $[0, 1]$, lowering the upper bound when the residual is positive and raising the lower bound when it is negative. The next candidate is a false-position step when both endpoint residuals are available and the proposed step lies sufficiently far inside the bounds. Otherwise, the algorithm uses the midpoint.

Value functions and optimal policies. Human capital is discretized on a common uniform grid $x_j = (j-1)\Delta x$, $j = 1, \ldots, N_x$, with $x_{N_x} = x_{\max}$. The stationary equations corresponding to the worker problem in Section 2.2 are

$$\begin{aligned} (\rho+\delta)U &= b + \mathcal{L}_u U + \mathcal{H}_u, \\ (\rho+\delta)V_\ell &= w_\ell - F\mathbf{1}\{\ell = f\} + \mathcal{L}_\ell V_\ell + \mathcal{H}_\ell, \qquad \ell \in \{i, f\}, \end{aligned} \tag{C.1}$$

where the employment equation holds on $\mathcal{C}_\ell$. The human-capital generator is $\mathcal{L}_\ell = \mu_\ell \partial_x + (\sigma_\ell^2/2)\partial_{xx}$. The Hamiltonians $\mathcal{H}_\ell$ collect optimized search returns net of search costs and, for employed workers, exogenous separation returns. They are evaluated at the current values using equation (5).

The spatial operator A_ℓ approximates diffusion with central differences and drift with an upwind difference. At an interior node, its off-diagonal rates are

$$A_{\ell,j,j+1} = \frac{\sigma_\ell^2}{2(\Delta x)^2} + \frac{[\mu_\ell]^+}{\Delta x}, \qquad A_{\ell,j,j-1} = \frac{\sigma_\ell^2}{2(\Delta x)^2} + \frac{[-\mu_\ell]^+}{\Delta x},$$

with $A_{\ell,j,j} = -(A_{\ell,j,j+1} + A_{\ell,j,j-1})$ and $[z]^+ = \max\{z, 0\}$. Removing the outward transition at each endpoint and adjusting the diagonal to preserve a zero row sum imposes reflection at zero and at the numerical upper boundary $x_{\max}$.

For an explicit pseudo-time step h_V, the undamped candidates at iteration n are

$$\begin{aligned}
\widehat{U}^{n+1} &= U^n + h_V\big[b + \mathcal{H}_u^n + A_u U^n - (\rho+\delta)U^n\big], \\
\widehat{V}_\ell^{n+1} &= \max\big\{U^n,\, V_\ell^n + h_V\big[w_\ell - F\mathbf{1}\{\ell = f\} + \mathcal{H}_\ell^n + A_\ell V_\ell^n - (\rho+\delta)V_\ell^n\big]\big\}, \qquad \ell \in \{i, f\}.
\end{aligned}$$

The employment obstacle uses the previous unemployment iterate U^n, with maxima taken node by node. The step satisfies $(\rho+\delta)h_V < 1$ and

$$h_V \max_{\ell \in \{u,i,f\}} \left\{\rho + \delta + \frac{\sigma_\ell^2}{(\Delta x)^2} + \frac{|\mu_\ell|}{\Delta x}\right\} \le 1,$$

so that the spatial and discount terms have nonnegative explicit weights. Damping combines these candidates with the current values when updating the iterate.

The final semi-implicit variational-inequality solve holds the Hamiltonians fixed at their current values and updates unemployment with pseudo-time step h_V:

$$\left[\left(\rho + \delta + \frac{1}{h_V}\right)\mathrm{Id} - A_u\right]\widehat{U}^{n+1} = b + \mathcal{H}_u^n + \frac{U^n}{h_V}.$$

For employment, let $B_\ell = [1+(\rho+\delta)h_V]\mathrm{Id} - h_V A_\ell$. The candidate solves the linear complementarity problem

$$\begin{gathered}
\widehat{V}_\ell^{n+1} \ge U^n, \\
B_\ell \widehat{V}_\ell^{n+1} \ge V_\ell^n + h_V\big[w_\ell - F\mathbf{1}\{\ell = f\} + \mathcal{H}_\ell^n\big], \\
(\widehat{V}_\ell^{n+1} - U^n) \odot \Big\{B_\ell \widehat{V}_\ell^{n+1} - V_\ell^n - h_V\big[w_\ell - F\mathbf{1}\{\ell = f\} + \mathcal{H}_\ell^n\big]\Big\} = 0.
\end{gathered}$$

Here $\ell \in \{i, f\}$, inequalities hold node by node, and $\odot$ denotes elementwise multiplication. The candidates are damped before the next iteration. At convergence, unemployment satisfies equation (C.1) on the grid, and the employment surplus $D_\ell = V_\ell - U$ satisfies the nodewise condition

$$\min\{D_\ell,\, (\rho+\delta)V_\ell - w_\ell + F\mathbf{1}\{\ell = f\} - A_\ell V_\ell - \mathcal{H}_\ell\} = 0, \qquad \ell \in \{i, f\}.$$

Stationary distribution. The KFE uses the same grid and spatial transition rates as the worker problem. Let $m_\ell = (m_\ell(x_1), \ldots, m_\ell(x_{N_x}))'$ denote the density vector in state ℓ. With columns indexing origins and rows indexing destinations, the forward spatial operator is A'_ℓ. At an interior node away from a quitting boundary,

$$(A'_\ell m_\ell)_j = A_{\ell,j-1,j} m_\ell(x_{j-1}) + A_{\ell,j,j} m_\ell(x_j) + A_{\ell,j+1,j} m_\ell(x_{j+1}).$$

The neighboring-node inflows and the outflow from node j discretize $-\partial_x J_\ell$ in equation (7). Using the transpose therefore preserves the spatial transition rates from the worker problem.

The variational inequality determines where workers remain employed and where they quit. For each employed state, the solver retains employment density only on the continuation nodes,

$$j = k_\ell, \ldots, N_x, \qquad m_\ell(x_j) = 0 \quad \text{for } j < k_\ell, \qquad \ell \in \{i, f\},$$

where k_ℓ is the first node with a numerically positive value gap $V_\ell - U$. When a positive quitting threshold is present, a worker quits upon first reaching that boundary. The continuous boundary condition and stationary quit flow are

$$\lim_{x \downarrow \bar{x}_u^\ell} m_\ell(x) = 0,$$

$$Q_\ell = -\lim_{x \downarrow \bar{x}_u^\ell} J_\ell(x) = \frac{\sigma_\ell^2}{2} \lim_{x \downarrow \bar{x}_u^\ell} m'_\ell(x).$$

As Proposition 7 shows, the density vanishes at the absorbing employment boundary while a positive outward flow can still enter unemployment. The numerical KFE must preserve this transfer.

To implement absorption, retain only transitions between employment continuation nodes in the forward spatial operator:

$$(A_\ell^K)_{jr} = \begin{cases} A_{\ell,r,j}, & j, r \geq k_\ell, \\ 0, & \text{otherwise}, \end{cases} \qquad \ell \in \{i, f\}.$$

This removes transitions into inactive employment nodes while retaining the original diagonal at the first active node. For $k_\ell > 1$, the downward rate $A_{\ell,k_\ell,k_\ell-1}$ therefore remains a loss from employment. Let e_j be the jth coordinate vector. The matrix T_ℓ transfers this loss into the adjacent unemployment node:

$$\mathsf{T}_\ell = A_{\ell,k_\ell,k_\ell-1} e_{k_\ell-1} e'_{k_\ell}, \qquad \mathsf{T}_\ell m_\ell = A_{\ell,k_\ell,k_\ell-1} m_\ell(x_{k_\ell}) e_{k_\ell-1},$$
$$Q_\ell^{\Delta x} = \Delta x \, A_{\ell,k_\ell,k_\ell-1} m_\ell(x_{k_\ell}), \quad \mathbf{1}' A_\ell^K m_\ell + \mathbf{1}' \mathsf{T}_\ell m_\ell = 0.$$

Here $\mathsf{T}_\ell m_\ell$ is a density inflow, while $Q_\ell^{\Delta x}$ is the aggregate mass of quitting workers per unit time. The transfer approximates unemployment entry at the continuous threshold, with the employment loss already counted in A_ℓ^K. If $k_\ell = 1$, the lower boundary reflects and $\mathsf{T}_\ell = 0$.

Search and exogenous separations complete the joint transition operator. Stack the densities as $\mathbf{m} = (m_u', m_i', m_f')'$. Before demographic exit and replacement, the operator is

$$G_0 = \begin{bmatrix} A_u' - \operatorname{diag}(\eta_u^i + \eta_u^f) & \delta_i \mathrm{Id} + \mathsf{T}_i & \delta_f \mathrm{Id} + \mathsf{T}_f \\ \operatorname{diag}(\eta_u^i) & A_i^K - \operatorname{diag}(\eta_i^f) - \delta_i \mathrm{Id} & \operatorname{diag}(\eta_f^i) \\ \operatorname{diag}(\eta_u^f) & \operatorname{diag}(\eta_i^f) & A_f^K - \operatorname{diag}(\eta_f^i) - \delta_f \mathrm{Id} \end{bmatrix}.$$

The first block row collects unemployment's spatial flows, search exits, and inflows from separations and quits, while the remaining rows give the corresponding employment flows. Search and separations preserve human capital and therefore have diagonal blocks, whereas T_i and T_f transfer boundary losses into unemployment. Together these transitions preserve worker mass, $\mathbf{1}'G_0 = 0$.

New workers enter unemployment with density g, normalized so that $\Delta x \sum_j g(x_j) = 1$. With $\mathbf{g} = (g', 0', 0')'$, the full operator including demographic exit and replacement is

$$G = G_0 - \delta \mathrm{Id} + \delta \mathbf{g}\, \Delta x\, \mathbf{1}', \qquad G\mathbf{m} = 0, \qquad \Delta x\, \mathbf{1}'\mathbf{m} = 1.$$

Demographic exits are replaced by entrants drawn from g, whereas quitting transfers an existing worker into unemployment at the boundary. At unit population mass, the stationary equations are equivalently

$$(G_0 - \delta \mathrm{Id})\mathbf{m} = -\delta \mathbf{g}.$$

The sparse linear solve imposes zero density at inactive employment nodes and replaces one stationarity row with population normalization. This modified solve matrix imposes the constraints, while G remains the transition operator used to evaluate the stationary residual. Entry normalization, stationary masses, and aggregate output all use rectangular quadrature, with weight Δx at every node.

Convergence and final refinement. The calculation has separate stopping criteria for worker values, the stationary distribution, and the aggregate fixed point. In both worker methods, the convergence statistic is the largest candidate change before damping,

$$\max_{\ell \in \{u,i,f\}} \|\widehat{W}_\ell^{n+1} - W_\ell^n\|_\infty \leq \epsilon_V.$$

When the spatial restriction reduces the explicit step, the stopping tolerance is rescaled with it so that a smaller step does not mechanically relax Bellman accuracy.

For the distribution, the stationarity diagnostic is

$$\left\|(G_0 - \delta \mathrm{Id})\mathbf{m} + \delta \mathbf{g}\,(\Delta x\, \mathbf{1}'\mathbf{m})\right\|_\infty.$$

The solver checks this residual and the minimum density against separate tolerances. It also records population normalization, densities in inactive nodes, and the signs and mass-preservation properties of the full transition operator. The outer root calculation stops when $|\mathcal{I} - T(\mathcal{I})| \leq \epsilon_{\mathcal{I}}$.

After the initial root calculation, I solve the worker problem again with the semi-implicit method from a fresh initialization and recompute the stationary distribution. Refinement can change the implied output share, so a residual above $\epsilon_{\mathcal{I}}$ restarts the root calculation using the refined worker method. These stopping criteria assess convergence on the chosen grid, while sensitivity to Δx and $x_{\max}$ requires a separate accuracy check. Numerical convergence does not by itself establish uniqueness of the economic equilibrium.

C.2 Numerical Calculation of Model Moments

I calculate model moments from the stationary equilibrium in Appendix C.1, using the same human-capital grid and rectangular quadrature as the equilibrium solver. Durations are expressed in model time, with calendar conversion given in Appendix C.3 and wage scaling in Appendix C.3.

Stationary aggregates and policy statistics. Stationary moments are weighted sums over the equilibrium densities. Define

$$m = m_u + m_i + m_f, \qquad \mathbf{M}_\ell = \Delta x\, \mathbf{1}' m_\ell, \qquad \mathbf{M} = \sum_{\ell \in \{u,i,f\}} \mathbf{M}_\ell.$$

The unemployment share is $\mathbf{M}_u$, and informal employment as a share of employment is $\mathbf{M}_i/(\mathbf{M}_i + \mathbf{M}_f)$. Benefit expenditure relative to output, $b\mathbf{M}_u/Y$, is not targeted in estimation. For a region $\mathcal{R}$ in state ℓ, its population share within that state and the conditional mean of a grid quantity $z(x)$ are, respectively,

$$\frac{\sum_j \mathbf{1}\{x_j \in \mathcal{R}\} m_\ell(x_j)\, \Delta x}{\mathbf{M}_\ell}, \qquad \frac{\sum_j z(x_j) \mathbf{1}\{x_j \in \mathcal{R}\} m_\ell(x_j)\, \Delta x}{\sum_j \mathbf{1}\{x_j \in \mathcal{R}\} m_\ell(x_j)\, \Delta x}.$$

Conditional means are calculated only for groups with positive mass. The discouraged share uses $\mathcal{R} = \{x : \eta_u^i(x) = \eta_u^f(x) = 0\}$ among unemployed workers. Comparative-advantage groups instead use the pooled stationary density m. For example, the economy-wide share with an informal advantage is

$$\frac{\sum_j \mathbf{1}\{V_i(x_j) > \max[U(x_j), V_f(x_j)]\} m(x_j)\, \Delta x}{\mathbf{M}}.$$

The formal-advantage group is constructed in the same way, using current-equilibrium values before subtracting the fixed cost of a move into formality. Prospective wages within either preference group are averaged with the pooled density.

For a search direction $\ell \to \ell'$, the active-search share uses $\mathcal{R} = \{x : \eta_\ell^{\ell'}(x) > 0\}$. The mean realized flow search cost among these workers is

$$\frac{\sum_j \frac{\kappa_\ell^{\ell'}}{2} [\eta_\ell^{\ell'}(x_j)]^2 \mathbf{1}\{\eta_\ell^{\ell'}(x_j) > 0\} m_\ell(x_j)\, \Delta x}{\sum_j \mathbf{1}\{\eta_\ell^{\ell'}(x_j) > 0\} m_\ell(x_j)\, \Delta x}.$$

This cost average conditions on active search, whereas the current wage by employment state averages over all workers in that state, including nonsearchers.

For $\ell \in \{u, i\}$, the formal-entry-constrained group has a positive gross value gain $V_f - W_\ell$ but zero formal-search intensity. Its share within the origin state is

$$\frac{\sum_j \mathbf{1}\{V_f(x_j) > W_\ell(x_j),\ \eta_\ell^f(x_j) = 0\} m_\ell(x_j)\,\Delta x}{\mathbf{M}_\ell}.$$

For unemployment, the numerical positive-gain test is implemented as $\text{round}[V_f(x) - U(x), 8] > 0$, together with $\eta_u^f(x) = 0$. Prospective formal wages for constrained workers use the conditional mean over that origin-specific group. These calculations give the populations and averages reported in Section 4.3.

Entry cohorts and numerical propagation. A stationary stock density is not the distribution of workers beginning a new spell. Using the transition blocks from Appendix C.1, the stationary inflow densities into new informal, formal, and unemployment spells are

$$\begin{aligned}
\mathbf{b}_i &= \eta_u^i \odot m_u + \eta_f^i \odot m_f, \\
\mathbf{b}_f &= \eta_u^f \odot m_u + \eta_i^f \odot m_i, \\
\mathbf{b}_u &= (\delta_i \mathrm{Id} + \mathsf{T}_i) m_i + (\delta_f \mathrm{Id} + \mathsf{T}_f) m_f + \delta g.
\end{aligned}$$

The unemployment inflow combines separations, quits, and replacement entrants. Normalizing each state's inflow gives the initial spell cohort,

$$\phi_\ell(0) = \frac{\mathbf{b}_\ell}{\Delta x\, \mathbf{1}'\mathbf{b}_\ell}.$$

Here $\phi_\ell(x, d) = S_\ell(d)\bar{m}_\ell(x, d)$ is the unnormalized density of workers whose original spell survives to duration d, with S_ℓ and $\bar{m}_\ell$ referring to the same cohort and labor-force-exit convention. A worker's first exit ends the spell, so subsequent returns do not enter this cohort.

The forward operator for surviving spells retains spatial movement and removes workers at their first exit. Without labor-force exit, its three state-specific versions are

$$\begin{aligned}
K_u^0 &= A_u' - \mathrm{diag}(\eta_u^i + \eta_u^f), \\
K_i^0 &= A_i^K - \mathrm{diag}(\eta_i^f) - \delta_i \mathrm{Id}, \\
K_f^0 &= A_f^K - \mathrm{diag}(\eta_f^i) - \delta_f \mathrm{Id}.
\end{aligned}$$

Including labor-force exit gives $K_\ell^\delta = K_\ell^0 - \delta \mathrm{Id}$. For the chosen operator K_ℓ, the numerical update is

$$\phi_\ell^{n+1} = P_\ell \phi_\ell^n, \qquad P_\ell = \mathrm{Id} + hK_\ell, \qquad d_n = nh. \tag{C.2}$$

I choose h to keep P_ℓ nonnegative and propagate the cohort until the requested horizon or until its remaining mass falls below tolerance. Inactive employment nodes retain zero density, while mass crossing a quitting boundary leaves the spell.

Duration profiles use K_ℓ^0, while completed-spell transitions and wages use K_ℓ^δ and treat labor-force exit as a competing event. Superscripts 0 and δ on ϕ_ℓ and S_ℓ distinguish these conventions when both are compared. At a fixed equilibrium, the corresponding continuous-time survivor densities satisfy

$$\phi_\ell^\delta(x,d) = e^{-\delta d}\phi_\ell^0(x,d).$$

This relation leaves instantaneous labor-market hazards unchanged after conditioning on survival. It does not generally leave interval probabilities or pooled destination shares unchanged, because the factor reweights exits within each interval.

Hazards, interval probabilities, and transitions. At each duration, survival is the remaining cohort mass. For a feasible search destination ℓ', the destination exit density and instantaneous hazard are

$$S_\ell(d_n) = \Delta x\, \mathbf{1}'\phi_\ell^n,$$
$$f_\ell^{\ell'}(d_n) = \Delta x \sum_j \eta_\ell^{\ell'}(x_j)\phi_\ell(x_j,d_n), \quad \lambda_\ell^{\ell'}(d_n) = \frac{f_\ell^{\ell'}(d_n)}{S_\ell(d_n)}.$$

For an employment origin, exits to unemployment include both exogenous separation and quitting,

$$f_\ell^u(d_n) = \delta_\ell S_\ell(d_n) + \Delta x\, \mathbf{1}'\mathsf{T}_\ell\phi_\ell^n, \qquad \ell \in \{i,f\}.$$

The second term equals $\Delta x\, A_{\ell,k_\ell,k_\ell-1}\phi_\ell(x_{k_\ell},d_n)$ when a quitting boundary is present. Exit densities and hazards are evaluated before the propagation step in equation (C.2).

For a duration interval $I_d = (d-\Delta, d]$, the algorithm sums its destination exit densities and divides by survival at the start of the interval:

$$E_\ell^{\ell'}(I_d) = h \sum_{n:\, d-\Delta \le d_n < d} f_\ell^{\ell'}(d_n), \qquad p_\ell^{\ell'}(d) \simeq \frac{E_\ell^{\ell'}(I_d)}{S_\ell(d-\Delta)}.$$

The time grid subdivides each interval, and the left-endpoint rectangular sum integrates exits across its durations.

For the population mapping, assume that inclusion in the risk set, conditional on origin and survival to $d-\Delta$, does not select spells by their subsequent exit time or destination. Assume also that each included spell is observed until its first exit or duration d, with its destination recorded if it exits. These sufficient observation conditions give the exact mapping below and are not required for the sampling results.

Survival in the origin state through duration t, conditional on survival to the beginning of the interval, is

$$S_\ell(t \mid d-\Delta) = \frac{S_\ell(t)}{S_\ell(d-\Delta)} = \exp\left\{ -\int_{d-\Delta}^{t} \sum_k \lambda_\ell^k(v)\, dv \right\},$$

The sum includes all competing exits from the origin. For hazards and probabilities defined on the same transition process, integrating the survival-weighted destination hazard gives

$$p_\ell^{\ell'}(d) = \int_{d-\Delta}^{d} S_\ell(t \mid d-\Delta)\lambda_\ell^{\ell'}(t)\, dt. \tag{C.3}$$

The interval width Δ aggregates variation across durations. For intervals over which survival remains close to one and the hazard varies little, $p_\ell^{\ell'}(d)/\Delta$ approximates the instantaneous hazard. An exact model comparison instead integrates the exit density over the same duration interval.

For destinations r and s with $\lambda_\ell^s(t) > 0$ throughout the interval, equation (C.3) implies

$$\frac{p_\ell^r(d)}{p_\ell^s(d)} = \frac{\displaystyle\int_{d-\Delta}^{d} S_\ell(t \mid d-\Delta)\lambda_\ell^s(t)\frac{\lambda_\ell^r(t)}{\lambda_\ell^s(t)}\, dt}{\displaystyle\int_{d-\Delta}^{d} S_\ell(t \mid d-\Delta)\lambda_\ell^s(t)\, dt}.$$

The interval relative hazard is therefore a positive weighted average of the instantaneous ratio, bounded by its infimum and supremum within the interval and equal to it when constant. If the instantaneous ratio is monotone across duration, its values in later nonoverlapping intervals are ordered relative to those in earlier intervals, and their weighted averages retain that ordering.

This relative-hazard ordering also permits incomplete observation when the risk set at the beginning of the interval remains representative. Conditional on actual exit duration, require that the probability of recording the exit and its destination is common across destinations. That probability multiplies the common within-bin weight, so the positive-average argument continues to apply whenever the denominator is positive. Under this weaker condition, individual interval probabilities need not equal their completely observed counterparts in Equation (C.3).

The share of labor-market exits going to a particular destination is calculated from the same interval exit masses,

$$\chi_\ell^{\ell'}(I_d) = \frac{E_\ell^{\ell'}(I_d)}{\sum_{k\neq\ell} E_\ell^k(I_d)}.$$

The smooth profiles use the instantaneous ratio $\lambda_\ell^{\ell'}(d)/\sum_{k\neq\ell}\lambda_\ell^k(d)$, whereas the binned shares pool exit masses across the interval, conditional on a labor-market exit.

For the completed-spell transition matrix in Table **??**, the cohort is propagated with K_ℓ^δ until its remaining mass is negligible. Each origin-to-destination probability is calculated as

$$\frac{h\sum_n f_\ell^{\ell'}(d_n)}{\sum_{k\neq\ell} h\sum_n f_\ell^k(d_n)}.$$

Because labor-force exit removes workers from the cohort but is excluded from the denominator, the matrix conditions on the first labor-market spell completion occurring before labor-force exit. For employment-to-unemployment transitions, completion includes both separation and quitting.

Entry-cohort state shares. The labor-force-entry cohort begins with unemployed entrants drawn from g and follows them through repeated state changes. Stacking its three state densities in $z(t)$ gives the initial condition and numerical update

$$z^0 = (g', 0', 0')', \qquad z^{n+1} = z^n + h_n(G_0 - \delta \mathrm{Id})z^n.$$

Excluding replacement births keeps the calculation confined to the original entrants, who can change state repeatedly until labor-force exit. The final step to each requested observation time is shortened to land on that time. Conditional on continued labor-force participation, the state share is

$$\frac{\Delta x\, \mathbf{1}' z_\ell(t)}{\Delta x\, \mathbf{1}' z(t)}, \qquad \ell \in \{u, i, f\}.$$

For a reported interval, I average these monthly conditional shares using the supplied exposure weights. The reporting durations and their calendar conversion determine where the paths are evaluated.

Completed-spell wage moments. For an employment spell lasting t, the average wage is $t^{-1}\int_0^t w_\ell(x(s))\,\mathrm{d}s$. I calculate its first two moments by following new spells from $\phi_\ell(0)$ and propagating surviving mass together with the density vectors $R_\ell^{(1)}$ and $R_\ell^{(2)}$, weighted by accumulated wage income and its square. Starting both reward vectors at zero, the update with the killed propagator $P_\ell = \mathrm{Id} + hK_\ell^\delta$ is

$$\begin{aligned}
\phi_\ell^{n+1} &= P_\ell \phi_\ell^n, \\
R_\ell^{(1),n+1} &= P_\ell\big(R_\ell^{(1),n} + h w_\ell \odot \phi_\ell^n\big), \\
R_\ell^{(2),n+1} &= P_\ell\big(R_\ell^{(2),n} + 2h w_\ell \odot R_\ell^{(1),n} + h^2 w_\ell^2 \odot \phi_\ell^n\big).
\end{aligned}$$

Products and squares are elementwise, and the update adds the current wage increment before propagating mass and rewards through the same transition matrix.

For an employment state ℓ, let $-\ell$ denote the other employment state. The linear completion functional is

$$\mathcal{E}_\ell[q] = \Delta x\, \mathbf{1}'[\mathrm{diag}(\eta_\ell^{-\ell}) + \delta_\ell \mathrm{Id} + \mathsf{T}_\ell]q.$$

It counts direct employment moves, exogenous separations, and quits. Labor-force exit censors the wage spell and is excluded from this functional, while remaining a loss in P_ℓ. Completion flows are evaluated on the propagated mass and reward vectors at the positive right endpoint $t_{n+1} = (n+1)h$:

$$e_{0\ell}^{n+1} = \mathcal{E}_\ell[\phi_\ell^{n+1}], \qquad e_{r\ell}^{n+1} = \mathcal{E}_\ell[R_\ell^{(r),n+1}], \quad r \in \{1, 2\}.$$

The raw wage moments use the same duration-weighted completion mass in their denominators:

$$L^M_{1,\ell} = \frac{h\sum_n e^{n+1}_{1\ell}}{h\sum_n t_{n+1}e^{n+1}_{0\ell}}, \quad L^M_{2,\ell} = \frac{h\sum_n e^{n+1}_{2\ell}/t_{n+1}}{h\sum_n t_{n+1}e^{n+1}_{0\ell}}.$$

Stock weighting gives a completed spell of duration t weight t, so it contributes accumulated wage income to the first numerator and accumulated wage income squared divided by t to the second. Only positive durations enter the second calculation, and propagation stops once the remaining cohort mass falls below tolerance. The reported standard deviation is $\sqrt{L^M_{2,\ell} - (L^M_{1,\ell})^2}$, with scale conversions given in Appendix C.3.

C.3 External Calibration and Normalizations

I set the external parameters and labor-force exit rate before solving the model, convert the inputs to the model clock, and recover dollar units from the formal wage level after estimation.

External parameters. I calibrate ρ using annual average yields on five- and ten-year BCU bonds issued by the Banco Central de Chile. These bonds are denominated in the inflation-indexed Unidad de Fomento (Banco Central de Chile, 2026). Let $r^{\text{BCU}}_{m,y}$ denote the annual yield in decimal form for maturity m and year y. Treating each yield as an effective annual real return, I convert it to a continuously compounded rate and give each year and maturity equal weight:

$$\rho^{\text{yr}} = \frac{1}{28}\sum_{m\in\{5,10\}}\sum_{y=2002}^{2015}\log\bigl(1 + r^{\text{BCU}}_{m,y}\bigr) = 0.02597.$$

The resulting discount rate is 2.60 percent per year and is held fixed during estimation.

The OECD average tax wedge, t^{OECD}, expresses the tax burden for a single worker without children earning the average wage as a percentage of the employer's labor cost (OECD, 2026). I use Chile's `AV_TW` series from the `DSD_TAX_WAGES_COU@DF_TW_COU` country dataflow, version 2.1, downloaded on August 10, 2026. The series is constant at 7 percent over 2000–2015. I convert this labor-cost share to the model parameter τ using

$$\tau = \frac{t^{\text{OECD}}}{1 - t^{\text{OECD}}} = \frac{0.07}{1 - 0.07} = 0.07527.$$

I hold this value fixed during estimation.

I set $\varepsilon = 6.48$, the median elasticity of substitution across product groups estimated by Redding & Weinstein (2020) from barcode data.

Labor-force exit calibration. I construct the EPS age CDF used to calibrate δ from the same person-month stock sample as the aggregate labor-market moments, comprising men ages 15–64 with valid employment histories and positive sampling weights. Age a is measured continuously in years using age at the start of the spell and elapsed months within it. For each calendar month, I calculate a weighted empirical age CDF. At each observed age, I take the unweighted mean of these CDF values across person-month observations to obtain $\widehat{F}_{\text{age}}(a)$. Each month therefore contributes in proportion to its number of observations at that age.

Let $a_{\min}$ denote the minimum observed age. For ages at which the empirical CDF is below one, the fitted specification is

$$\log\left[1 - \widehat{F}_{\text{age}}(a)\right] = -\delta^{\text{yr}}(a - a_{\min}) + \nu_a,$$

where ν_a is the regression residual. I estimate the slope by weighted least squares without an intercept, weighting each age by its total person-month stock weight, and take its negative as the annual calibration $\widehat{\delta}$. Interpreting the age profile as stationary, I assign this rate to labor-force exit and replacement. The input also absorbs permanent panel attrition, which the data do not separately identify. The regression uses the linear log-survival specification on the restricted sample without an upper-truncation correction.

Time units. One unit of numerical time is one calendar year. I divide empirical durations recorded in months by 12 before evaluating the model profiles. Both the discount-rate calibration and the age-profile slope are already continuous-time annual rates, so $\rho = \rho^{\text{yr}}$ and $\delta = \widehat{\delta}$. Search intensities and separation rates are measured per year. The units of μ_ℓ and σ_ℓ are human capital per year and per square-root year, respectively. Wages, benefits, the recurring formal-state cost, and search costs are annual flows, whereas Ψ_f is a one-time cost. Appendix A.6 gives the transformations under alternative units of time.

Formal-stock wage scale and recovery of A. The empirical formal-stock scale s_f^D is the geometric mean monthly wage over the observed formal months in the wage sample of Appendix B.3. I average component log wages with weights equal to job duration times the survey weight and exponentiate the mean. The sample requires full wage coverage but includes left-censored and incomplete formal spells, subject to the wage sample's age and occupation restrictions. I compute its model counterpart from the equilibrium solved with $A = 1$:

$$s_f^M = a_f \exp\left\{\frac{\int_0^\infty \log x \, m_f(x)\,\mathrm{d}x}{\mathbf{M}_f}\right\}.$$

At positive grid points I evaluate the integral by rectangular quadrature. If the zero node carries formal mass, I retain that mass and approximate the density as constant over its cell $[0, \Delta x]$. The cell's average log human capital is

$$\frac{1}{\Delta x}\int_0^{\Delta x} \log x \, \mathrm{d}x = \log \Delta x - 1.$$

The calculation does not condition on spell completion.

The empirical scale s_f^D is monthly. Matching its annual value to the model scale gives

$$\widehat{A} = \frac{12 s_f^D}{s_f^M}.$$

This implements Equation (17) and expresses A in real 2010 U.S. dollars per year per unit of human capital, using the same solved equilibrium.

For wage comparisons with the data, I divide completed-spell means and standard deviations by the formal-stock wage scale, and variances by its square. Let $\widehat{L}_{k,\ell}$ denote the stock-weighted empirical kth raw moment of the arithmetic spell-average wage in sector ℓ, using completed spells that do not begin at the January 1980 left boundary. The normalized empirical mean and variance are

$$\frac{\widehat{L}_{1,\ell}}{s_f^D}, \qquad \frac{\widehat{L}_{2,\ell} - \widehat{L}_{1,\ell}^2}{(s_f^D)^2}, \qquad \ell \in \{f, i\}.$$

I apply the same conversion to the model moments using s_f^M.

D Additional Results

D.1 Additional Descriptive Evidence

This subsection collects supplementary sample and mobility evidence. Labor histories and analysis weights are constructed as described in Appendix B.

D.1.1 Additional Sample Descriptives

Placeholder. Select and assemble the existing descriptive profiles by age, education, industry, occupation, and self-employment status. For each exhibit, state the unit of observation, sample restrictions, and weight.

D.1.2 First-Time Job-Search Cohort Descriptives

Placeholder. Present the existing cohort descriptives and employment-state paths.

D.1.3 Wages by Completed Duration

Table D.1 reports the regressions behind Figure 3a. I regress the arithmetic spell-average wage constructed in Appendix B.3 on an intercept and completed duration, separately by sector, using the sector-spell flow weights in Appendix B.2. The figure's markers show weighted mean wages within pooled weighted deciles of duration, while its lines fit the individual spells. Shaded bands are 95 percent confidence intervals for the fitted means, clustered by worker.

Taking conditional expectations in Equation (15) relates these wage profiles to the model for employment paths without reflection. With Brownian increments measured from spell entry, $\overline{W}_T = T^{-1}\int_0^T W_t\,\mathrm{d}t$. All expectations below use flow weights and condition on the regression sample, with time and wage units matched to the regression:

$$\mathbb{E}[\overline{w}_\ell \mid T, \ell] = \mathbb{E}[w_0 \mid T, \ell] + a_\ell \left[\frac{\mu_\ell T}{2} + \sigma_\ell \mathbb{E}[\overline{W}_T \mid T, \ell]\right].$$

Completed duration can select both initial wages and shock histories, since search and quitting depend on the evolving human-capital path. In particular, the selected mean shock need not be zero. The fitted slope therefore combines deterministic wage growth with duration-related differences in entry wages and realized shocks.[30]

[30] If a worker remains employed after reaching zero human capital, reflection adds $a_\ell T^{-1}\int_0^T (L_t - L_0)\,\mathrm{d}t$ to Equation (15), where $L_t - L_0$ is the regulator accumulated since spell entry. This term vanishes when the worker quits at or before zero.

Table D.1: Spell-average wages and completed duration

	Formal	Informal
Duration (months)	2.92	1.38
	(1.22)	(1.88)
Constant	440.57	393.17
	(26.73)	(26.81)
Spells	3,705	2,372
Workers	2,197	1,639
R^2	0.004	0.001

Notes: Sector-specific flow-weighted regressions of the standardized monthly wage equivalent, in real 2010 USD, on completed duration in months. The sample comprises completed formal and informal sector-spells with valid positive wages throughout, for men ages 15–64 at spell start, beginning directly out of unemployment and lasting at most 36 months. January 1980 starts and spells containing armed-forces or unclassifiable occupations are excluded. SEs clustered by worker are in parentheses. Source: EPS 2002–2015, author's calculations.

D.1.4 Transition Matrices at Alternative Horizons

Table D.2 follows the person-month stock in each origin state and records its state one, six, or twelve months later, conditional on continuous observation. Workers may change states several times, including returning to the origin, before the destination is recorded. A spell's first-exit destination alone does not determine the state observed at the horizon.

Table D.2: Employment-state transition matrices at alternative horizons

Origin state	Formal	Informal	Unemployed
Panel A. One-month horizon			
Formal	99.31	0.20	0.49
Informal	0.62	98.79	0.59
Unemployed	1.71	0.77	97.52
Panel B. Six-month horizon			
Formal	96.76	1.17	2.07
Informal	3.49	94.01	2.50
Unemployed	7.89	3.35	88.76
Panel C. Twelve-month horizon			
Formal	95.06	2.13	2.81
Informal	6.33	90.05	3.62
Unemployed	11.77	4.98	83.25

Notes: Rows report employment state at month t, and columns report employment state at $t + h$. Panels A–C set $h = 1$, 6, and 12 and contain 12,539, 12,538, and 12,536 workers, respectively. Each panel requires continuous observation through its horizon, so intervening gaps are treated as censoring. Entries are row percentages and use person-month stock weights. The sample consists of men ages 15–64 at origin month t; armed-forces occupations are excluded.

D.1.5 Cross-Country Labor-Market Transitions

Table D.3 compares Chile with annual transitions obtained from the continuous-time generators in Bosch & Maloney (2010) for Argentina, Brazil, and Mexico. Under their broad Definition 1 classification, unemployment and time out of the labor force form a common non-employment state, while informal self-employment and informal salaried employment form a common informal state. The foreign state space is therefore (F, INF, N). Chile retains its baseline (F, I, U) classification, with its histories sampled on each country's interview schedule.

The synthetic panels use four interviews six months apart for Argentina, four consecutive monthly interviews repeated one year later for Brazil, and five quarterly interviews for Mexico. Interview timing is anchored at the start of the first non-left-censored spell, and a candidate window is retained only when the worker's state is observed at every required interview. In each of 1,000 Monte Carlo draws, one admissible window is selected uniformly per worker. Annual transitions are pooled using the person-month weight at the origin date and averaged across draws.

Table D.3: Cross-country labor-market states and annual transitions

	Argentina	Brazil	Mexico
Panel A. Aggregate state shares			
Formal	36.8	51.7	55.0
	(54.2)	(54.1)	(54.1)
Informal	39.8	28.2	31.5
	(19.0)	(19.2)	(19.0)
Non-employment	23.4	20.1	13.5
	(26.8)	(26.7)	(26.9)
Panel B. Annual transition probabilities			
$F \to F$	82.1	82.2	82.8
	(94.3)	(94.1)	(94.1)
$F \to I$	10.4	9.1	11.8
	(2.4)	(2.4)	(2.4)
$F \to U$	7.5	8.7	5.3
	(3.3)	(3.5)	(3.5)
$I \to F$	10.9	18.8	22.5
	(7.2)	(7.1)	(7.1)
$I \to I$	71.7	67.6	68.9
	(87.8)	(87.9)	(87.8)
$I \to U$	17.4	13.6	8.6
	(5.0)	(5.0)	(5.1)
$U \to F$	9.6	19.4	17.5
	(10.6)	(10.5)	(10.8)
$U \to I$	31.8	21.9	24.3
	(4.7)	(4.9)	(4.7)
$U \to U$	58.6	58.6	58.2
	(84.7)	(84.7)	(84.5)

Notes: Entries are percentages. In each cell, the first line is the Definition 1 foreign benchmark and the smaller parenthetical second line is the corresponding country-specific synthetic Chilean weighted Monte Carlo mean. Parentheses do not contain standard errors. The foreign state space is (F, INF, N), while the Chilean state space remains (F, I, U).

D.2 Robustness of the Empirical Evidence

The comparisons below examine alternative measurement and censoring choices for the empirical objects constructed in Appendix B.

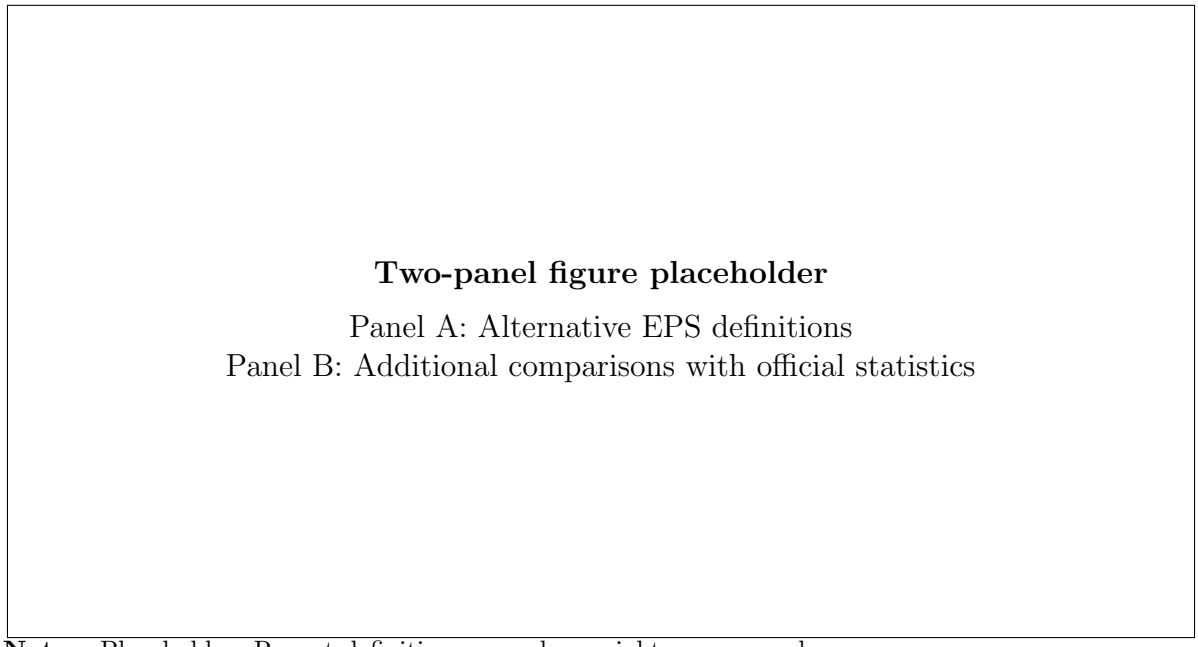
D.2.1 Alternative Informality Measures

Table D.4: Baseline and alternative definitions of informal employment

Definition	Classification rule	Empirical source or precedent
Baseline definition	—	—
Alternative definition 1	—	—
Alternative definition 2	—	—
Official-statistics definition	—	—

Notes: Placeholder. State the exact variables and coding rules used by each definition and explain how each maps to the model's formal and informal employment states.

Figure D.1: Robustness of the empirical informality measure

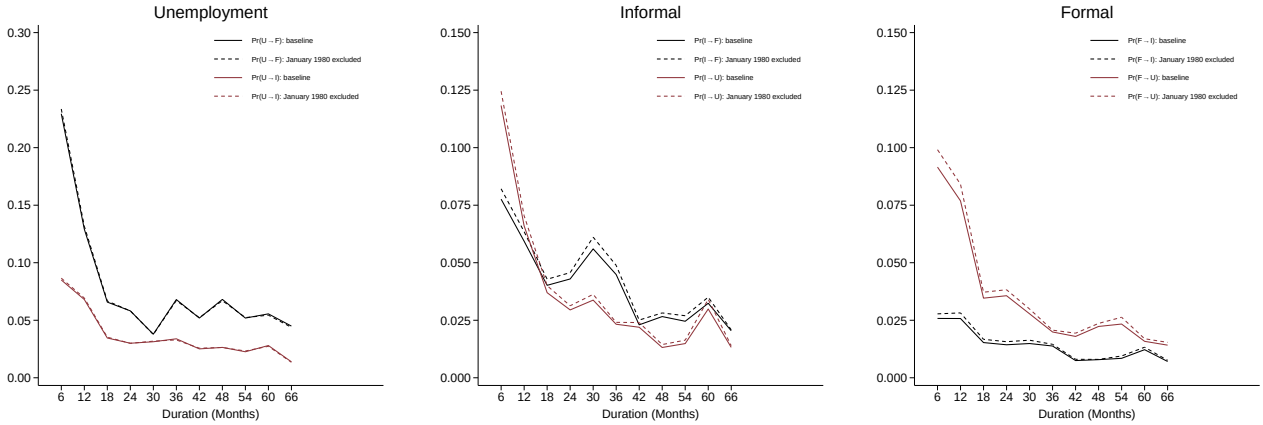


Notes: Placeholder. Report definitions, samples, weights, years, and sources.

D.2.2 Left-Censoring of Sector Spells

Figure D.2 recomputes the estimates after excluding spells beginning in January 1980, which may be left-censored at the beginning of the retrospective history window. The bins, weights, and inference conventions otherwise remain the same.

Figure D.2: Destination-specific hazards excluding January 1980 spells



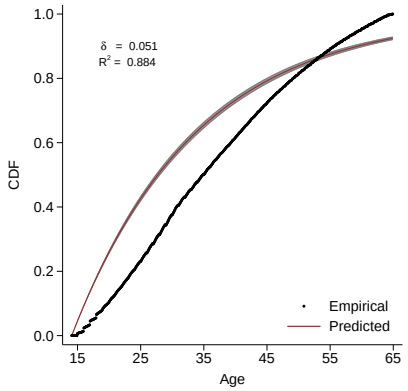
Notes: Solid lines report the baseline destination-specific hazards. Dashed lines exclude sector-spells first observed in January 1980, which are potentially left-censored because the retrospective labor-history window begins in that month. Excluding these spells removes observations whose elapsed duration at the beginning of the history window is unknown. Both specifications use six-month duration intervals and sector-spell flow weights. Panels are organized by origin state.

D.3 Additional Model and Counterfactual Results

D.3.1 Age Profile and Constant-Exit Fit

Figure D.3 reports the age profile and descriptive constant-exit fit. Appendix C.3 documents the calibration.

Figure D.3: Age Profile and Fitted Constant-Exit Model



Notes: Black markers report the empirical age CDF for all workers ages 15–64, constructed from person-month stock weights. The solid line is the corresponding descriptive constant-hazard fit, and the shaded region is its 95 percent confidence interval. The model's value of δ is estimated separately on men ages 15–64.

D.3.2 Income before Formal Entry or Labor-Force Exit

Table D.5 reports the stopped-income decomposition constructed in Appendix **??**.

Table D.5: Income before Formal Entry or Labor-Force Exit

Group	Informal wage premium	Stopping horizon	Total
Informality lowers access	+12,620	+38.9	+12,659
Informality raises access	+9,586	−5.22	+9,581
All entrants	+11,377	+20.8	+11,397

Notes: Entries decompose the with-minus-without difference in undiscounted wages and benefits accumulated until first formal entry or labor-force exit, in real 2010 USD, as in Equation (**??**). Workers start unemployed at the same initial human capital under baseline prices and policies, and rows group them by the sign of the access effect. The informal wage premium is informal earnings above benefits, and the stopping-horizon term values the difference in the length of the wait. Means include workers who exit without formalizing. Government transfers, search and entry costs, and earnings after formal entry are excluded.